

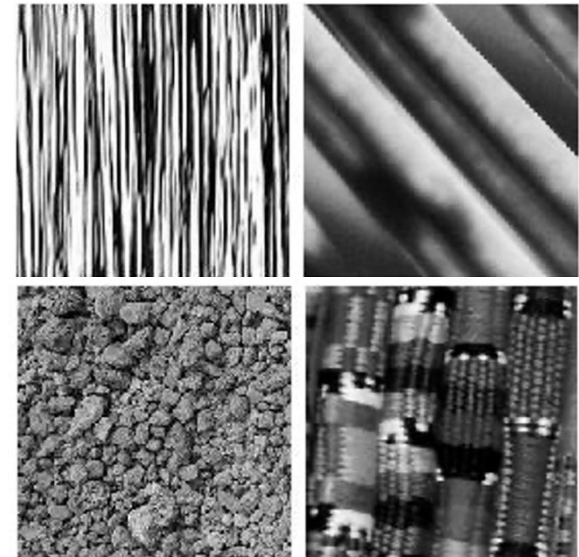


Extracción de Características de Intensidad

(CAPÍTULO 2 – PR02C)

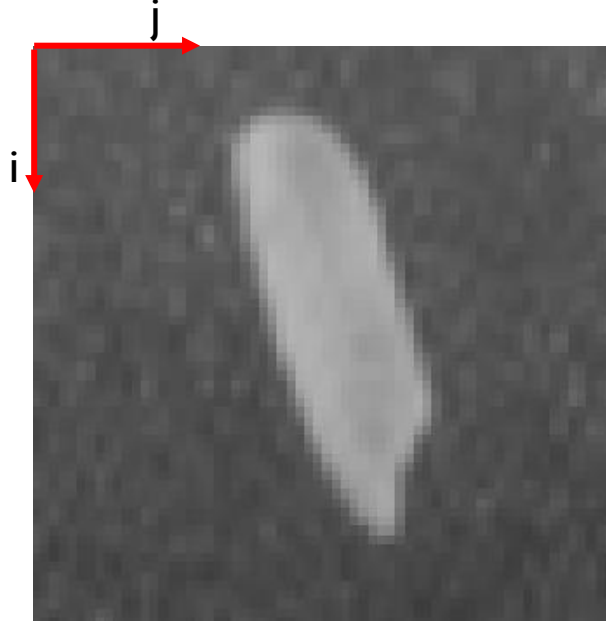
Domingo Mery 

Ciencia de la Computación
Universidad Católica – Chile
<http://domingomery.ing.uc.cl>

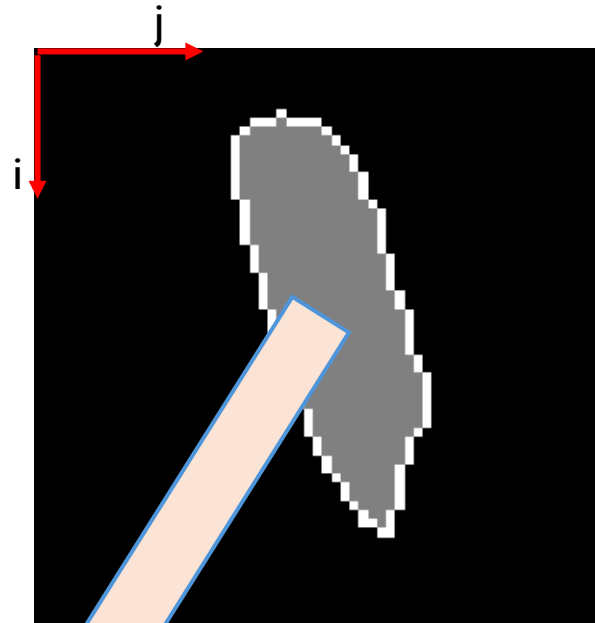


Introducción

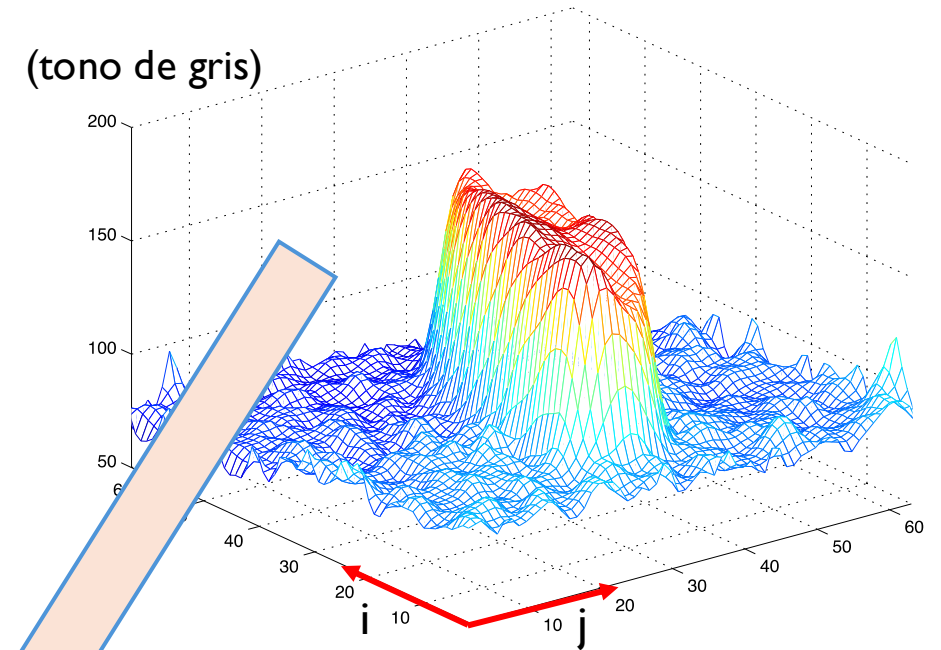
¿Qué es la Extracción de Características?



a) Imagen en escala de grises



b) Segmentación



c) Representación 3D de a)

Existen dos categorías de características:

1) Las **Características Geométricas** proporcionan información sobre traslación, rotación, forma y tamaño.

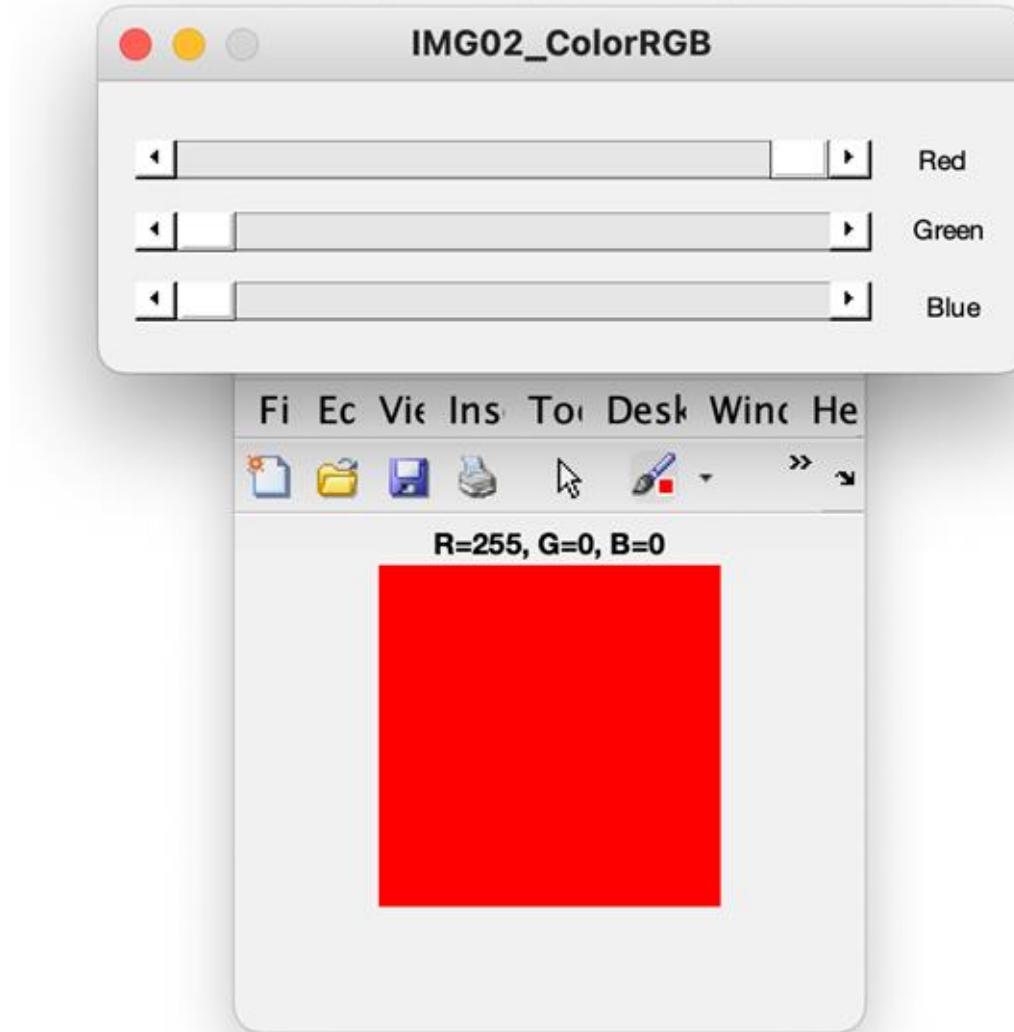
2) Las **Características de Intensidad** proporcionan información sobre cómo se distribuyen los tonos de grises (o los canales de color).

Espacio de Color RGB

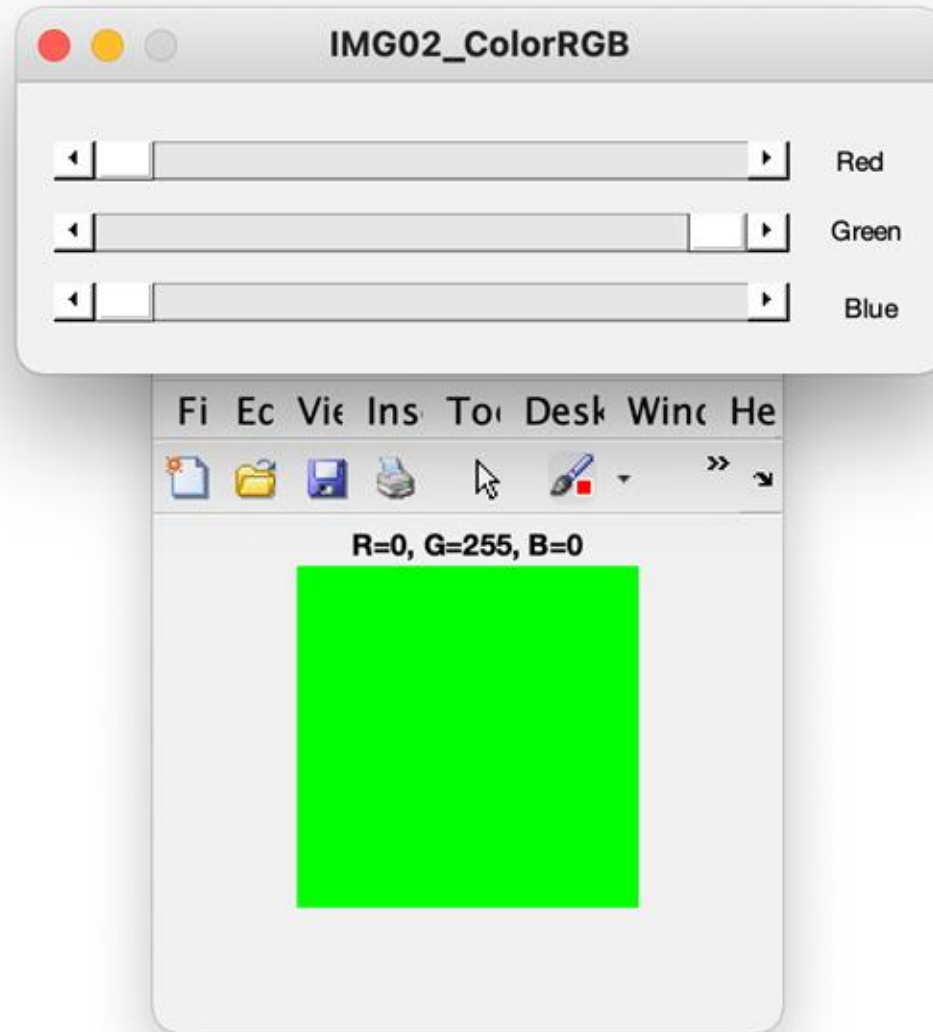
[Espacio de Color RGB]



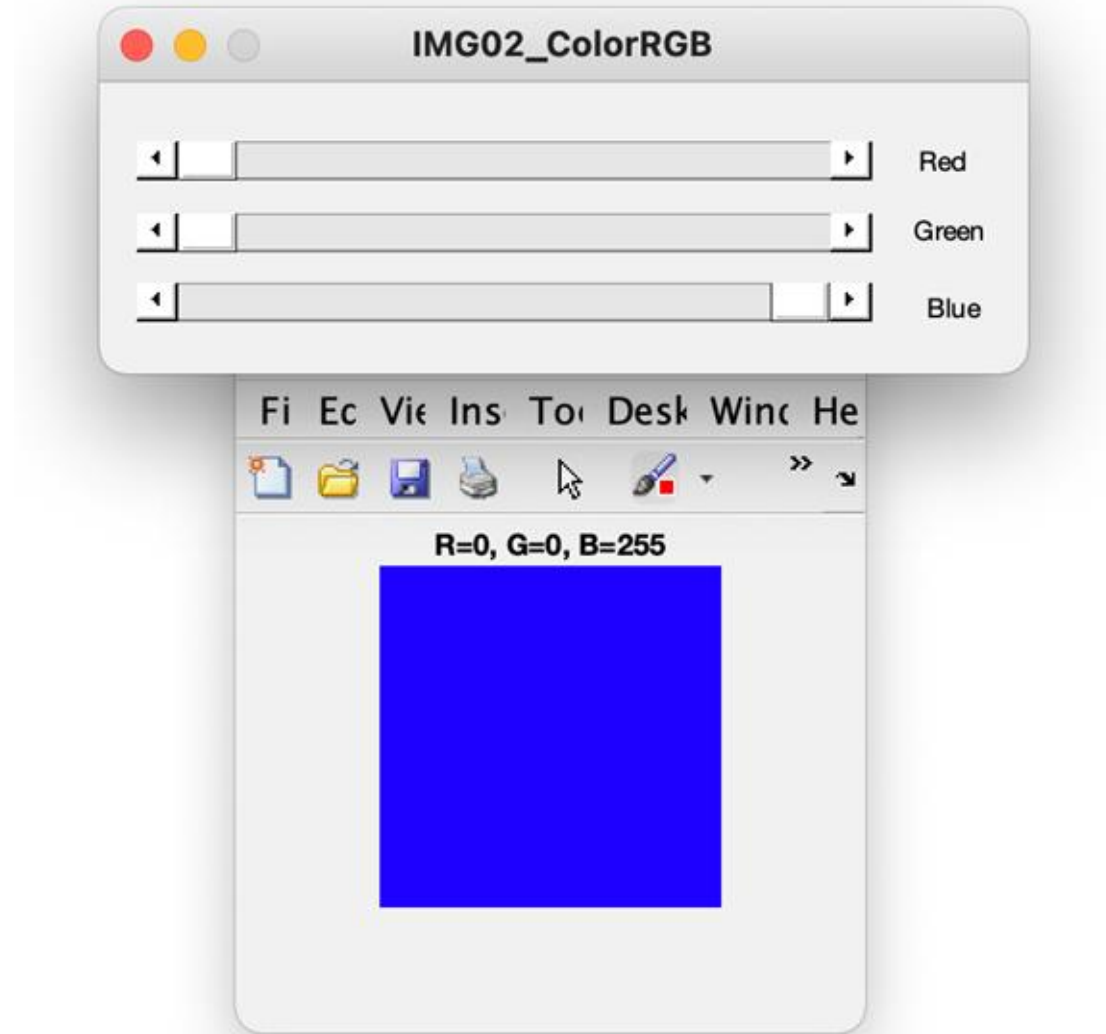
[Espacio de Color RGB]



[Espacio de Color RGB]



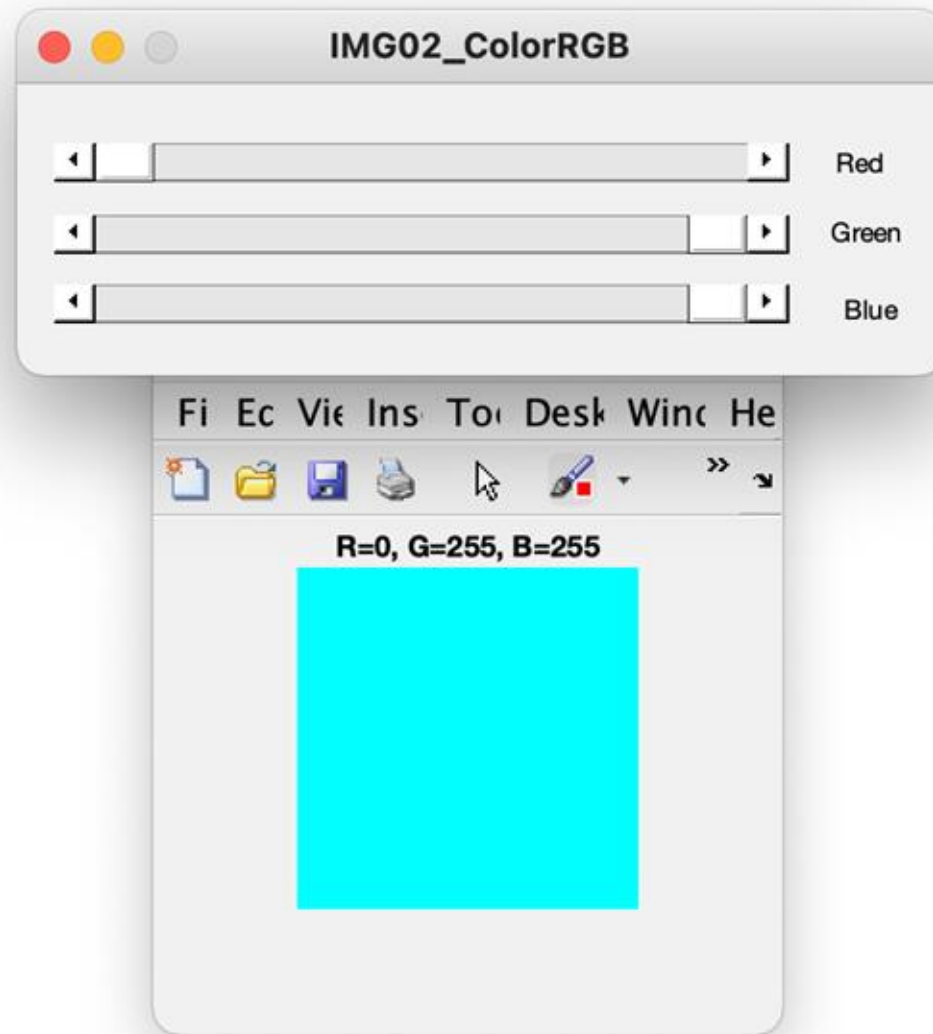
[Espacio de Color RGB]



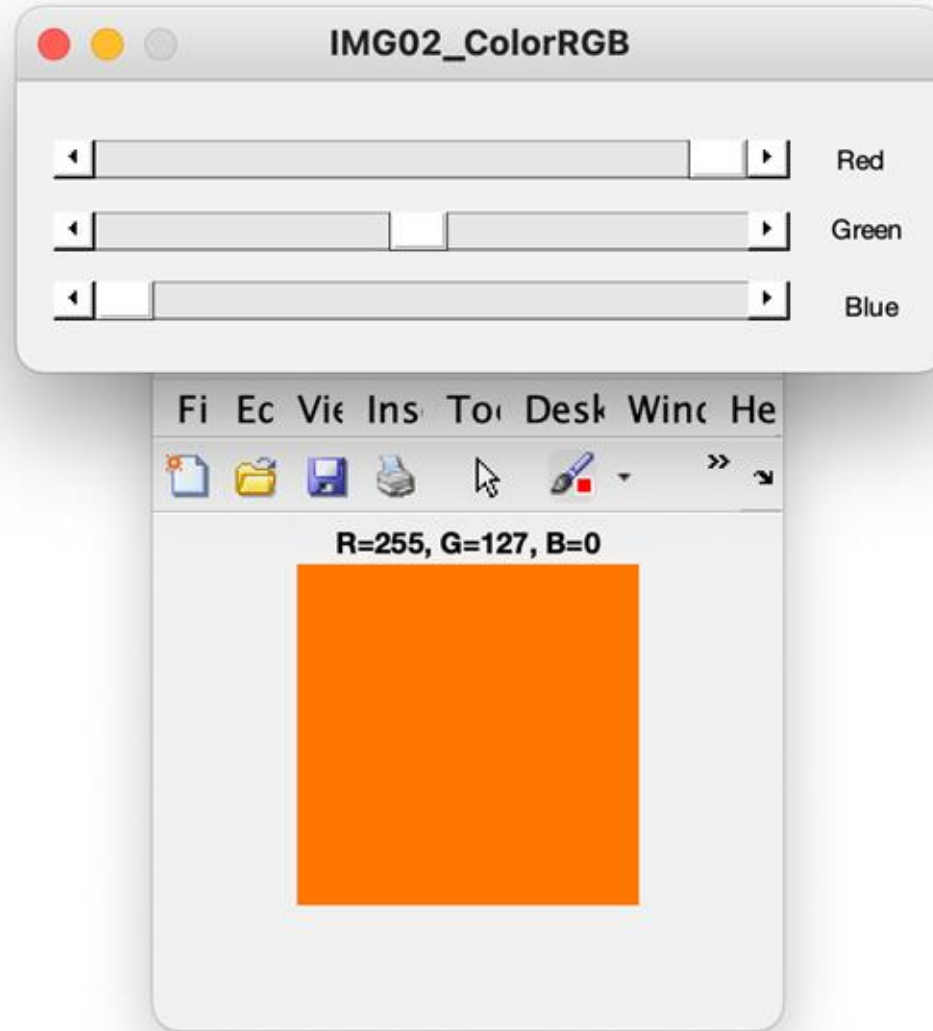
[Espacio de Color RGB]



[Espacio de Color RGB]



[Espacio de Color RGB]



[Espacio de Color RGB]



[Espacio de Color RGB]



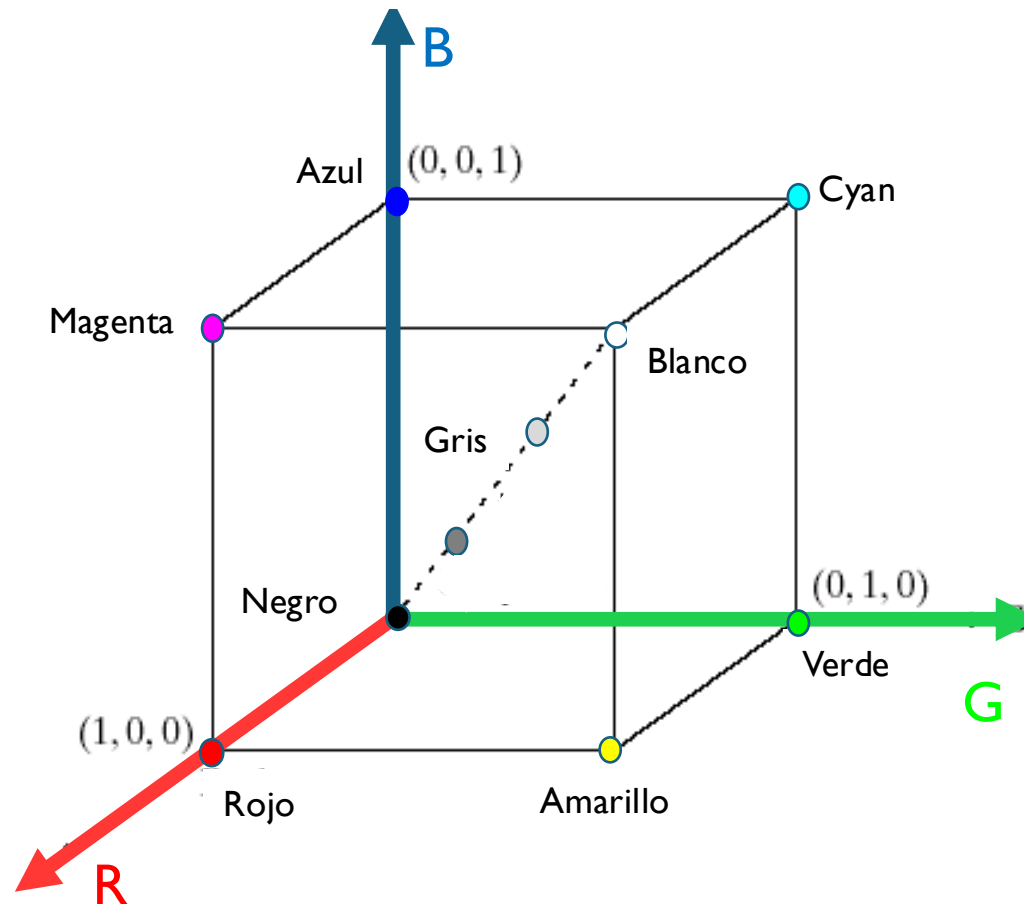
[Espacio de Color RGB]



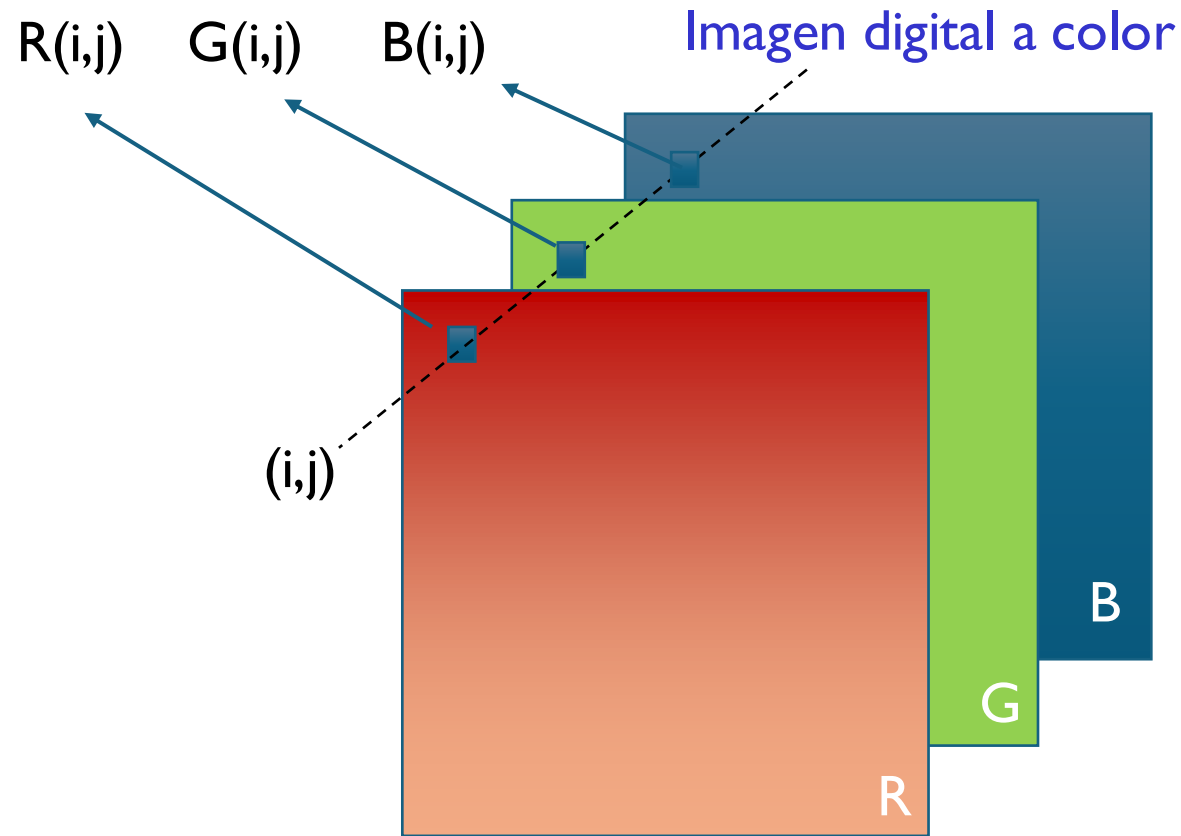
[Espacio de Color RGB]



[Espacio de Color RGB]



[Espacio de Color RGB: Cómo se almacena una imagen RGB]



La imagen a color de $N \times M$ pixeles es almacenada como tres matrices de $N \times M$ pixeles, una para cada color (R,G,B). Así, el color del pixel (i,j) queda definido por el color dado por $R(i,j)$, $G(i,j)$ y $B(i,j)$.

Ejemplo de Segmentación



Ejemplo de Segmentación



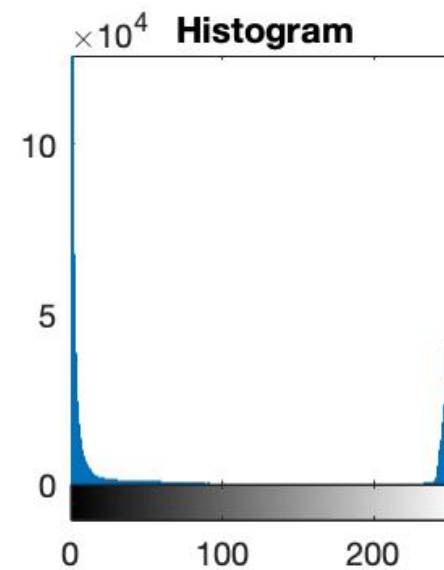
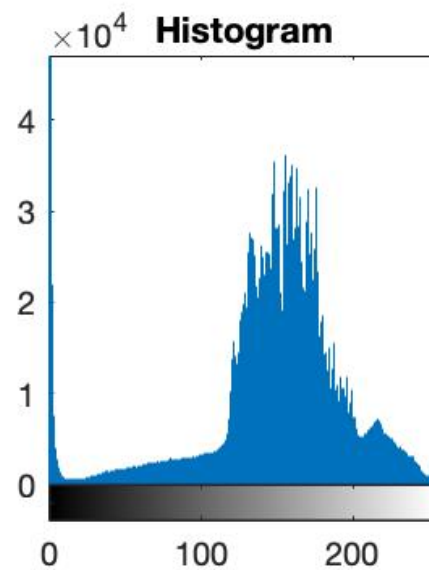
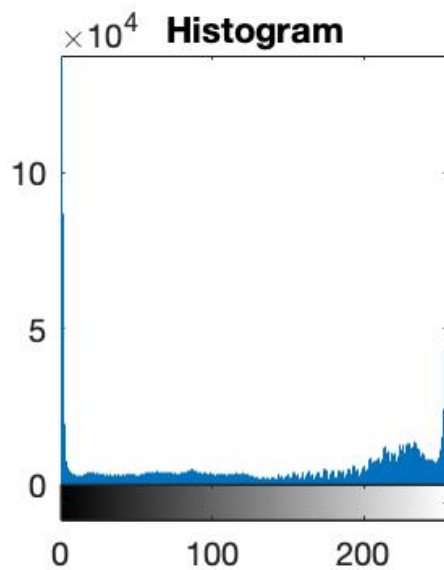
Red



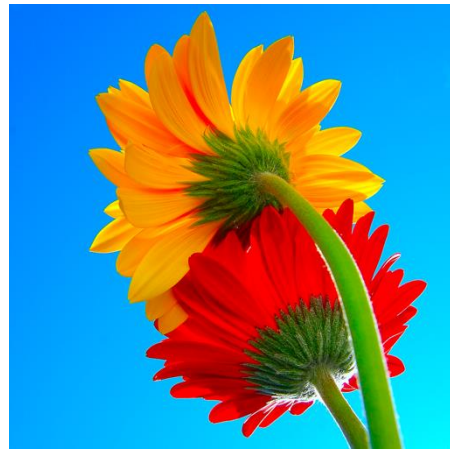
Green



Blue



Segmentación por canal de color



Descomposición de
Colores

R

Proc
R

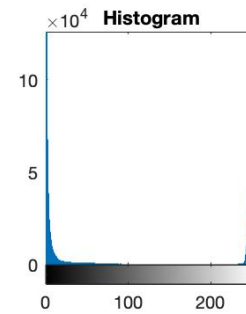
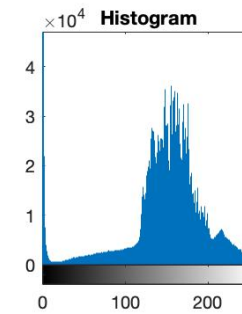
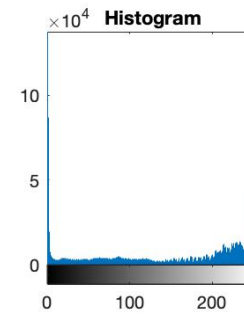
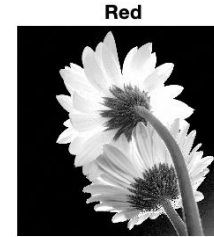
G

Proc
G

B

Proc
B

Composición



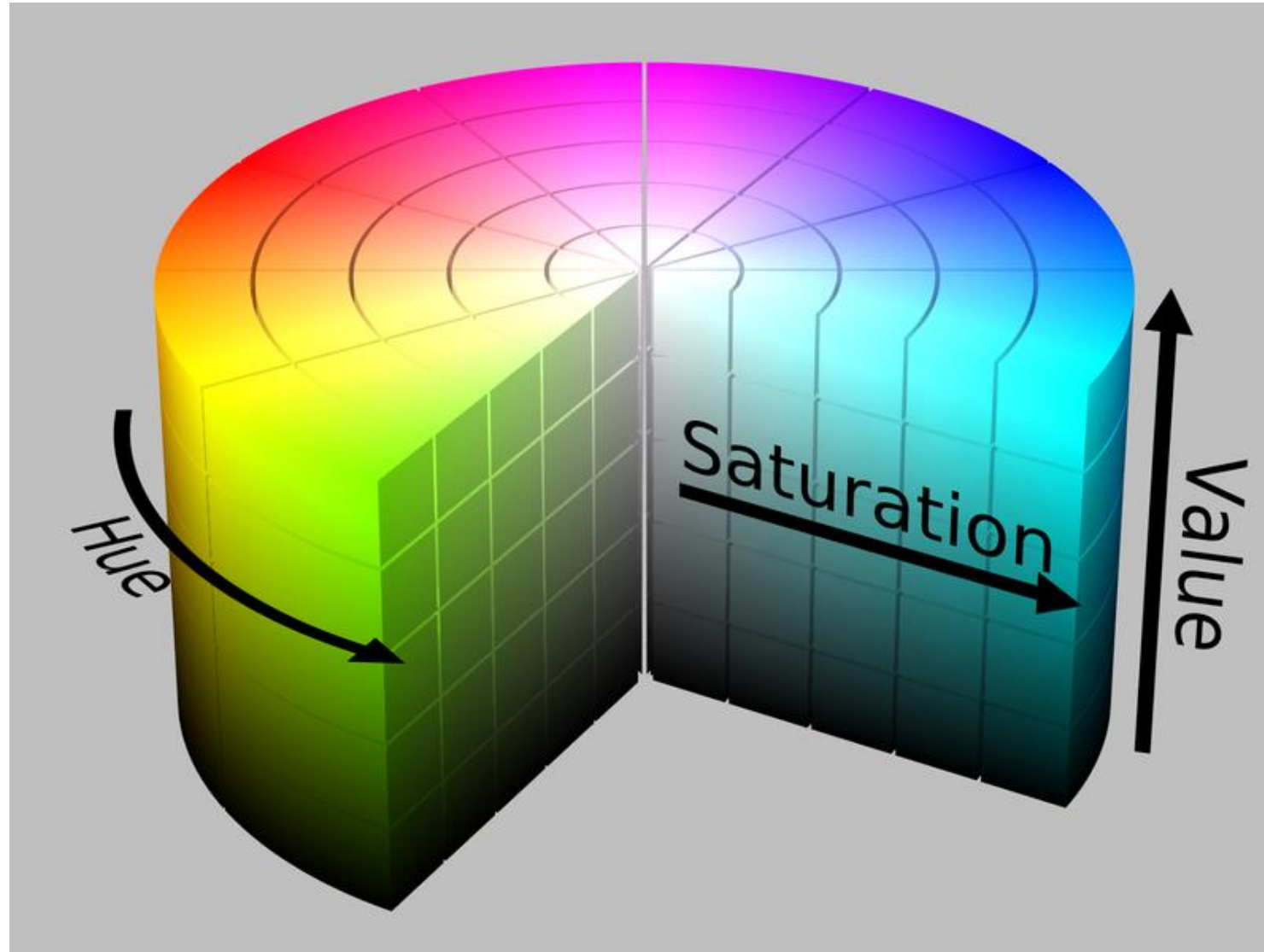






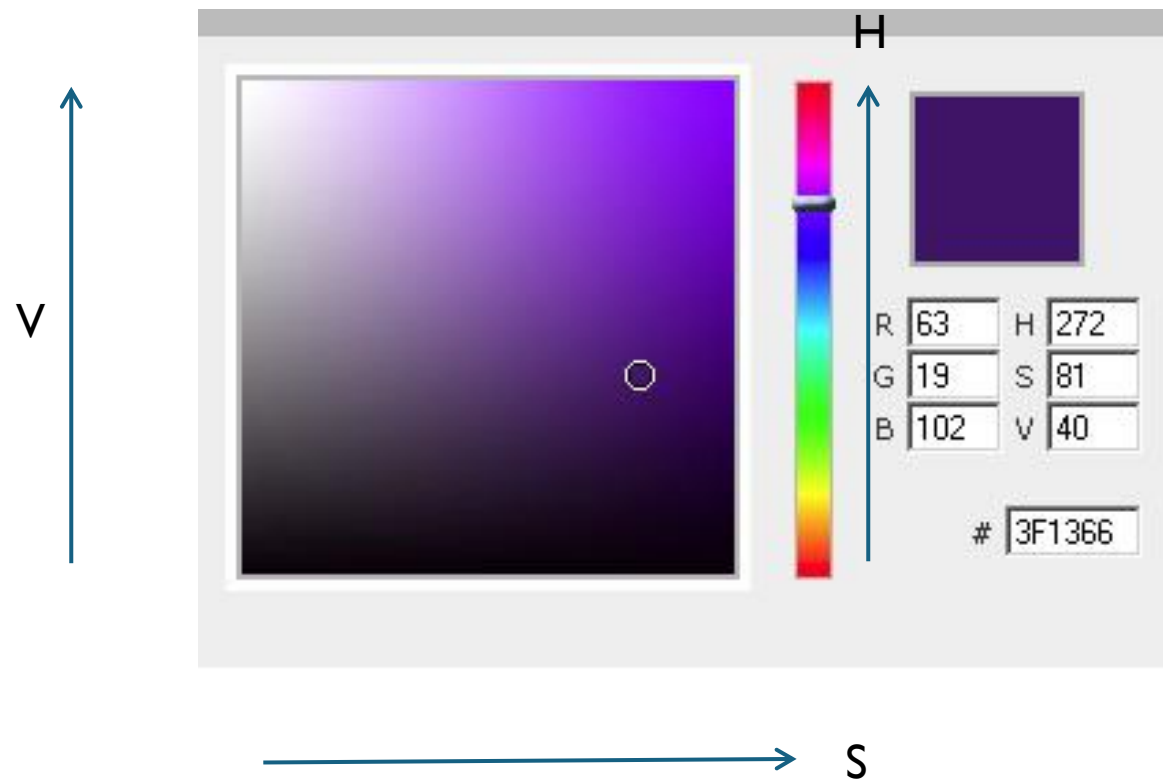


Espacio de Color HSV

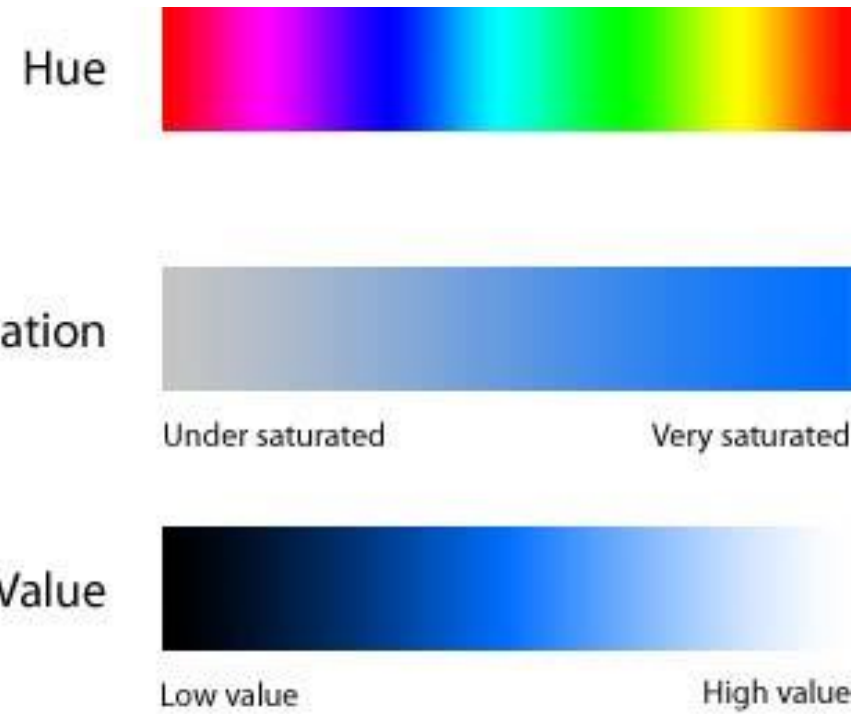


Color Picker Tool:

<https://url-decode.com/tool/hsv-to-rgb>



[Espacio de Color HSV]



Conversión RGB a HSV

The R, G, B values are divided by 255 to change the range from 0..255 to 0..1:

$$R' = R/255$$

$$G' = G/255$$

$$B' = B/255$$

$$C_{max} = \max(R', G', B')$$

$$C_{min} = \min(R', G', B')$$

$$\Delta = C_{max} - C_{min}$$

Hue calculation:

$$H = \begin{cases} 0^\circ & \Delta = 0 \\ 60^\circ \times \left(\frac{G' - B'}{\Delta} \bmod 6 \right) & , C_{max} = R' \\ 60^\circ \times \left(\frac{B' - R'}{\Delta} + 2 \right) & , C_{max} = G' \\ 60^\circ \times \left(\frac{R' - G'}{\Delta} + 4 \right) & , C_{max} = B' \end{cases}$$

Saturation calculation:

$$S = \begin{cases} 0 & , C_{max} = 0 \\ \frac{\Delta}{C_{max}} & , C_{max} \neq 0 \end{cases}$$

Value calculation:

$$V = C_{max}$$

Conversión HSV a RGB

When $0 \leq H < 360$, $0 \leq S \leq 1$ and $0 \leq V \leq 1$:

$$C = V \times S$$

$$X = C \times (1 - |(H / 60^\circ) \bmod 2 - 1|)$$

$$m = V - C$$

$$(R', G', B') = \begin{cases} (C, X, 0) & , 0^\circ \leq H < 60^\circ \\ (X, C, 0) & , 60^\circ \leq H < 120^\circ \\ (0, C, X) & , 120^\circ \leq H < 180^\circ \\ (0, X, C) & , 180^\circ \leq H < 240^\circ \\ (X, 0, C) & , 240^\circ \leq H < 300^\circ \\ (C, 0, X) & , 300^\circ \leq H < 360^\circ \end{cases}$$

$$(R, G, B) = ((R' + m) \times 255, (G' + m) \times 255, (B' + m) \times 255)$$

[Cómo mejorar una imagen a color]



[Cómo mejorar una imagen a color]

1. Tomar la imagen a color
2. Convertir RGB a HSV
3. Ecualizar sólo el canal V
4. Convertir el nuevo HSV a RGB
5. Fin



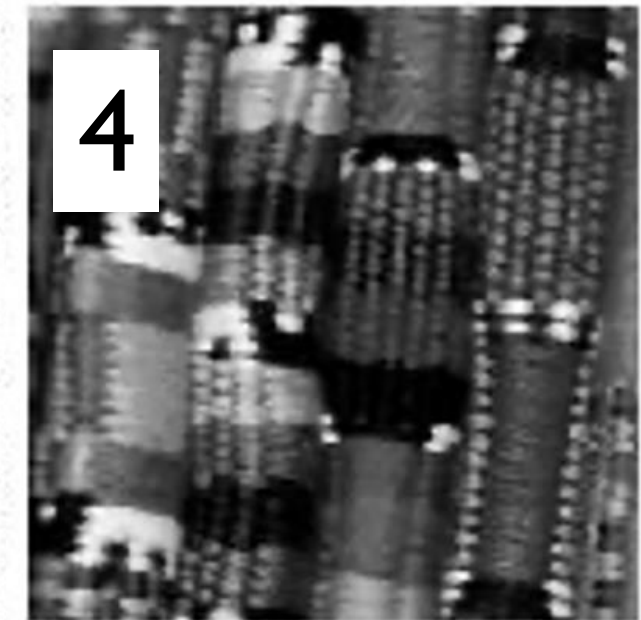
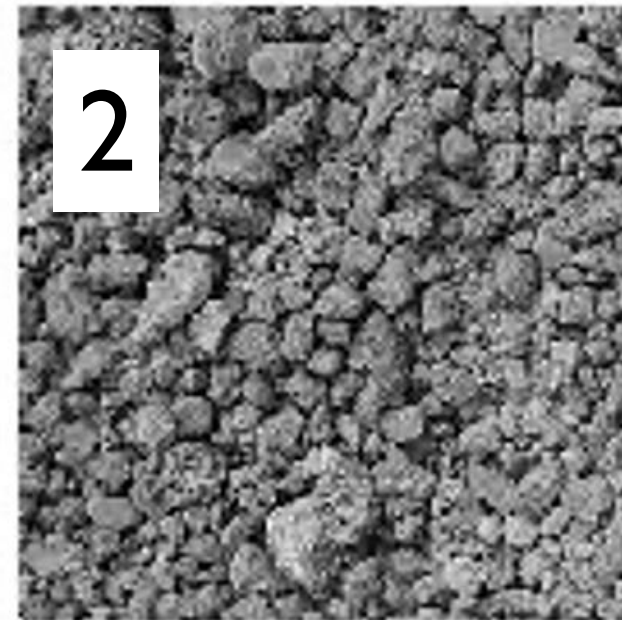
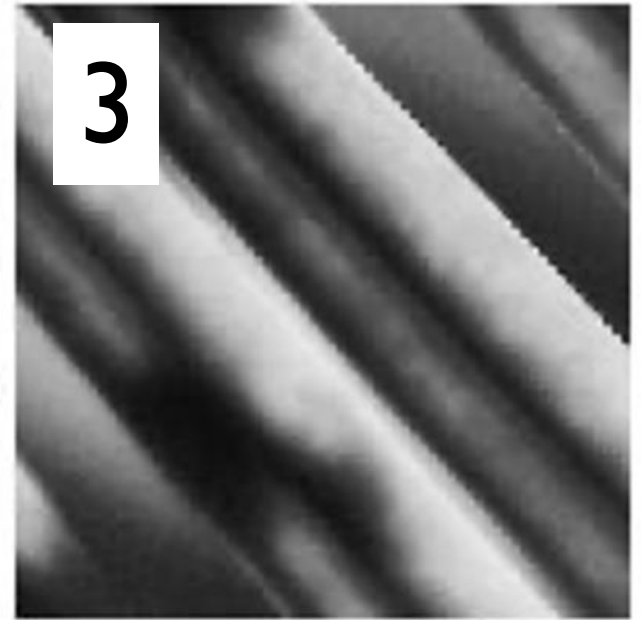
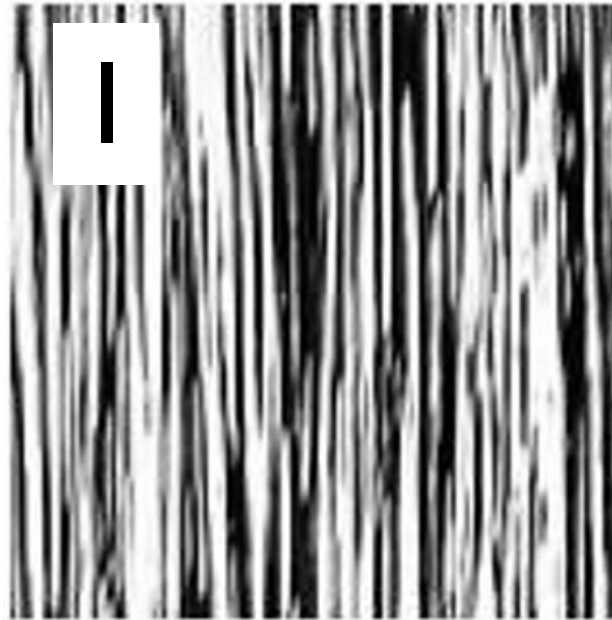


Características de Intensidad

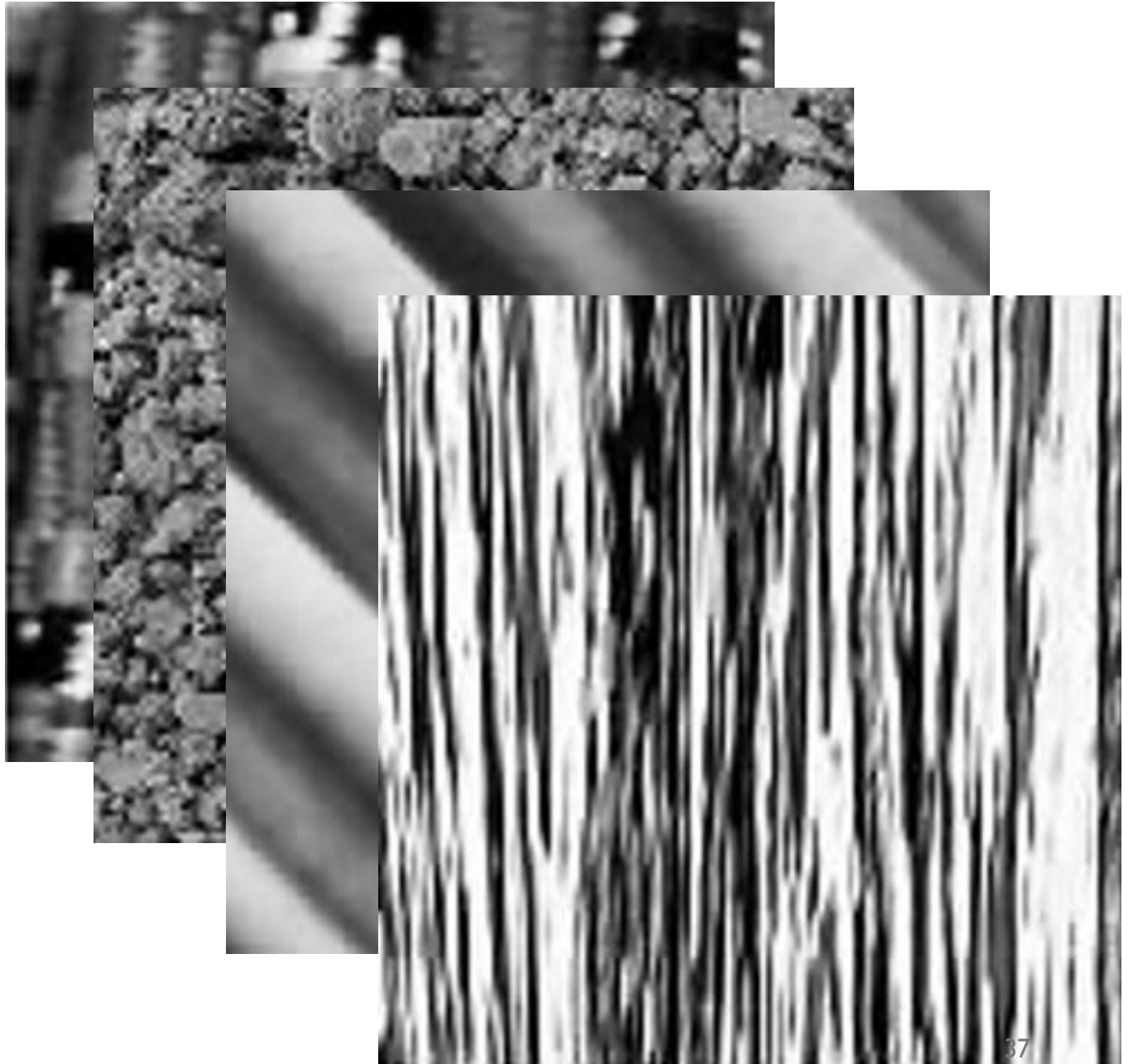
Características de Textura según Haralick

Haralick, R. M. (1979). Statistical and structural approaches to texture. *Proceedings of the IEEE*, 67(5), 786-804.

Memorice estas
cuatro texturas

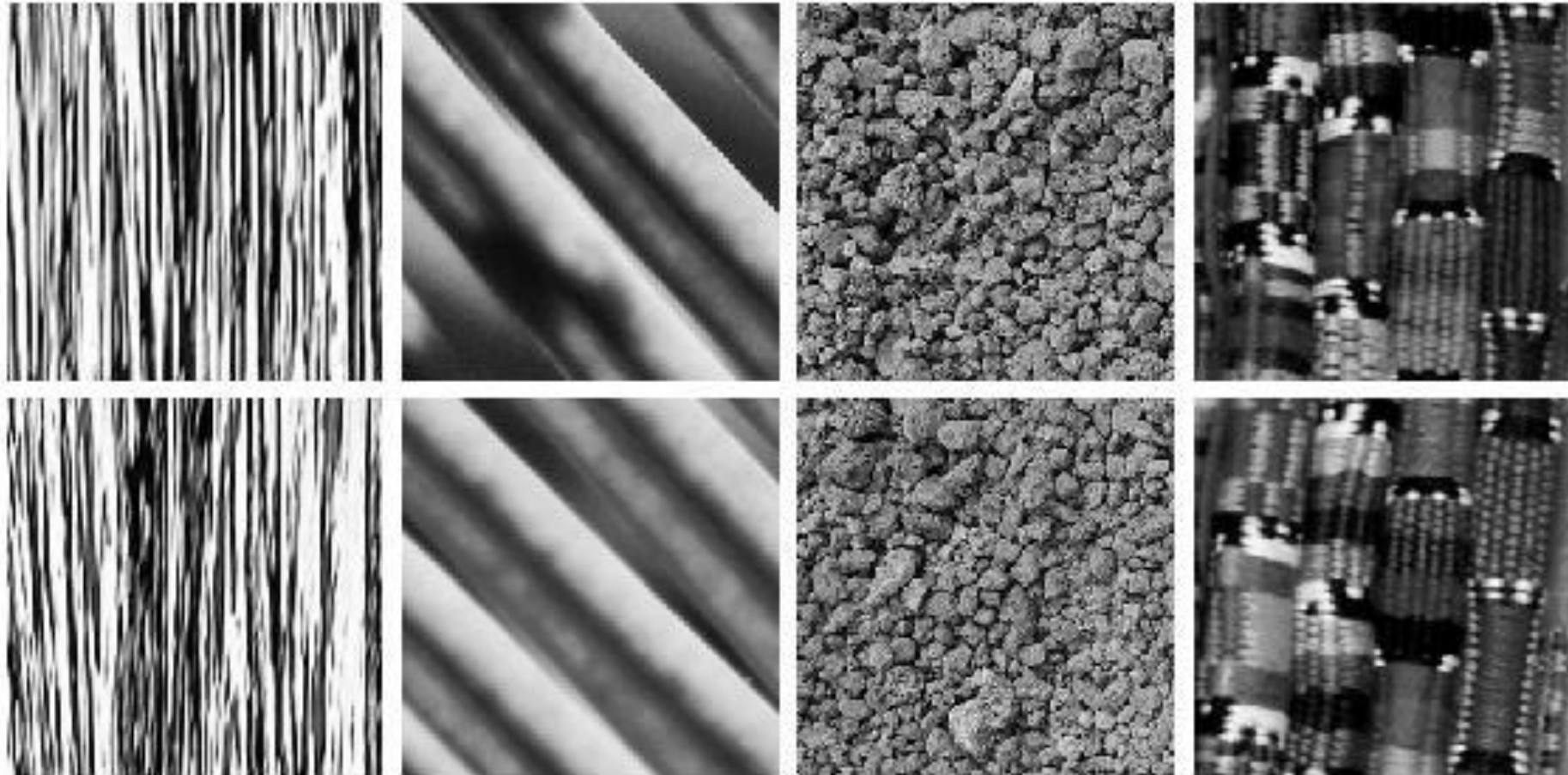


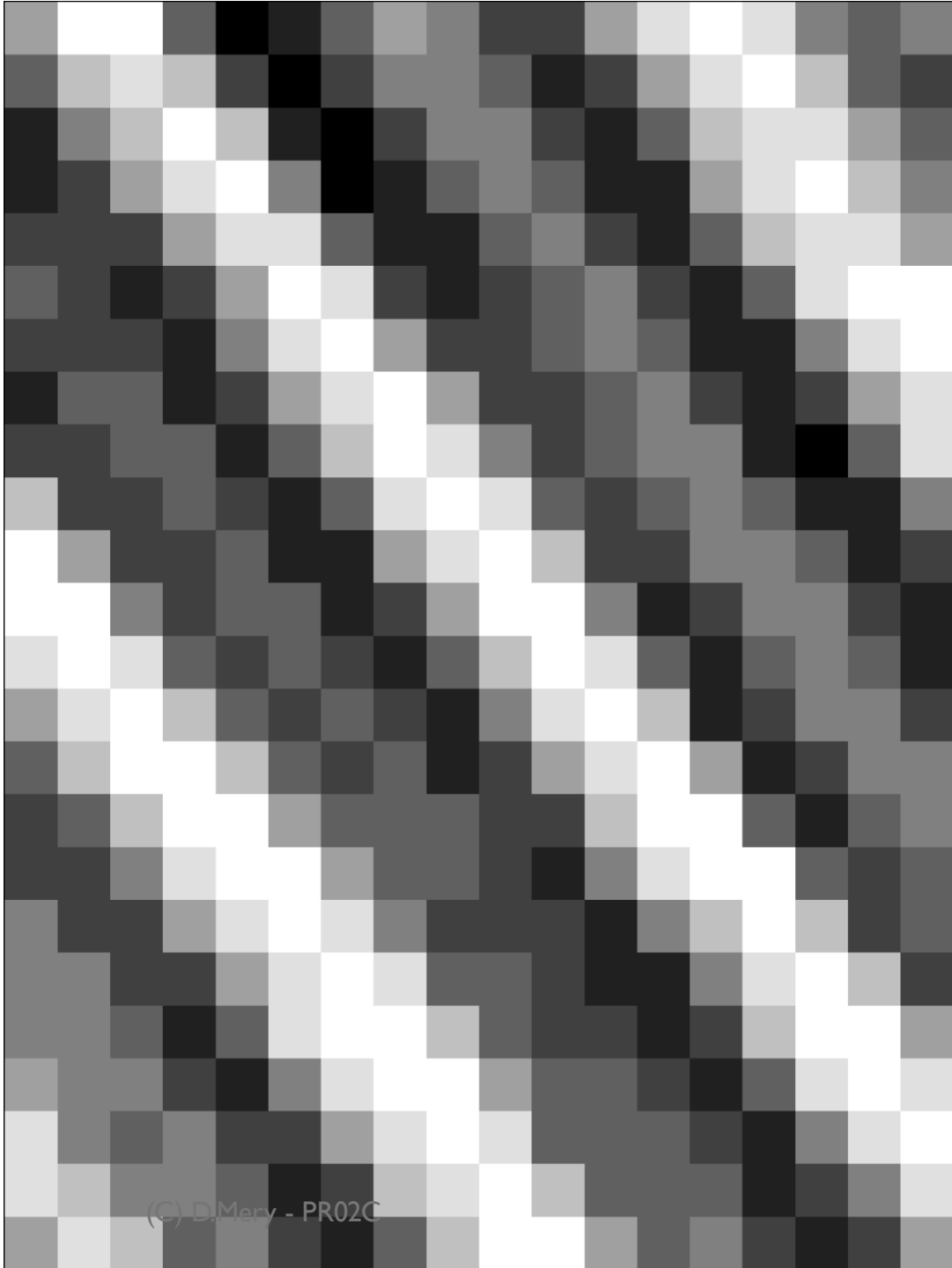
Diga a qué textura
pertence



Texturas Diferentes →

Texturas Similares →





Ejemplo:
¿Cómo caracterizar la textura
de esta imagen?

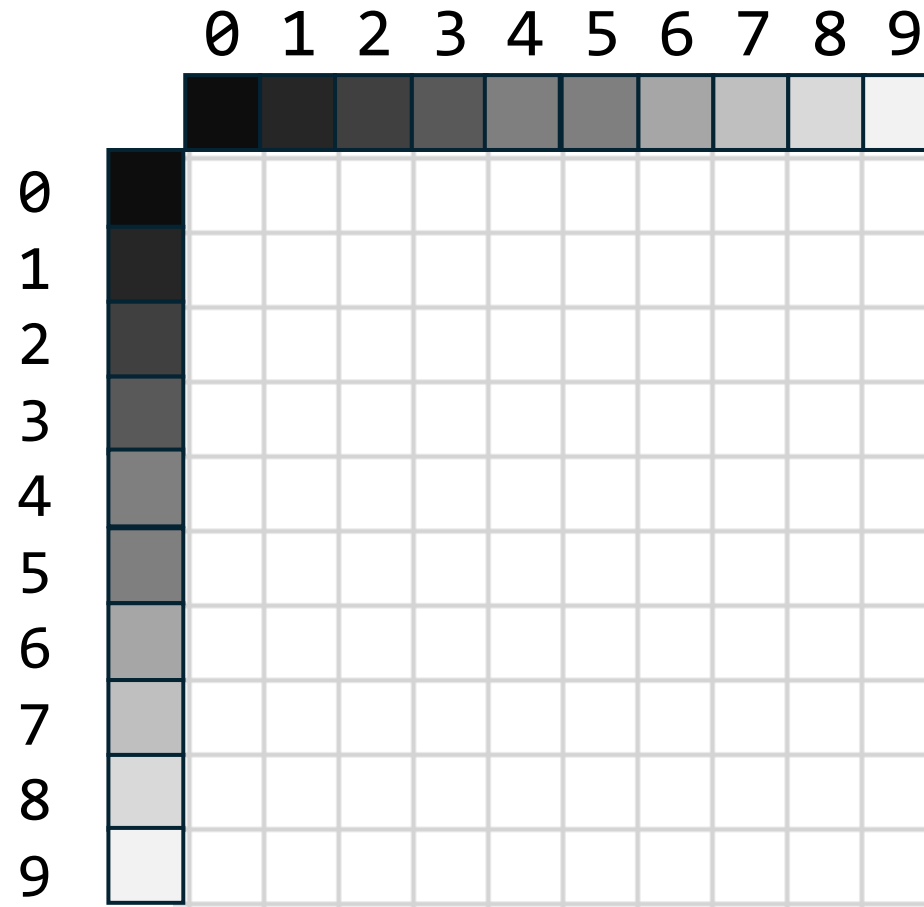


Ejemplo:

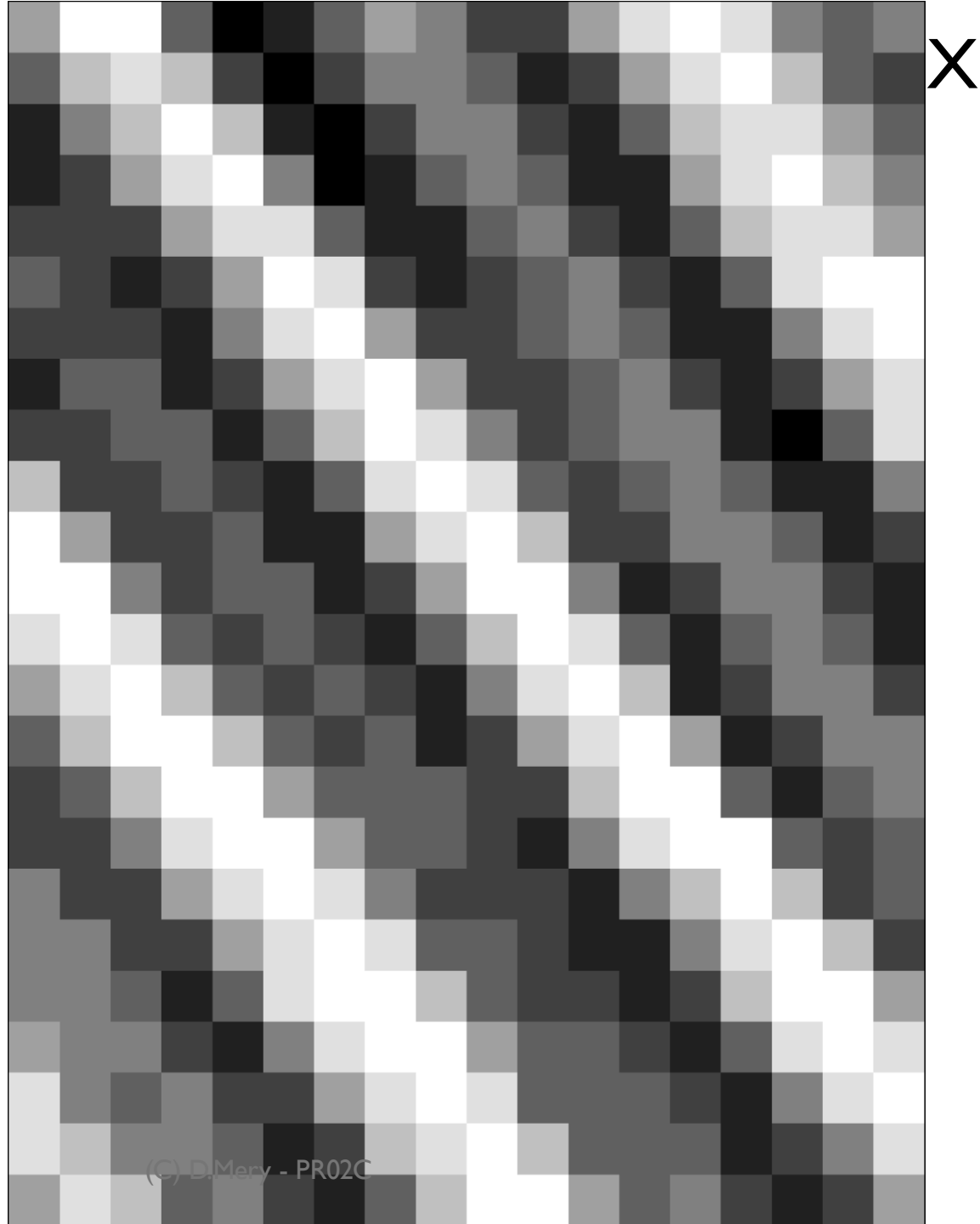
¿Cómo caracterizar la textura de esta imagen?

¿Y cómo diferenciarla de ésta?

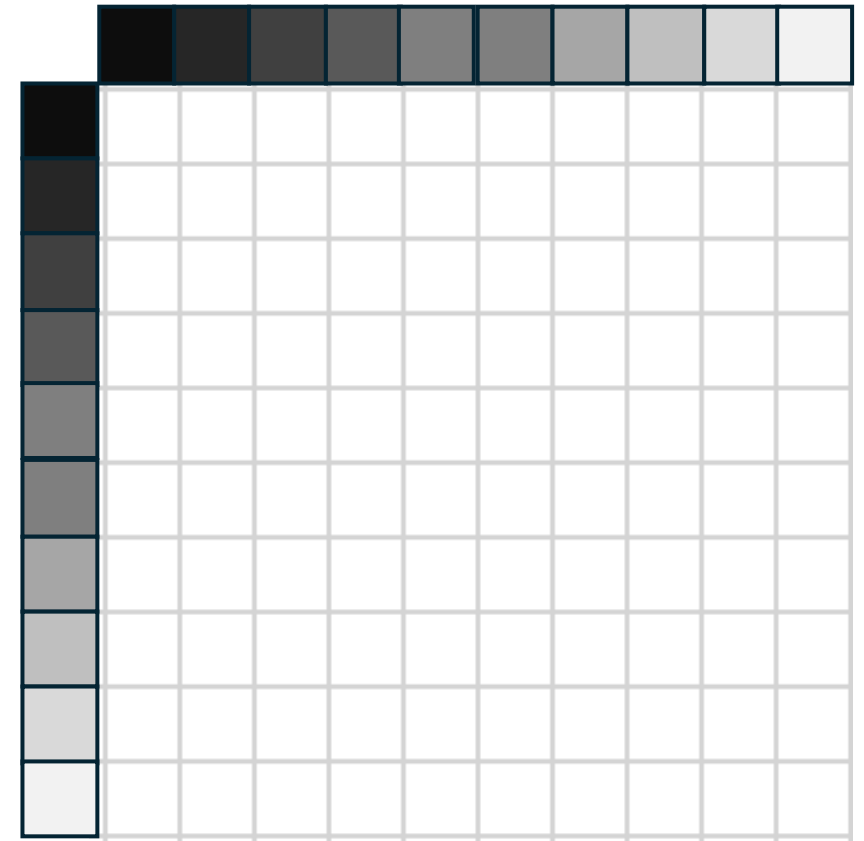
Matriz de Co-ocurrencia



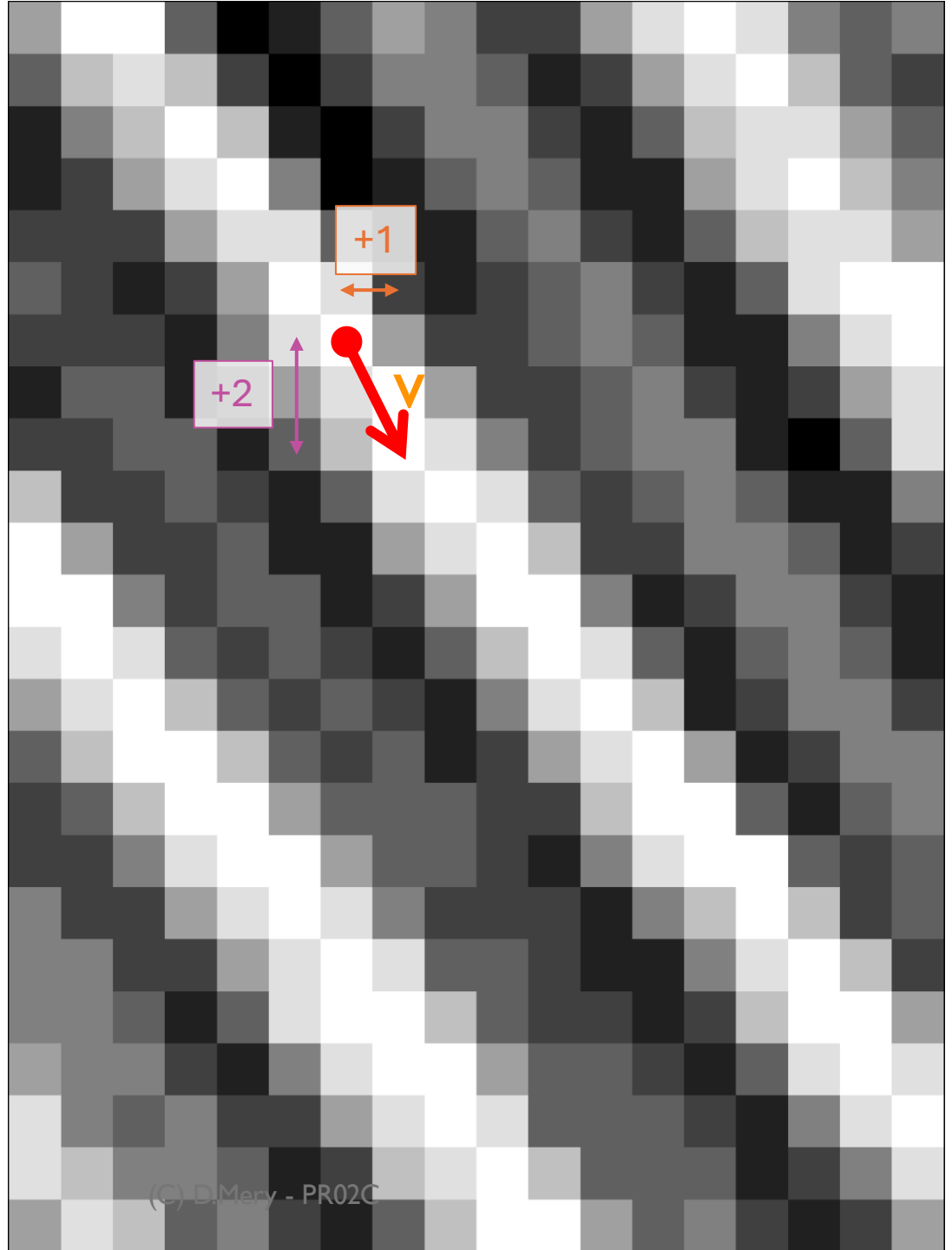
Ejemplo simple para una imagen de
10 tonos de gris



Matriz de Co-ocurrencia (para imagen X)

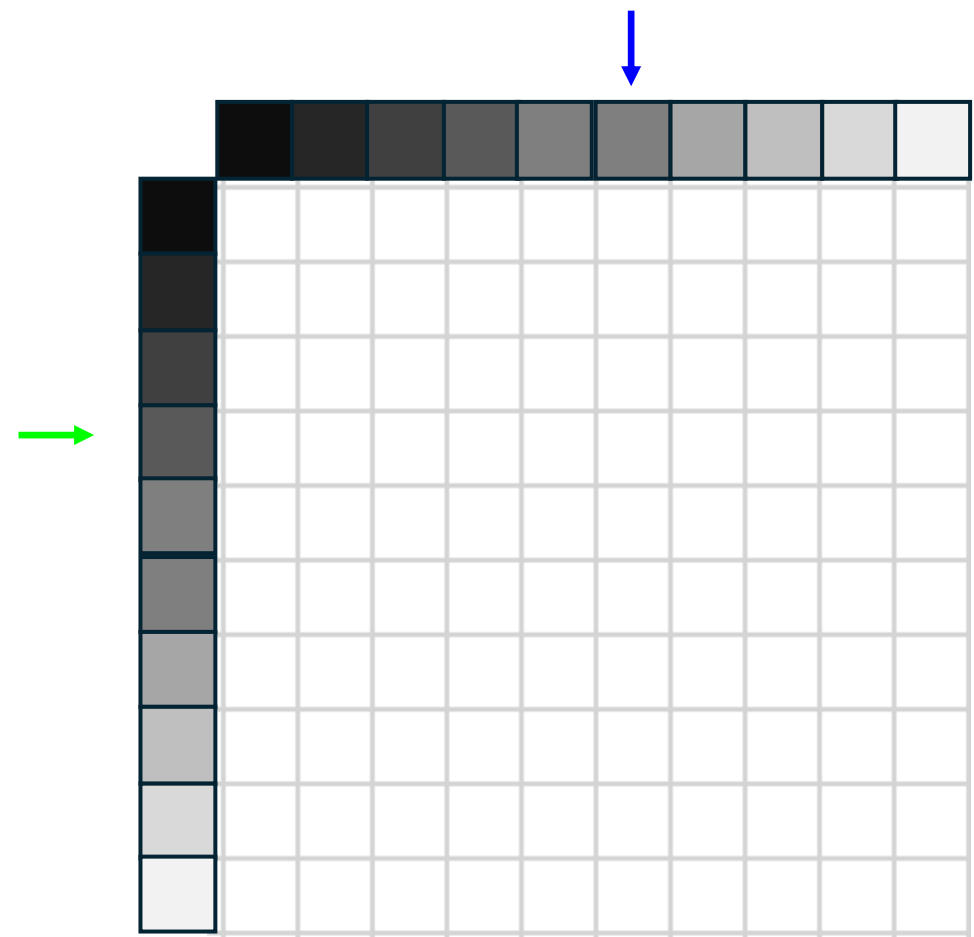


Contaremos ocurrencia
de pares de pixeles

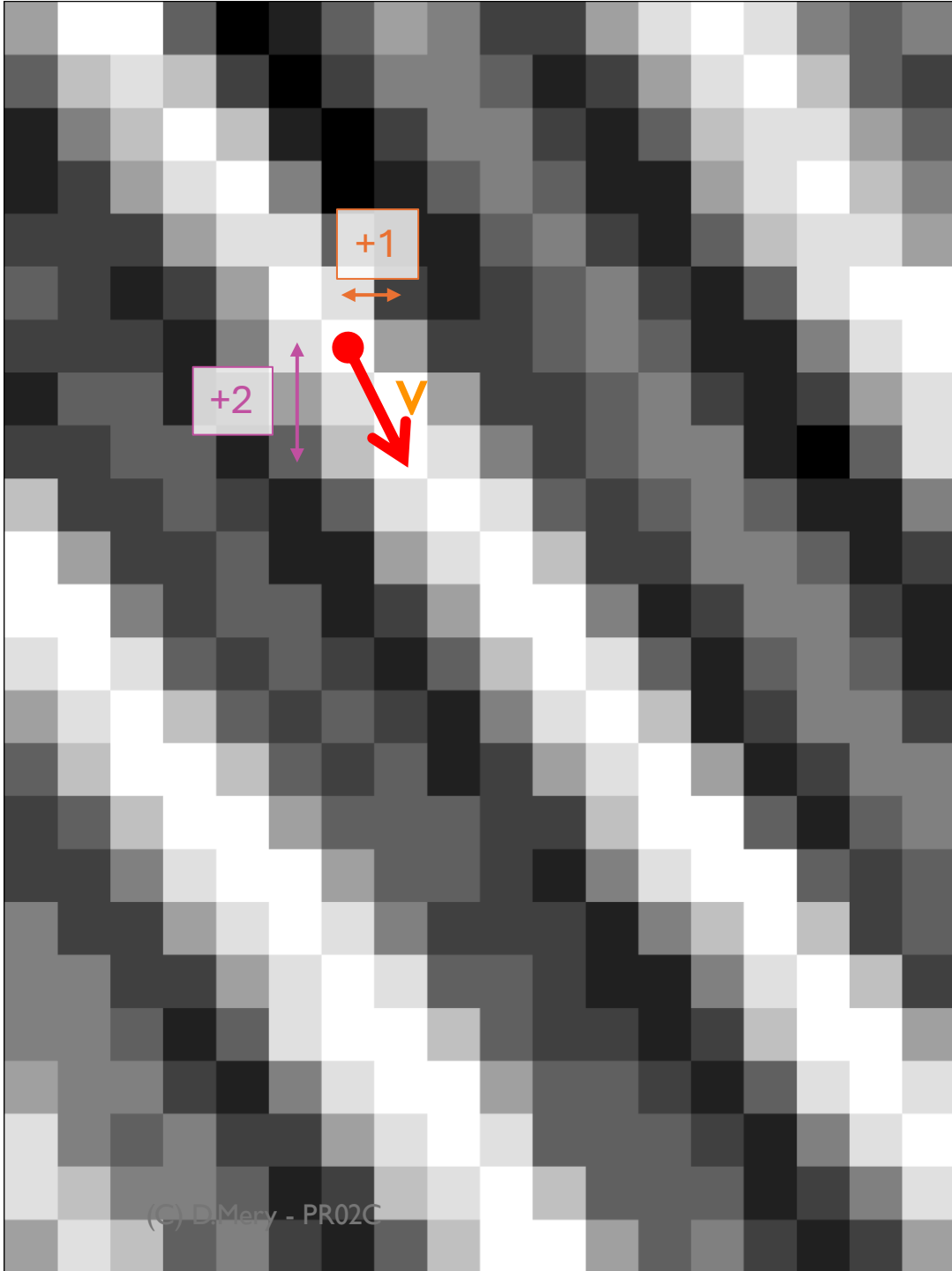


X

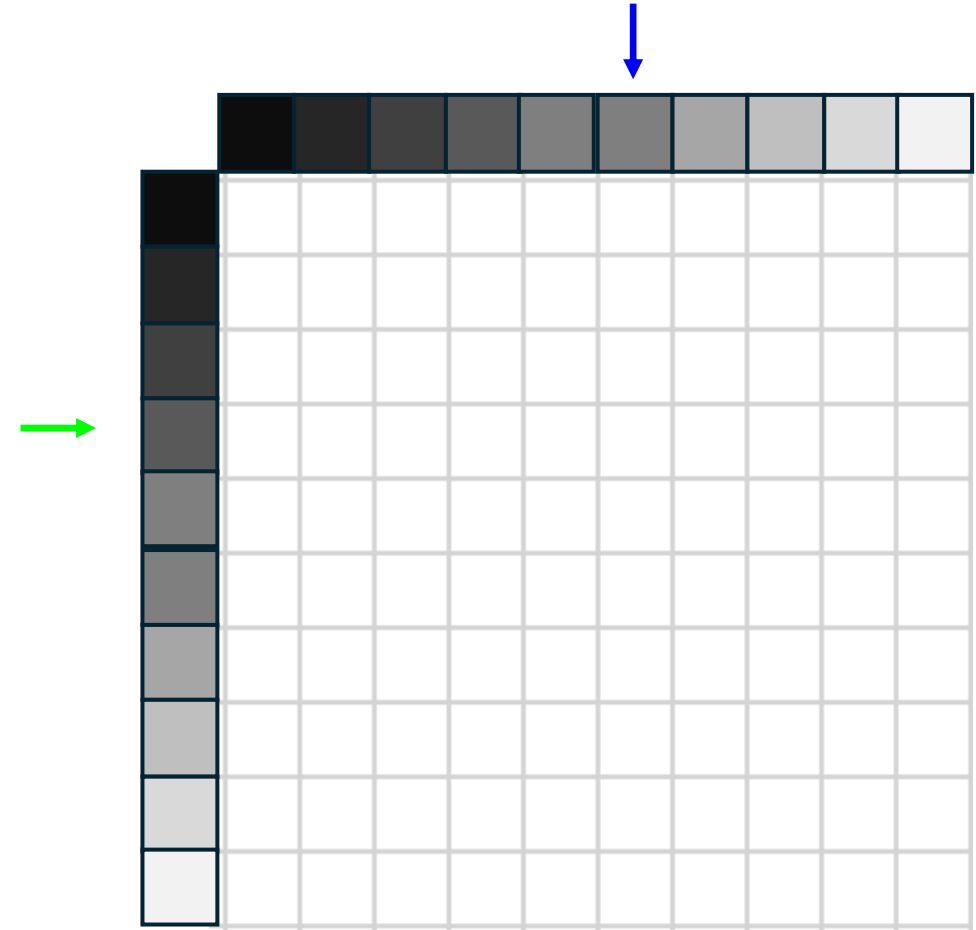
Contaremos ocurrencia
de pares de pixeles



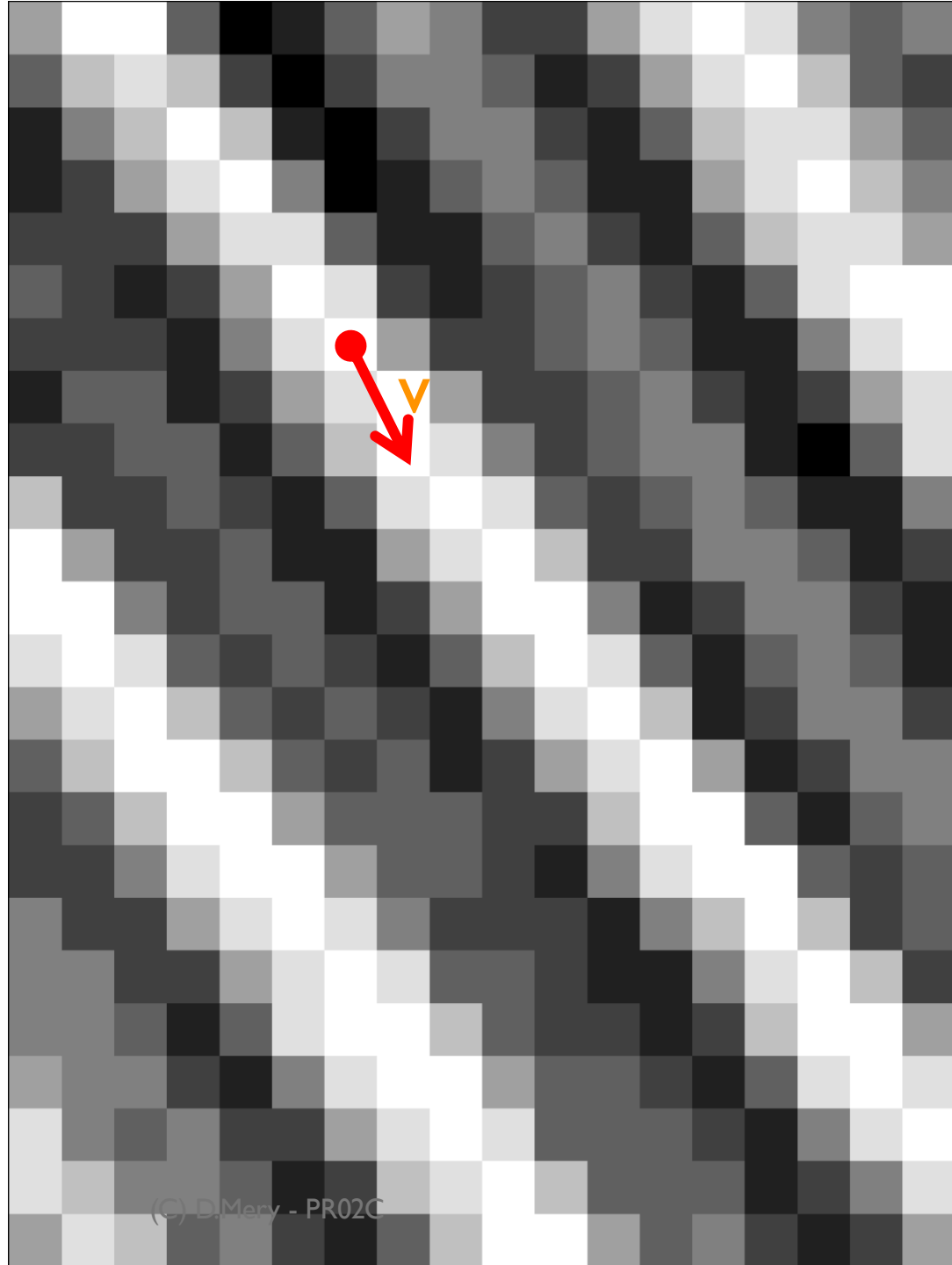
1. Se define **vector del par**: ej. $v = (+2, +1)$
2. Se define los tonos de gris del **par**: ej. $(3, 5)$



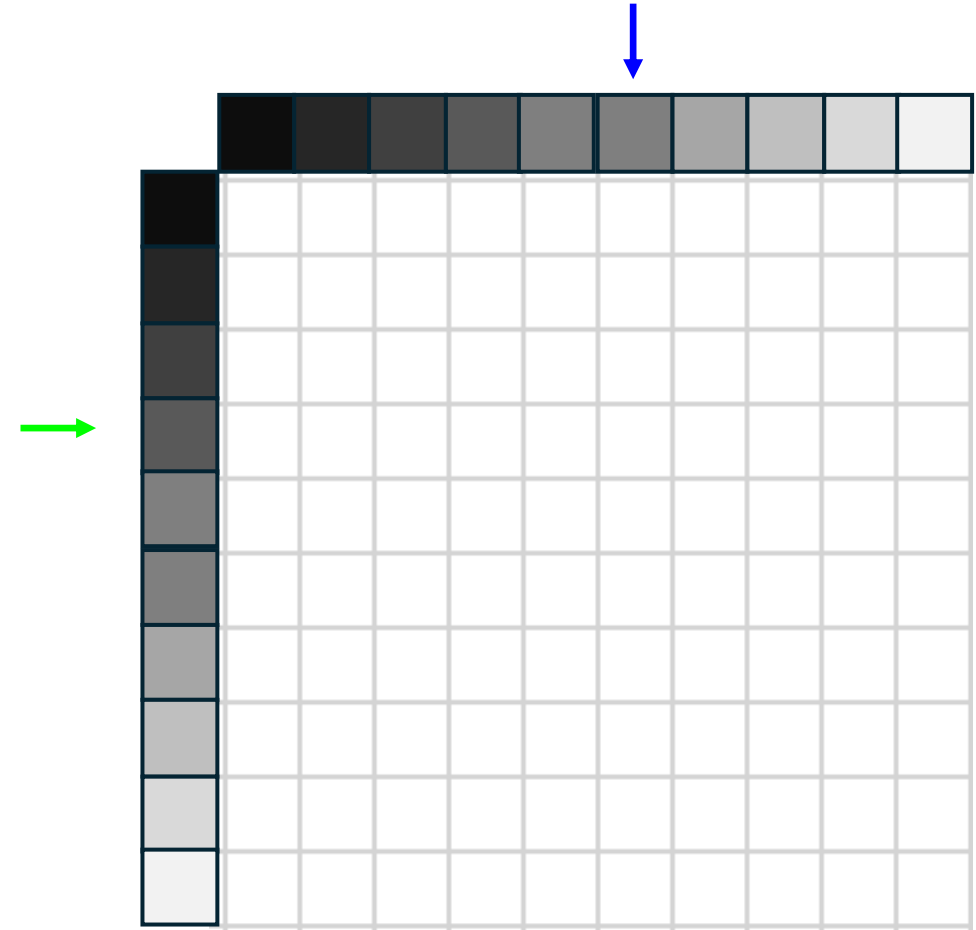
- X
1. Se define **vector del par**: ej. $v = (+2, +1)$
 2. Se define los tonos de gris del **par**: ej. $(3, 5)$



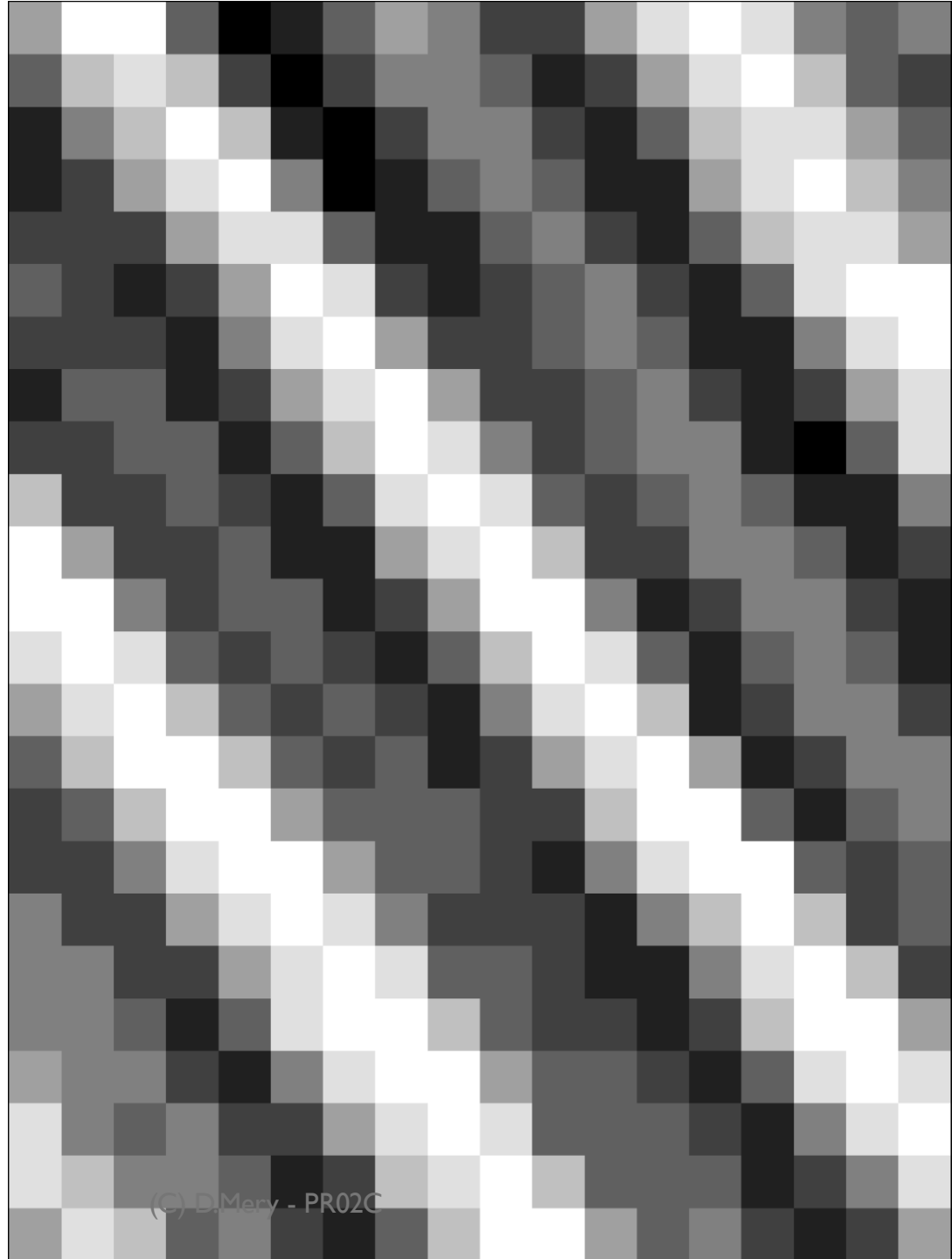
3. Contaremos todas las veces que ocurre
 $X(i, j) == 3 \text{ AND } X(i+2, j+1) == 5$



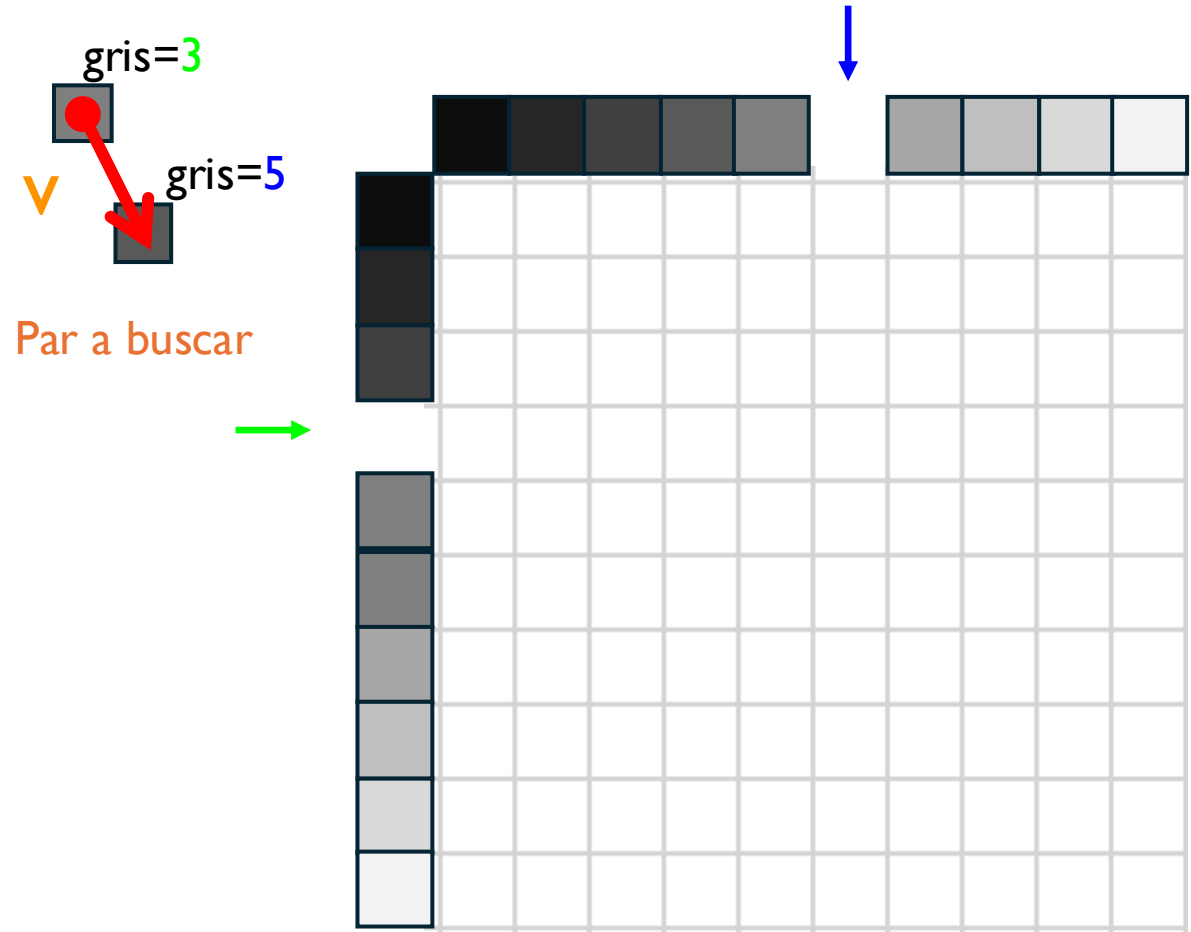
- X
1. Se define **vector del par**: ej. $v = (+2, +1)$
 2. Se define los tonos de gris del **par**: ej. $(3, 5)$



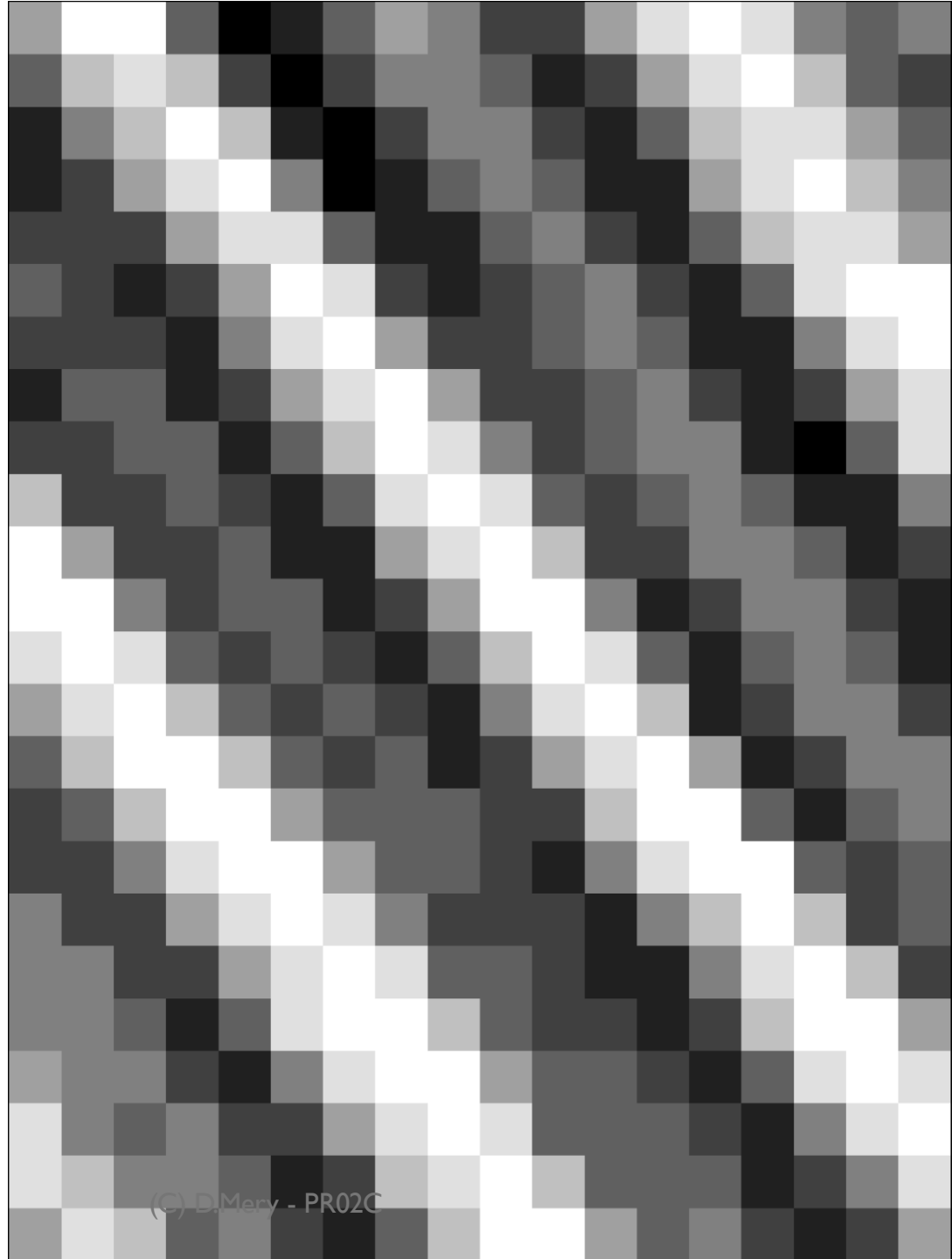
3. Contaremos todas las veces que ocurre
 $X(i, j) == 3 \text{ AND } X(i+2, j+1) == 5$



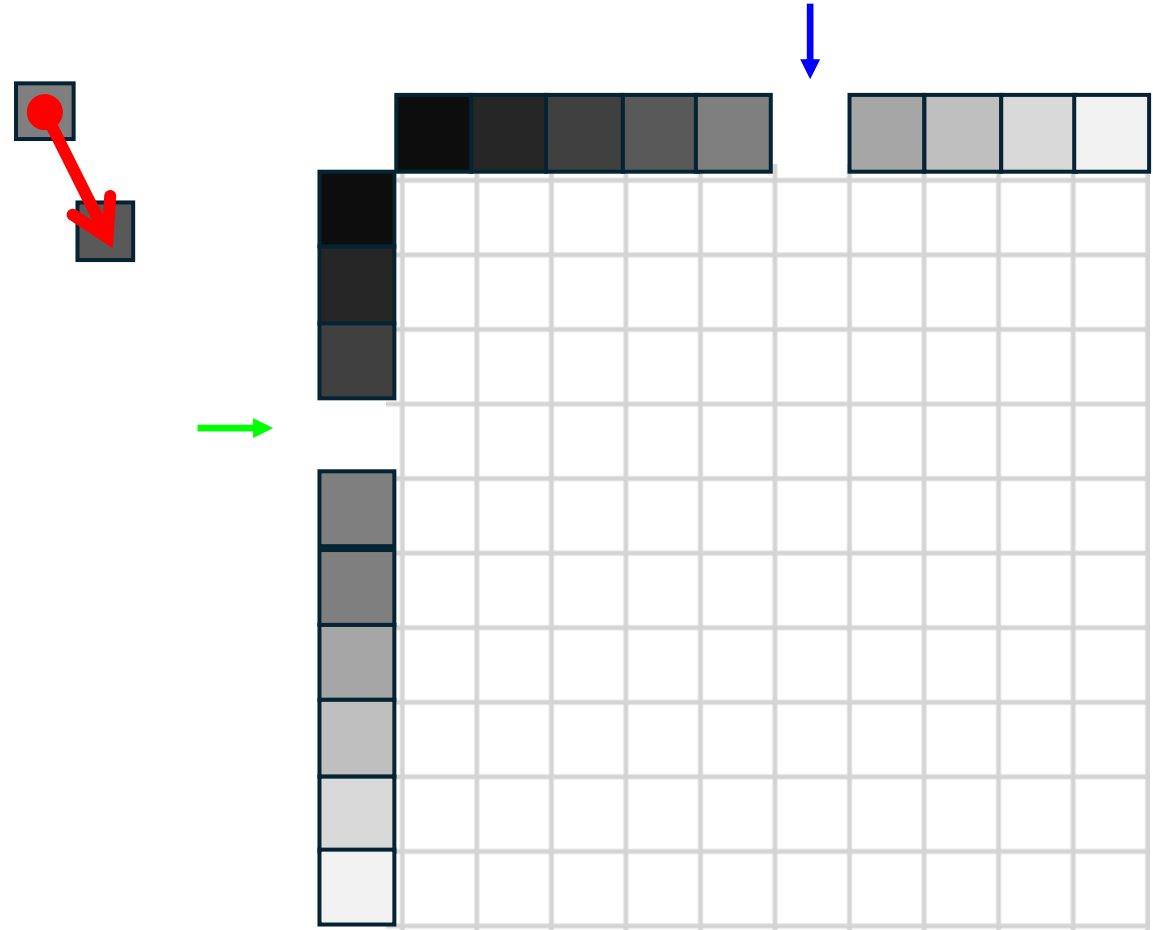
- X
1. Se define **vector del par**: ej. $v = (+2, +1)$
 2. Se define los tonos de gris del **par**: ej. $(3, 5)$



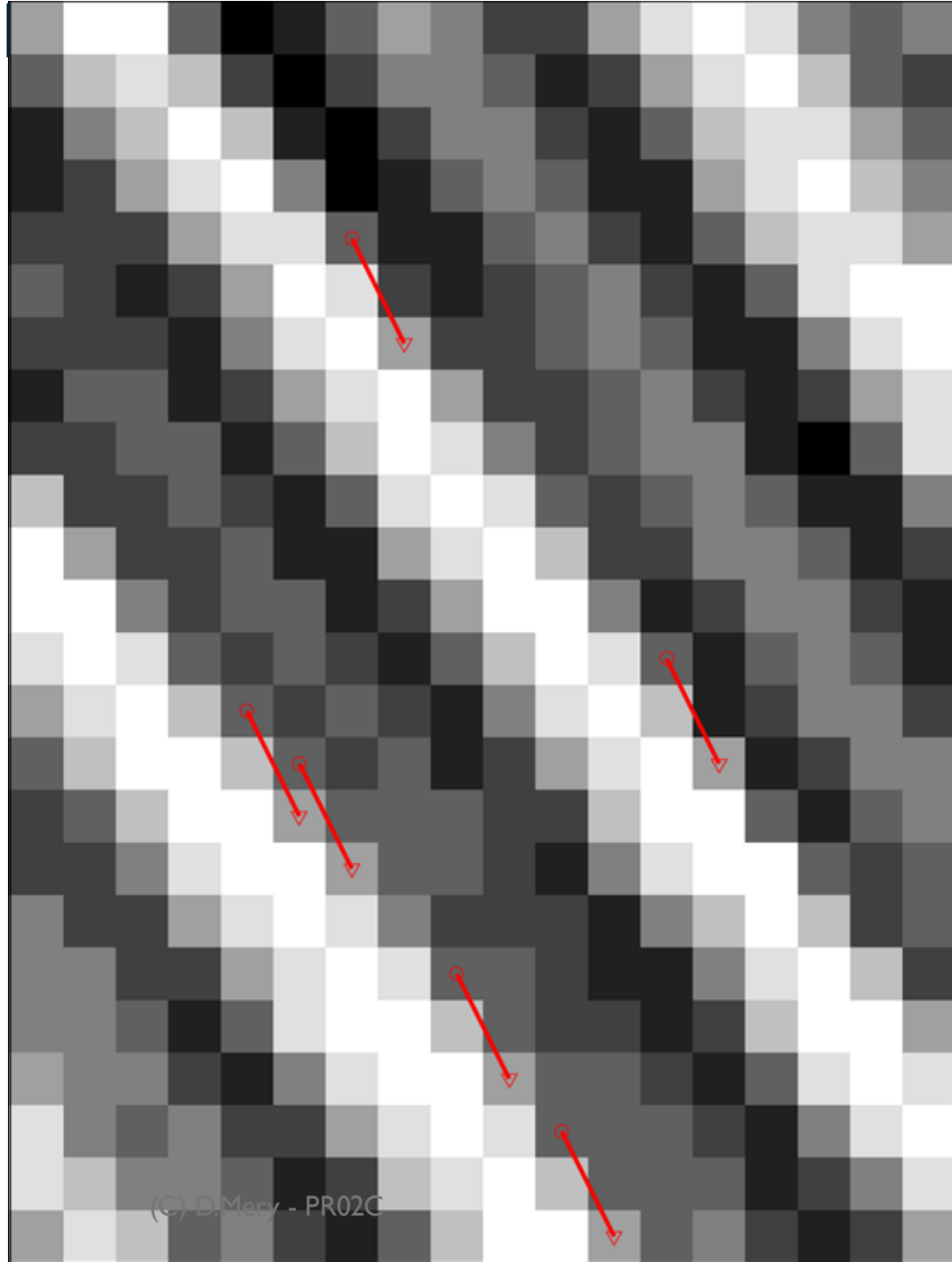
3. Contaremos todas las veces que ocurre
 $X(i,j) == 3 \text{ AND } X(i+2,j+1) == 5$



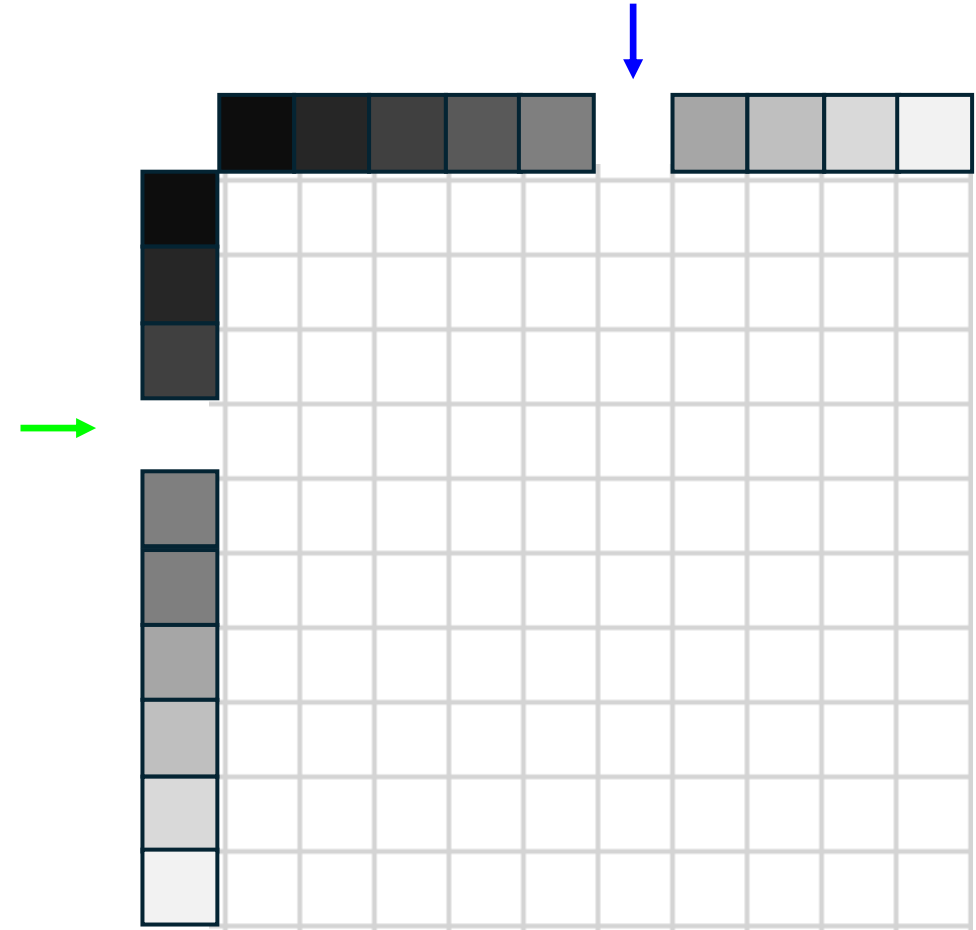
- X
1. Se define **vector del par**: ej. $v = (+2, +1)$
 2. Se define los tonos de gris del **par**: ej. $(3, 5)$



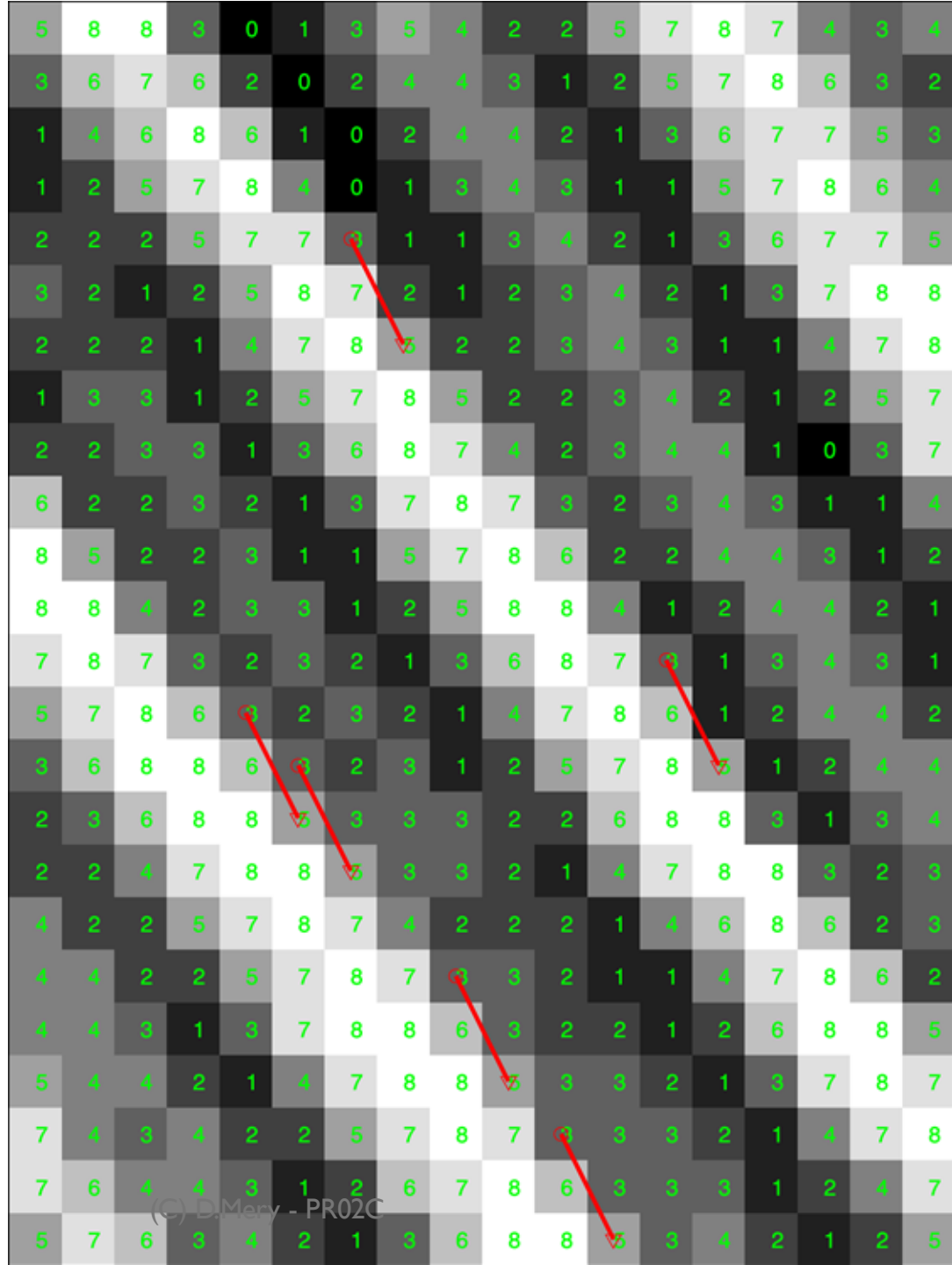
3. Contaremos todas las veces que ocurre
 $X(i, j) == 3 \text{ AND } X(i+2, j+1) == 5$



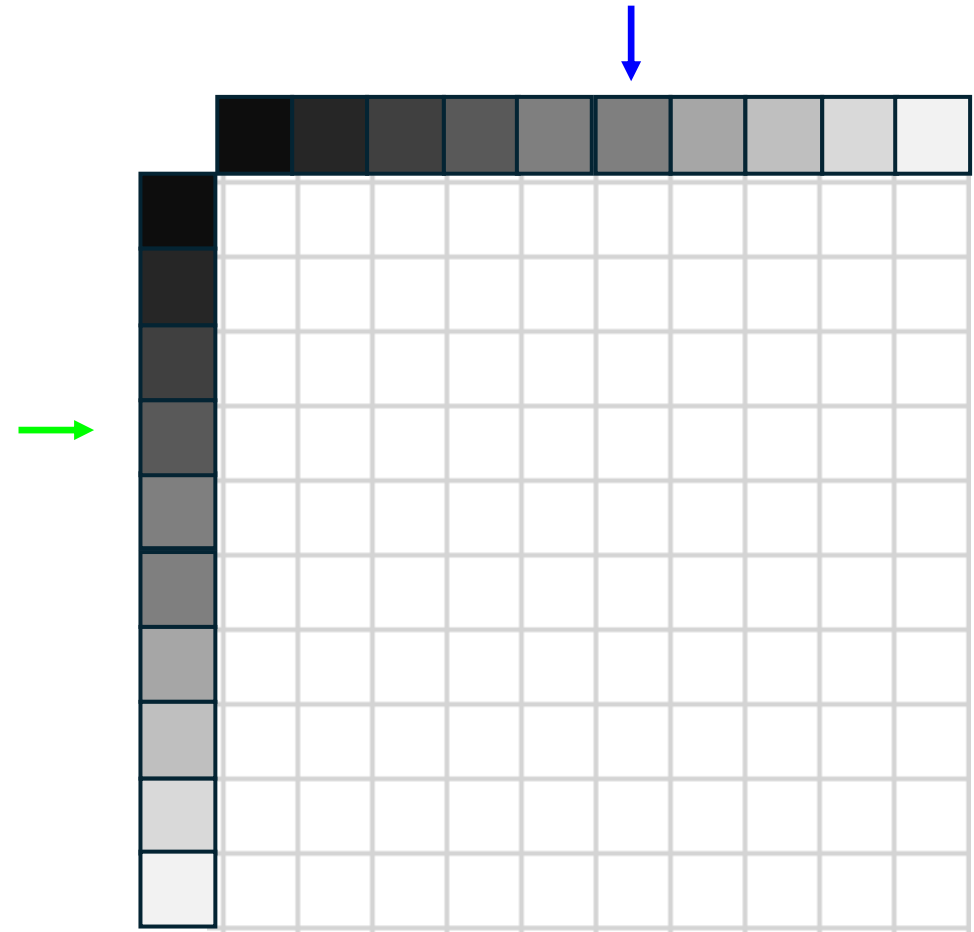
- X
1. Se define **vector del par**: ej. $v = (+2, +1)$
 2. Se define los tonos de gris del **par**: ej. $(3, 5)$



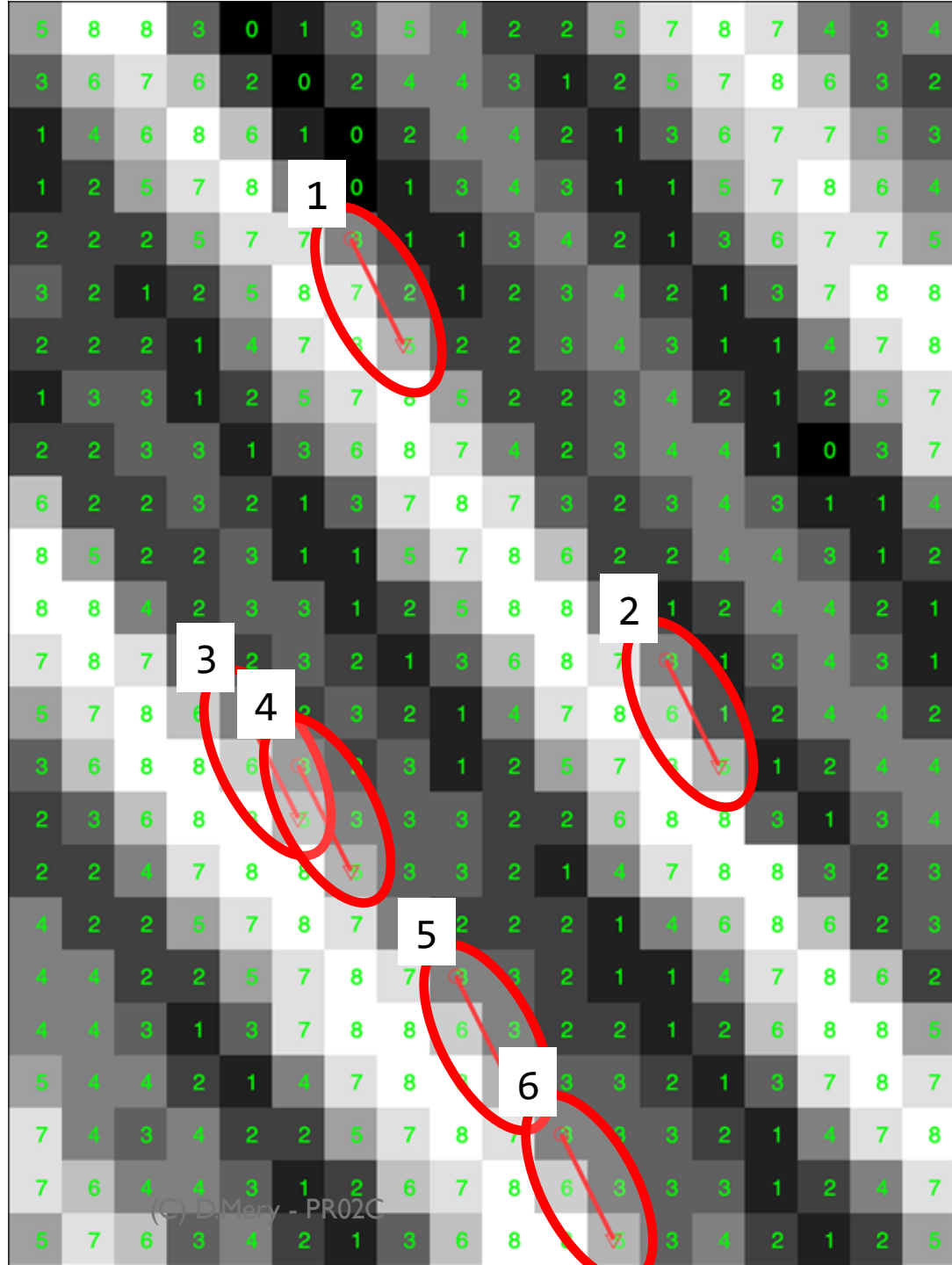
3. Contaremos todas las veces que ocurre
 $X(i, j) == 3 \text{ AND } X(i+2, j+1) == 5$



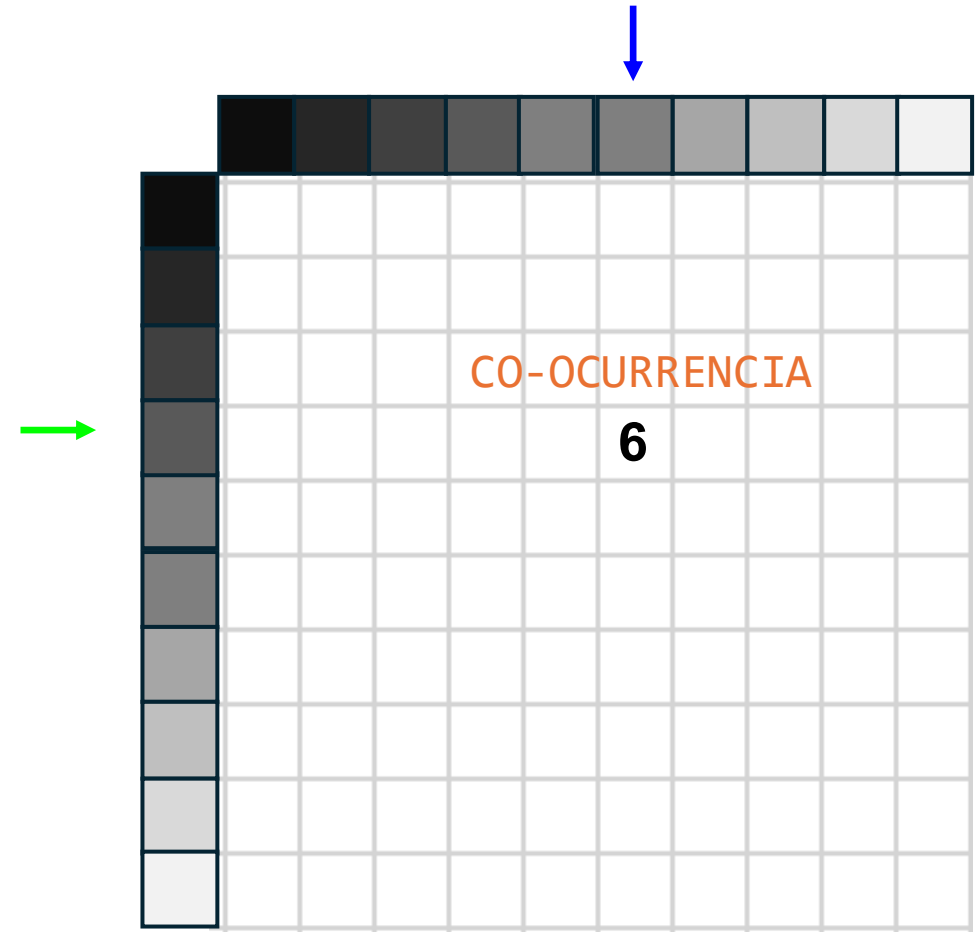
- X
1. Se define **vector del par**: ej. $v = (+2, +1)$
 2. Se define los tonos de gris del **par**: ej. $(3, 5)$



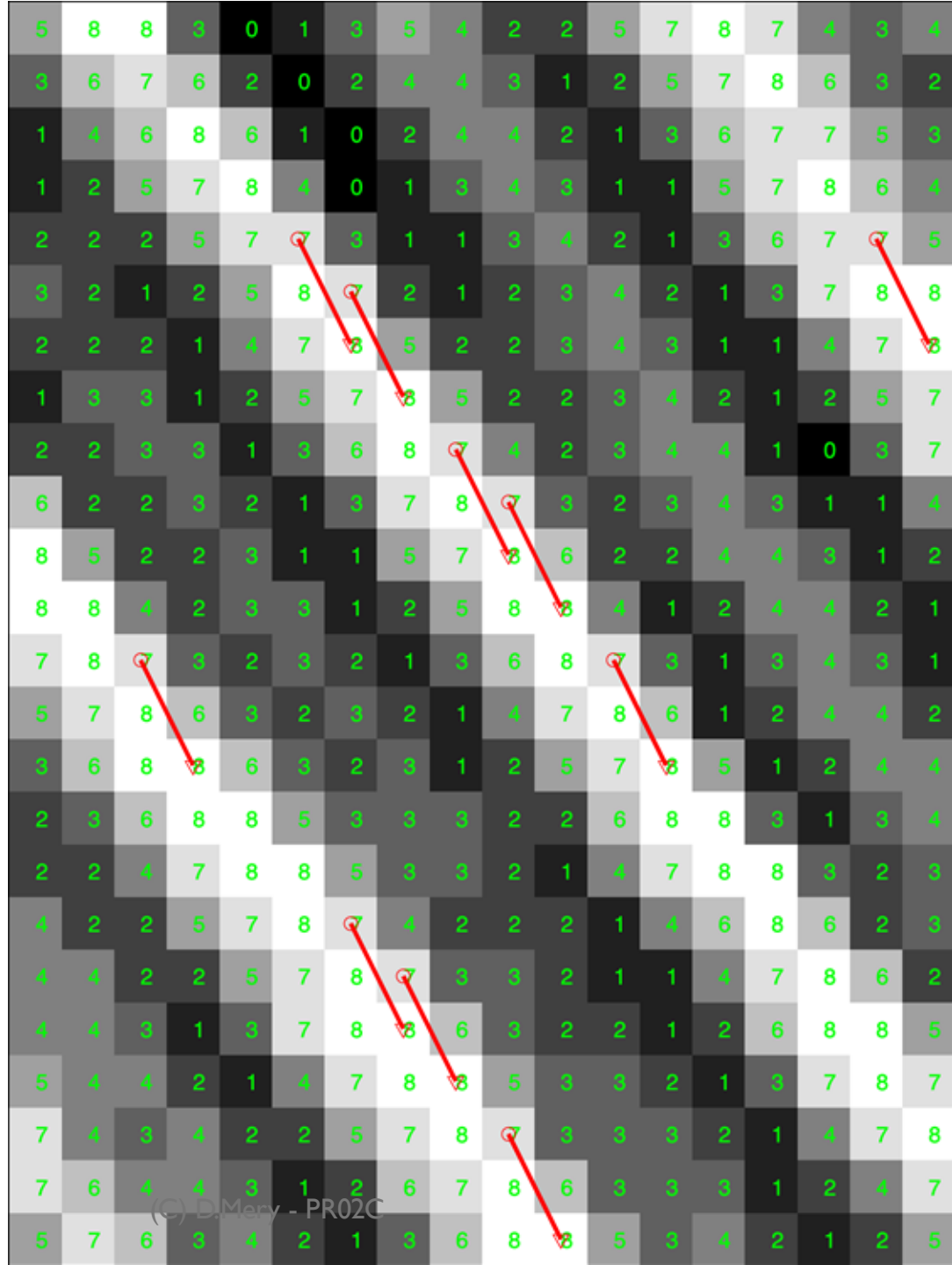
3. Contaremos todas las veces que ocurre $X(i, j) == 3 \text{ AND } X(i+2, j+1) == 5$



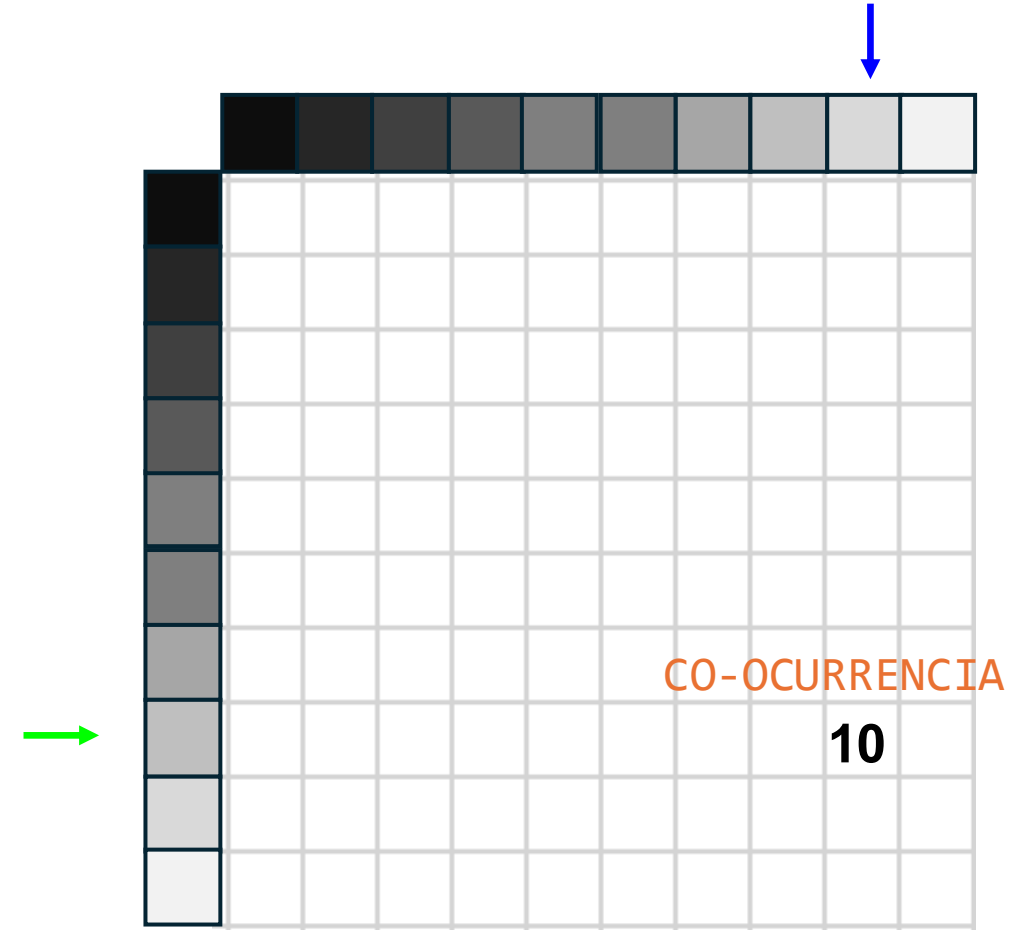
- X
1. Se define **vector del par**: ej. $v = (+2, +1)$
 2. Se define los tonos de gris del **par**: ej. $(3, 5)$



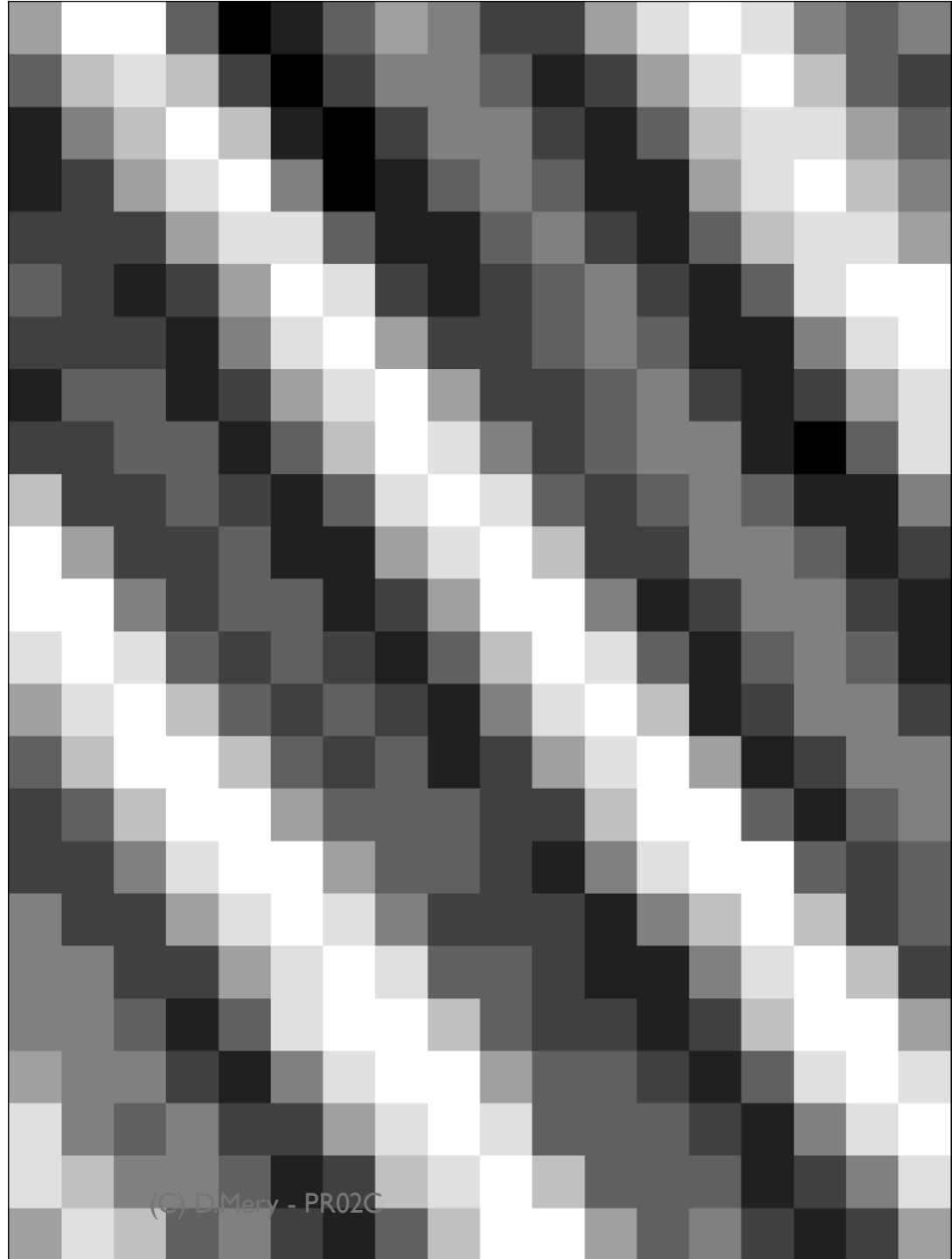
3. Contaremos todas las veces que ocurre $X(i, j) == 3 \text{ AND } X(i+2, j+1) == 5$



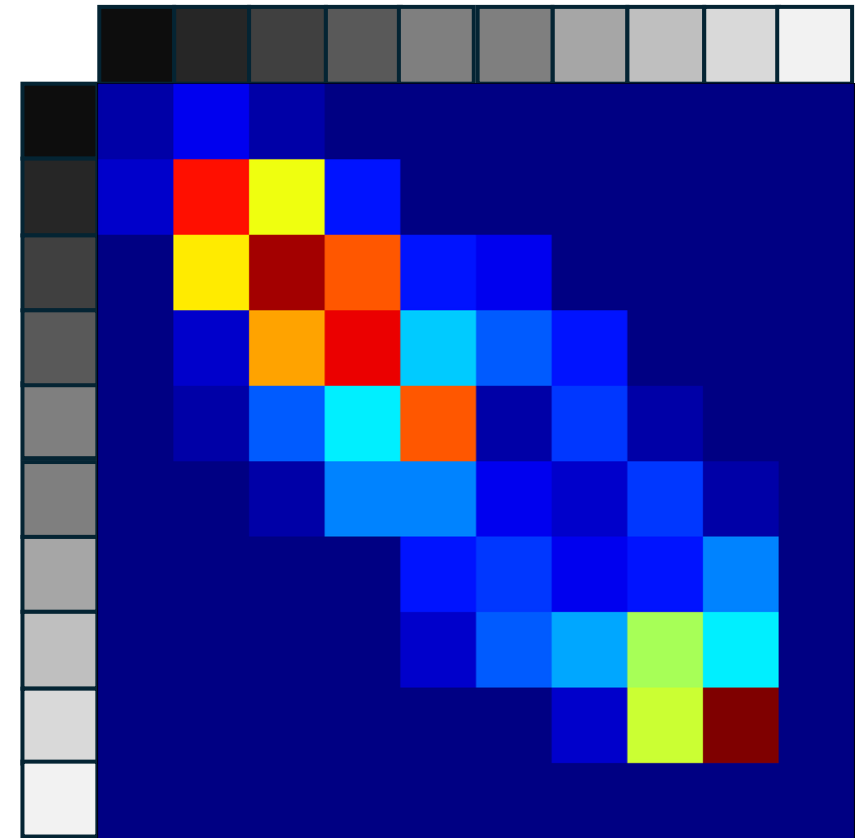
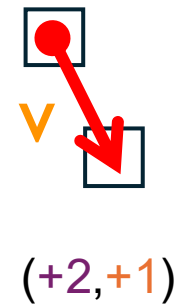
- X 1. Se define **vector del par**: ej. $v = (+2, +1)$
2. Se define los tonos de gris del **par**: ej. $(7, 8)$



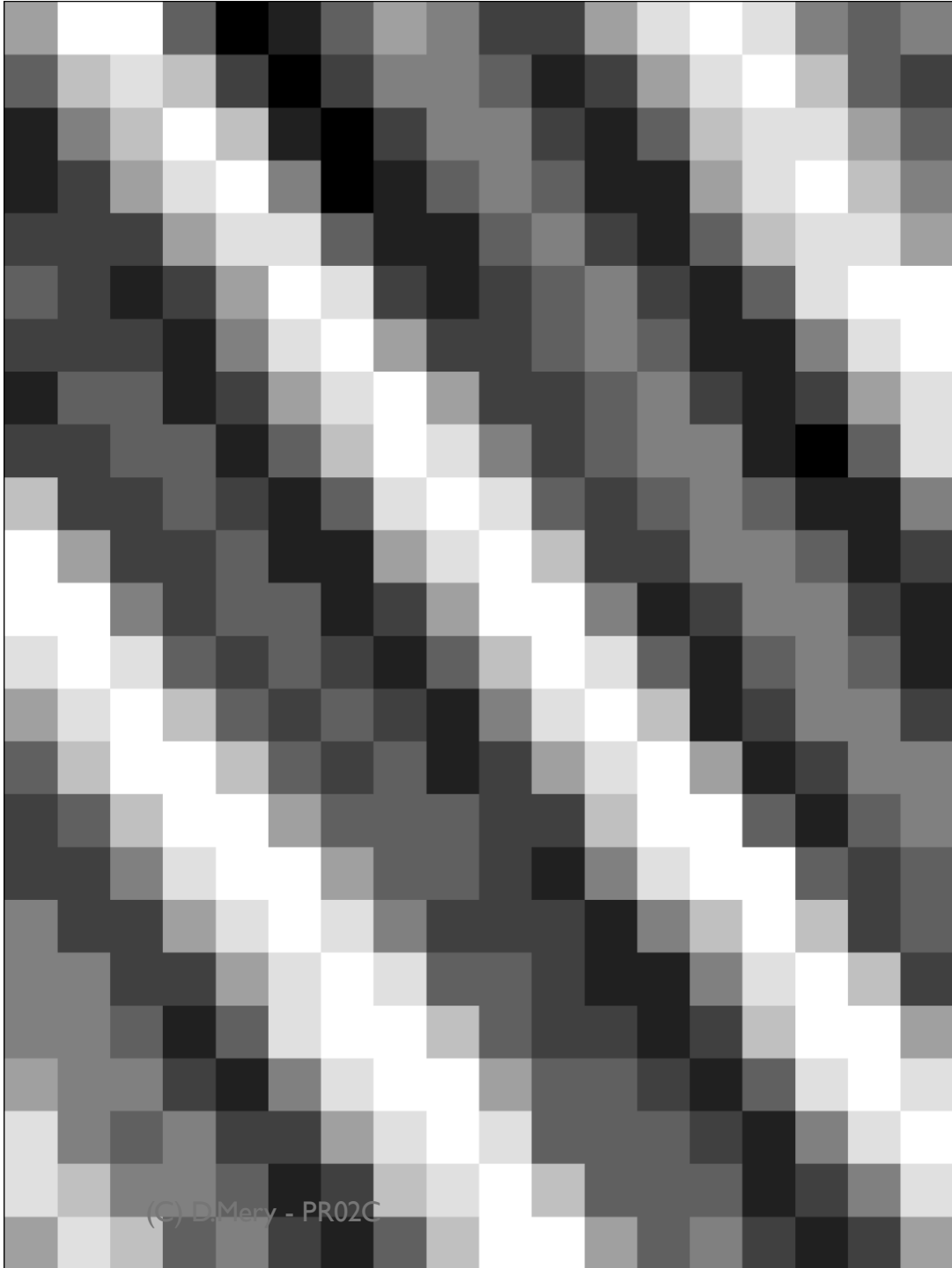
3. Contaremos todas las veces que ocurre
- $$X(i, j) == 7 \text{ AND } X(i+2, j+1) == 8$$



X for $p1 = 0:9$
for $p2 = 0:9$
 $P(p1, p2) = \text{CO-OCURRENCIA}$



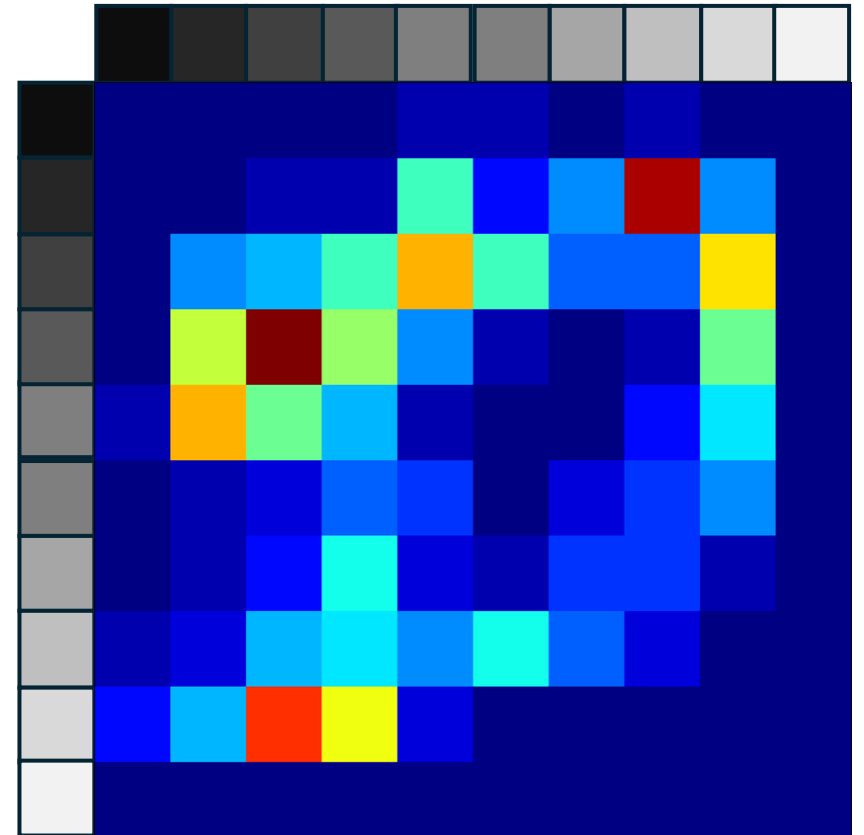
Matriz de Co-ocurrencia



X for p1 = 0:9
for p2 = 0:9
P(p1,p2) = CO-OCURRENCIA



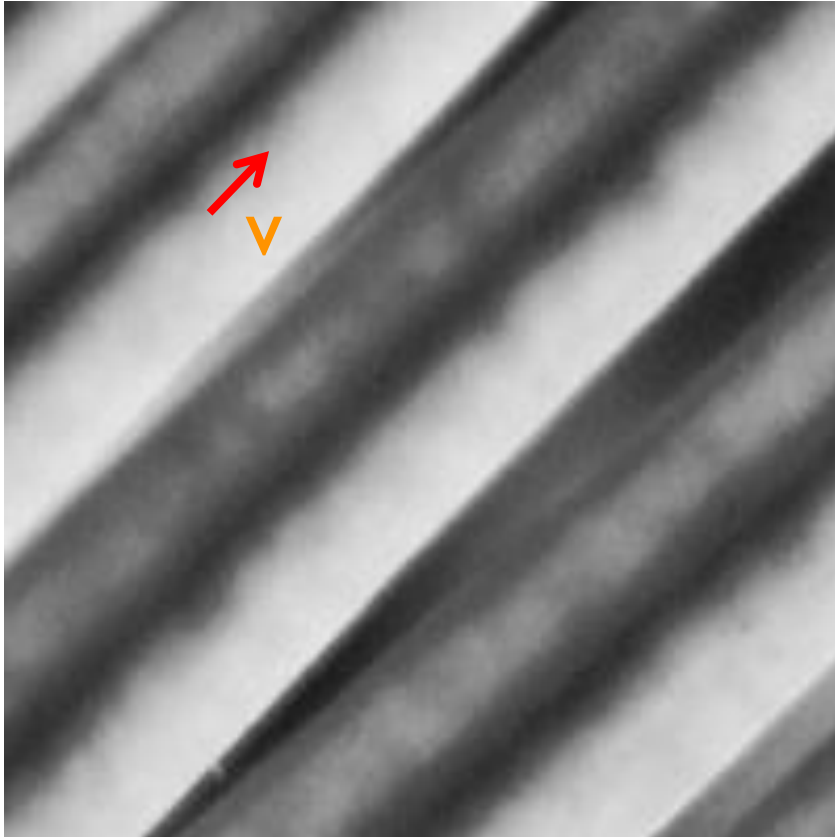
(-2,+1)



Matriz de Co-ocurrencia

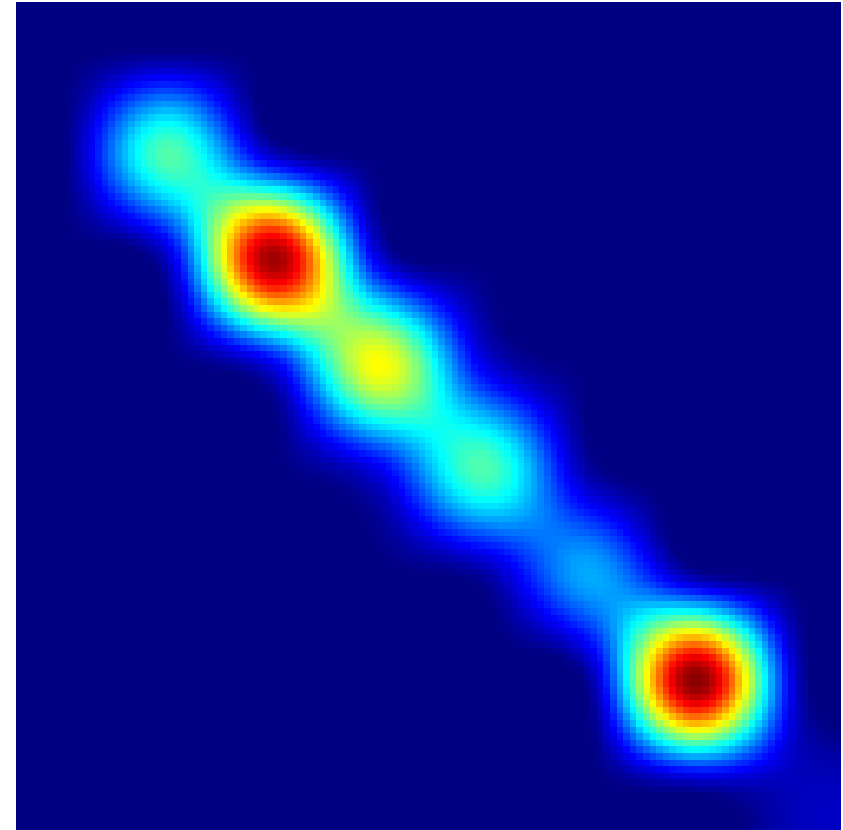
Texturas de Haralick (ejemplos reales)

X



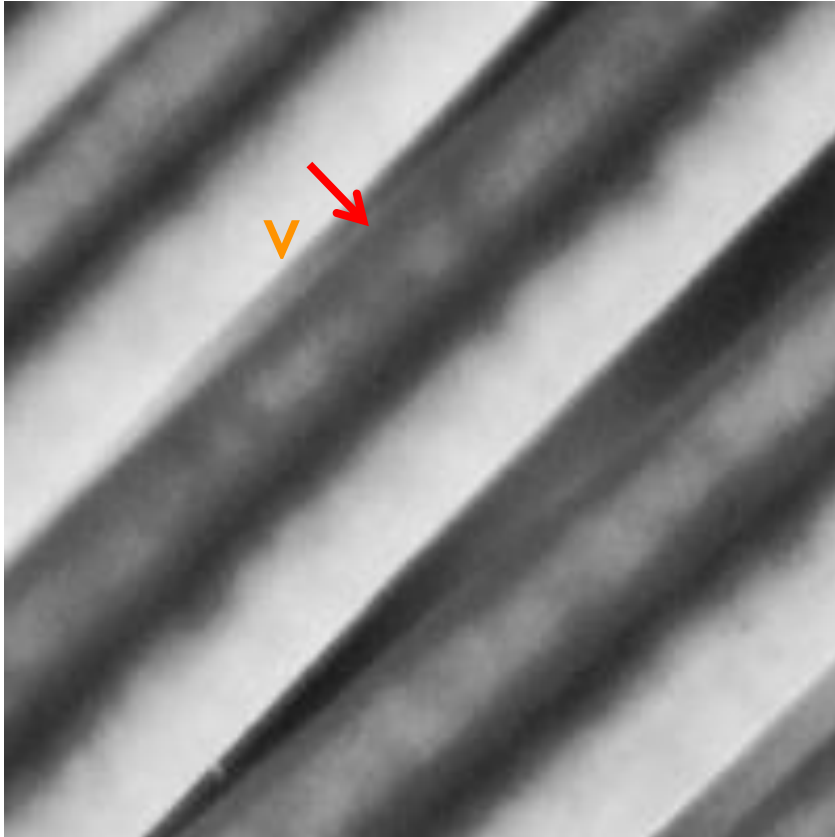
En esta dirección, los valores de píxeles no cambian significativamente.

P_v



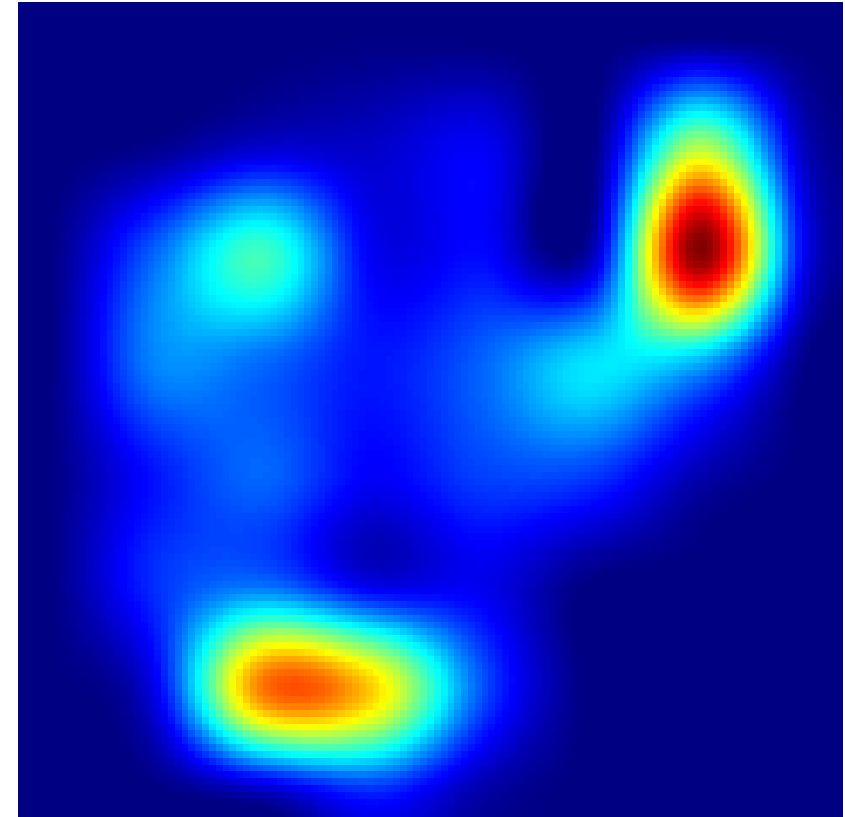
Matriz de Co-ocurrencia

X



En esta dirección, los valores de píxeles cambian significativamente de bajo a alto y de alto a bajo.

P_v



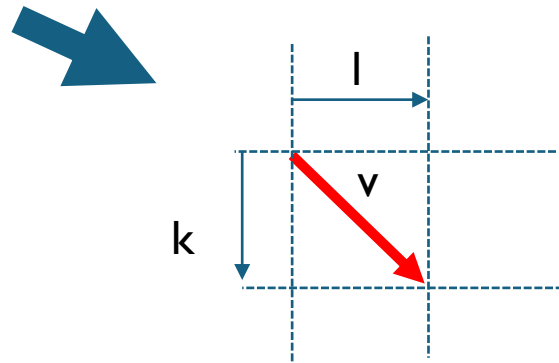
Matriz de Co-ocurrencia

Texturas de Haralick (definiciones matemáticas)

Matriz de Co-ocurrencia

$$P_{kl}(i, j) = \frac{1}{nm} \sum_{p=1}^n \sum_{q=1}^m \begin{cases} 1, & \text{if } I(p, q) = i \text{ and } I(p+k, q+l) = j \\ 0, & \text{otherwise} \end{cases}$$

$\downarrow$
 Vector v



Características basadas en la Co-ocurrencia

Contrast

$$I_{kl} = \sum_{i=1}^{N_x} \sum_{j=1}^{N_x} (i - j)^2 P_{kl}[i, j]$$

Moment of
inverse difference

$$Z_{kl} = \sum_{i=1}^{N_x} \sum_{j=1}^{N_x} \frac{P_{kl}[i, j]}{1 + (i - j)^2}$$

Energy

$$E_{kl} = \sum_{i=1}^{N_x} \sum_{j=1}^{N_x} [P_{kl}[i, j]]^2$$

Entropy

$$H_{kl} = \sum_{i=1}^{N_x} \sum_{j=1}^{N_x} P_{kl}[i, j] \log(P_{kl}[i, j])$$

Características basadas en la Co-ocurrencia

Angular second moment:

$$f_1 = \sum_{i=1}^{N_x} \sum_{j=1}^{N_x} [p(i, j)]^2$$

Contrast:

$$f_2 = \sum_{n=0}^{N_x-1} n^2 \sum_{i=1}^{N_x} \sum_{j=1}^{N_x} p(i, j) \text{ for } |i - j| = n$$

Correlation:

$$f_3 = \frac{1}{\sigma_x \sigma_y} \sum_{i=1}^{N_x} \sum_{j=1}^{N_x} [ij \cdot p(i, j) - \mu_x \mu_y]^2$$

Sum of squares:

$$f_4 = \sum_{i=1}^{N_x} \sum_{j=1}^{N_x} (i - j)^2 p(i, j)$$

Inverse difference moment:

$$f_5 = \sum_{i=1}^{N_x} \sum_{j=1}^{N_x} \frac{p(i, j)}{1 + (i - j)^2}$$

Sum average:

$$f_6 = \sum_{i=2}^{2N_x} i \cdot p_{x+y}(i)$$

Sum variance:

$$f_7 = \sum_{i=2}^{2N_x} (i - f_6) \cdot p_{x+y}(i)$$

Sum entropy:

$$f_8 = - \sum_{i=2}^{2N_x} p_{x+y}(i) \cdot \log(p_{x+y}(i))$$

Entropy:

$$f_9 = - \sum_{i=1}^{N_x} \sum_{j=1}^{N_x} p(i, j) \log(p(i, j))$$

Difference variance:

$$f_{10} = \text{var}(\mathbf{p}_{x+y})$$

Difference entropy:

$$f_{11} = - \sum_{i=0}^{N_x-1} p_{x-y}(i) \cdot \log(p_{x-y}(i))$$

Information measures of correlation 1: $f_{12} = \frac{f_9 - HXY1}{\max(HX, HY)}$

Information measures of correlation 2: $f_{13} = \sqrt{1 - \exp(-2(HXY2 - HXY))}$

Maximal correlation coefficient:

$$f_{14} = \sqrt{\lambda_2}$$

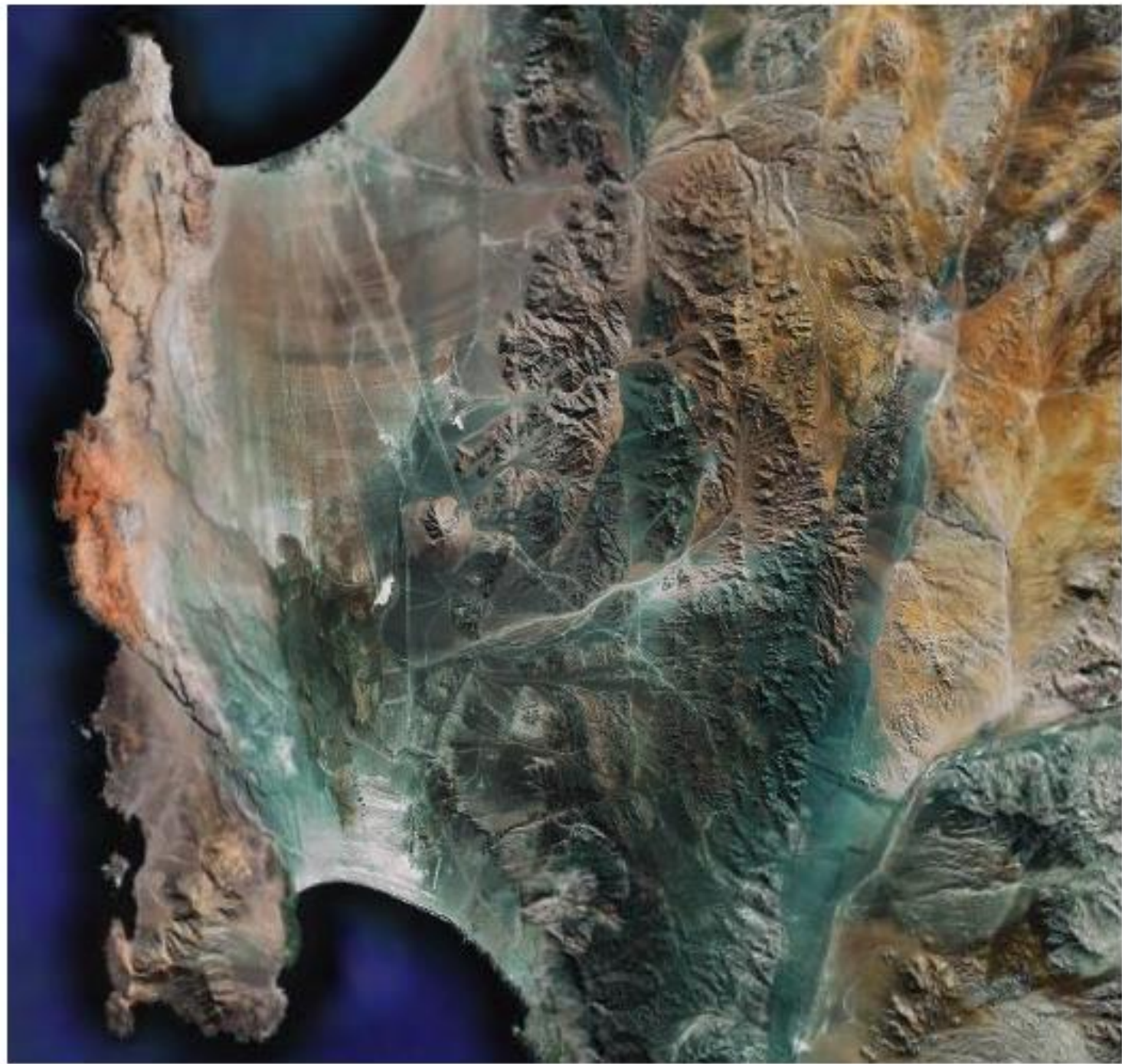
Texturas de Haralick (ejemplo en imagen satelital)

Ejemplo

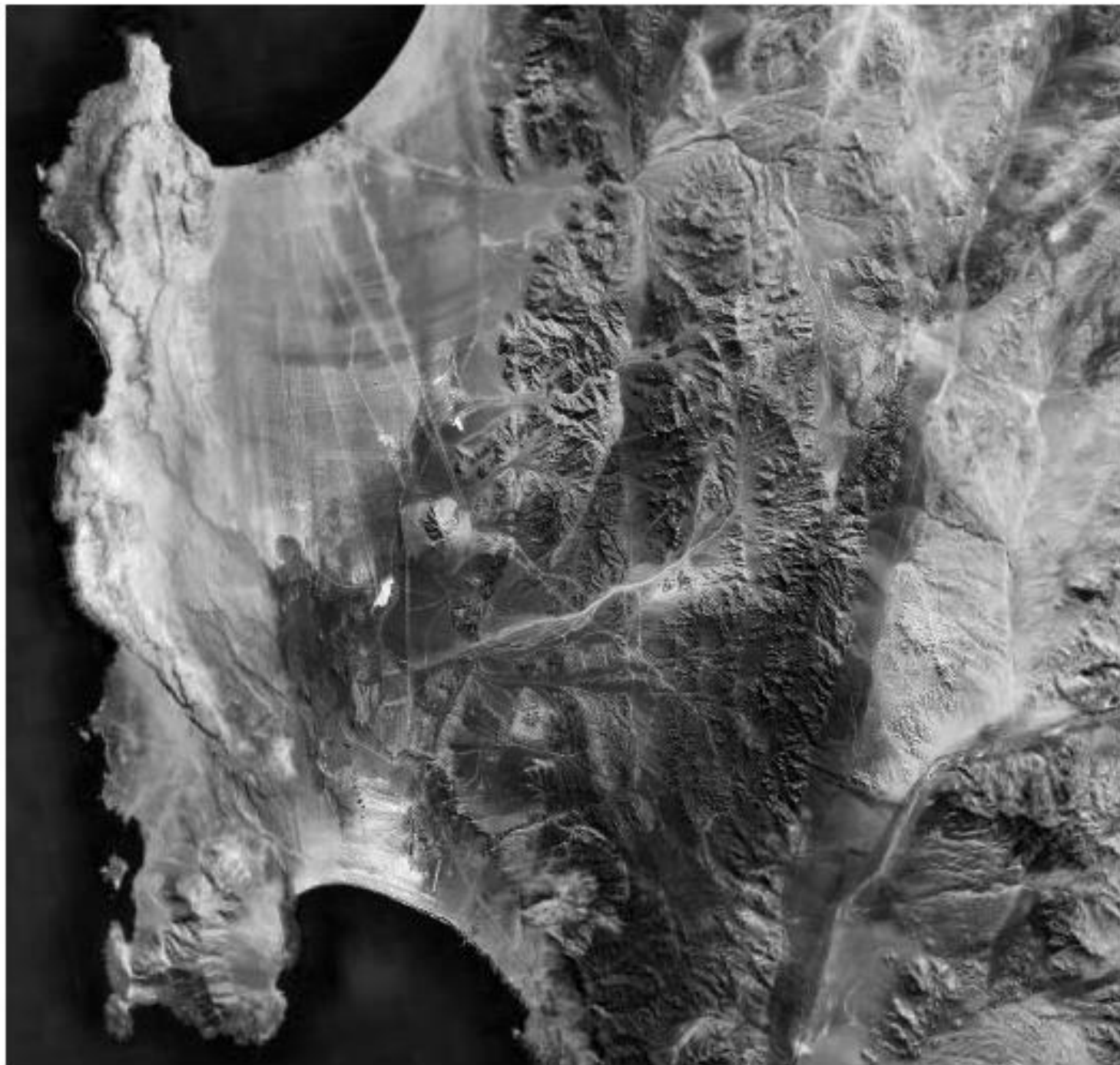
Detección de
Montañas y
zonas desérticas



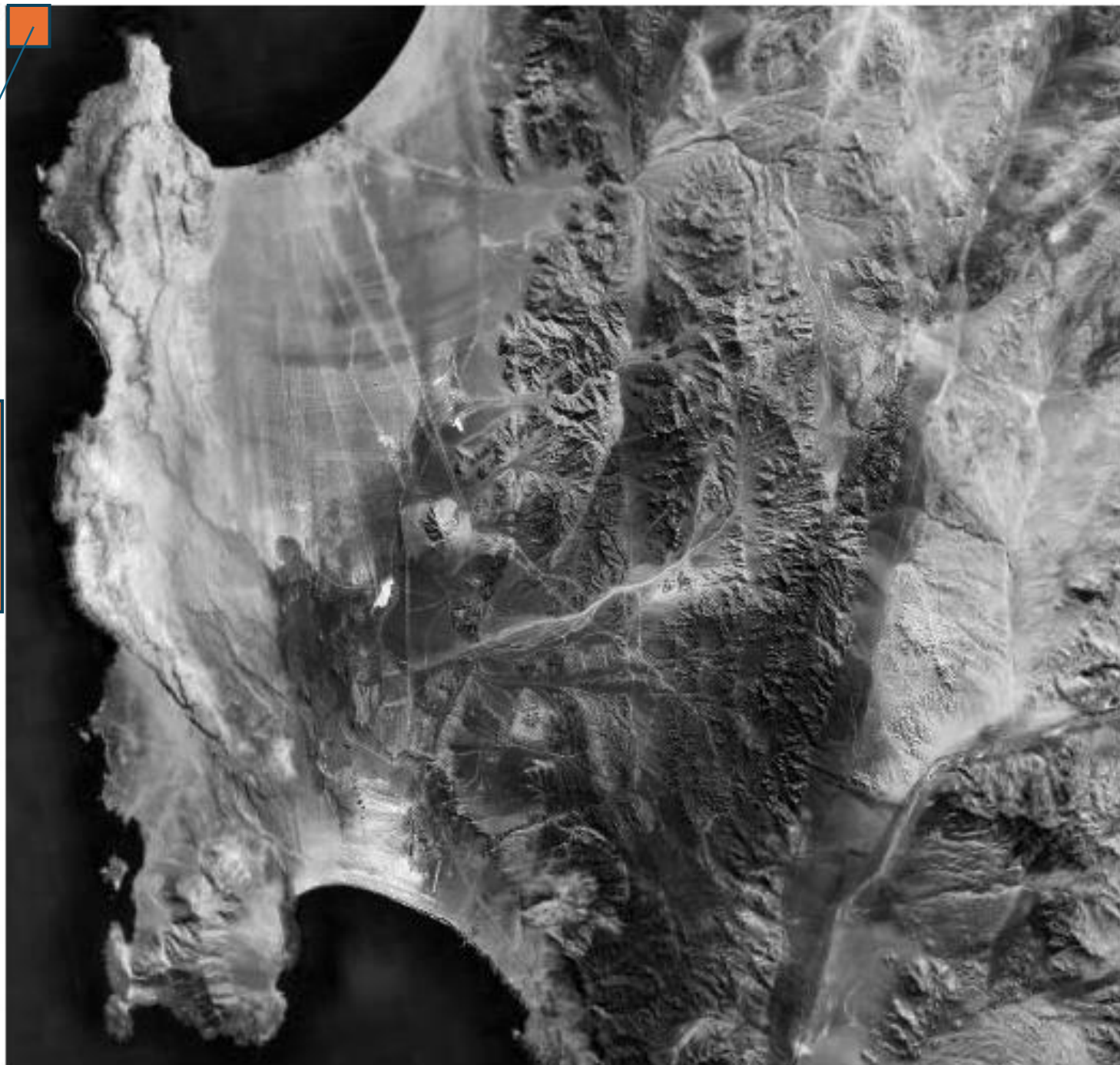
Original

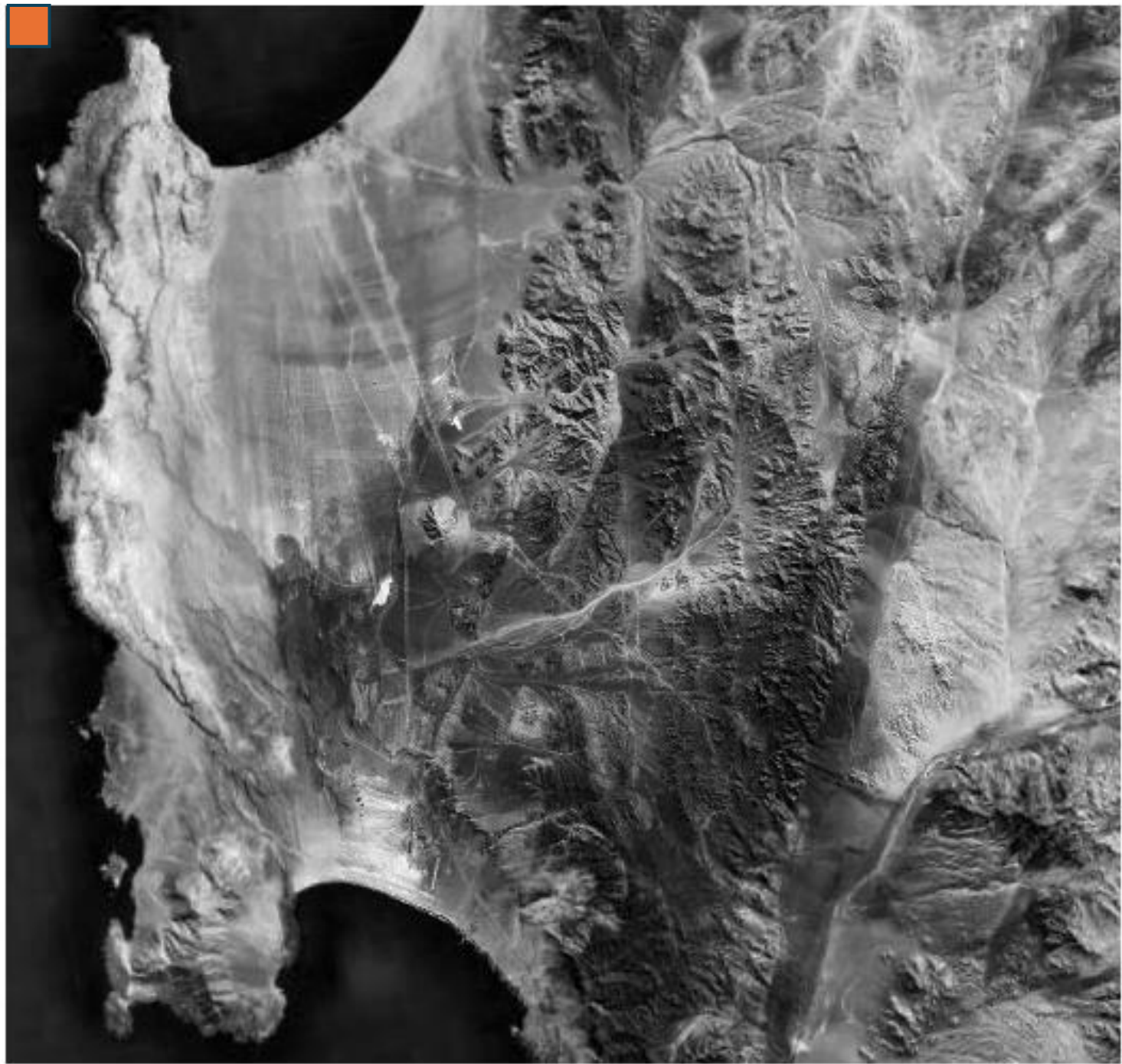


Canal rojo

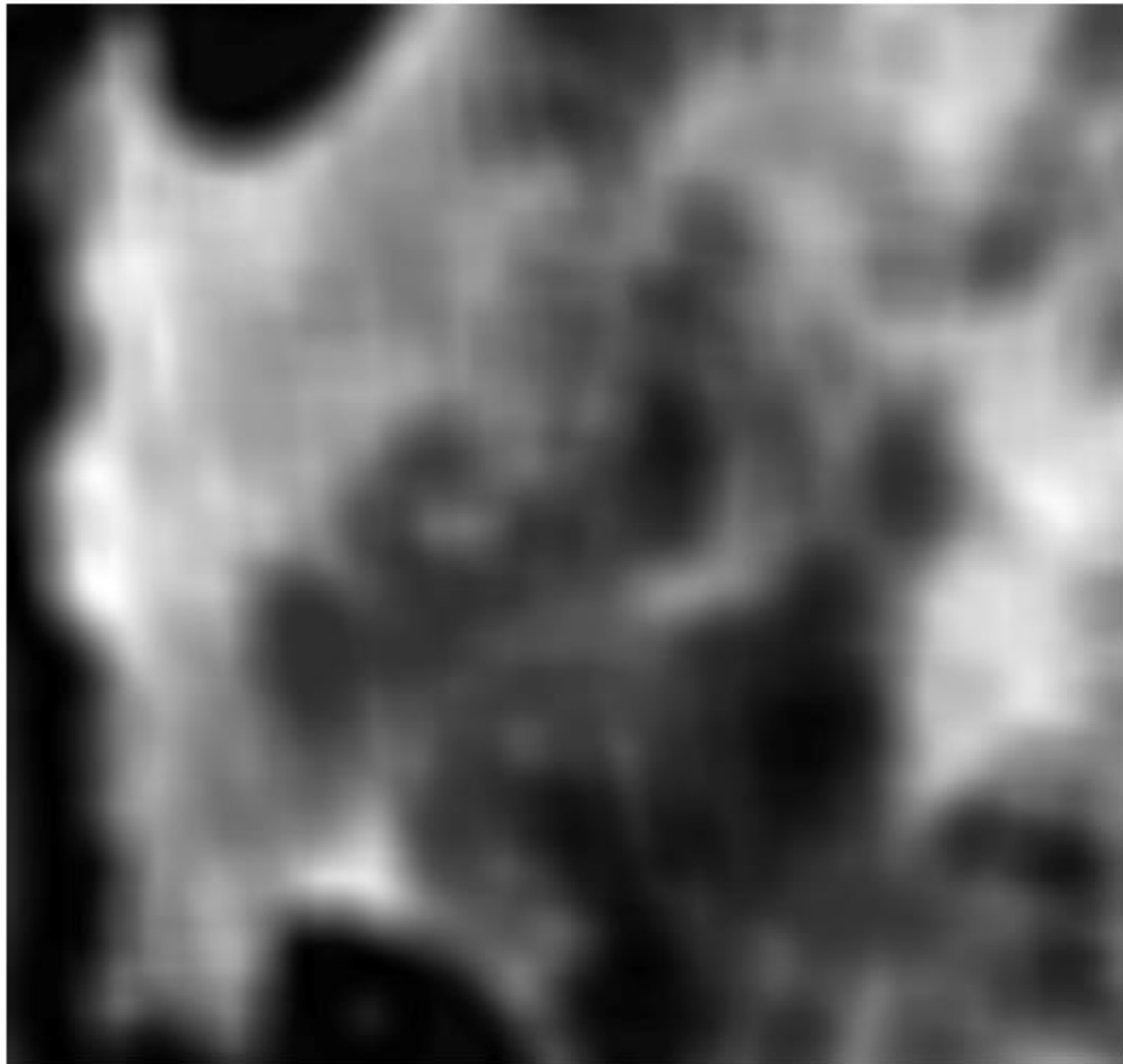


Extracción de
Haralick
Características





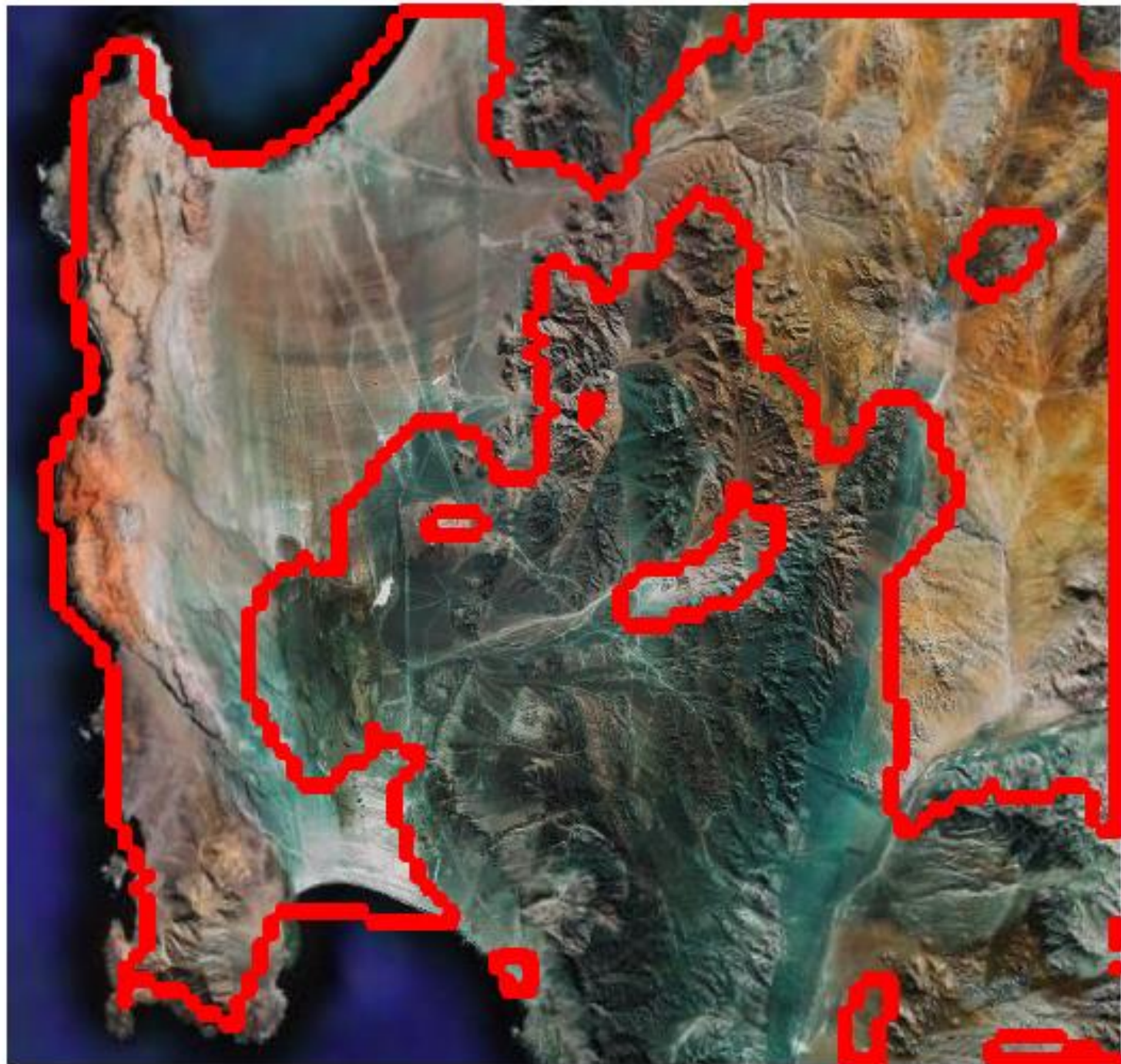
Sum
Variance



Sum
Variance
> Threshold

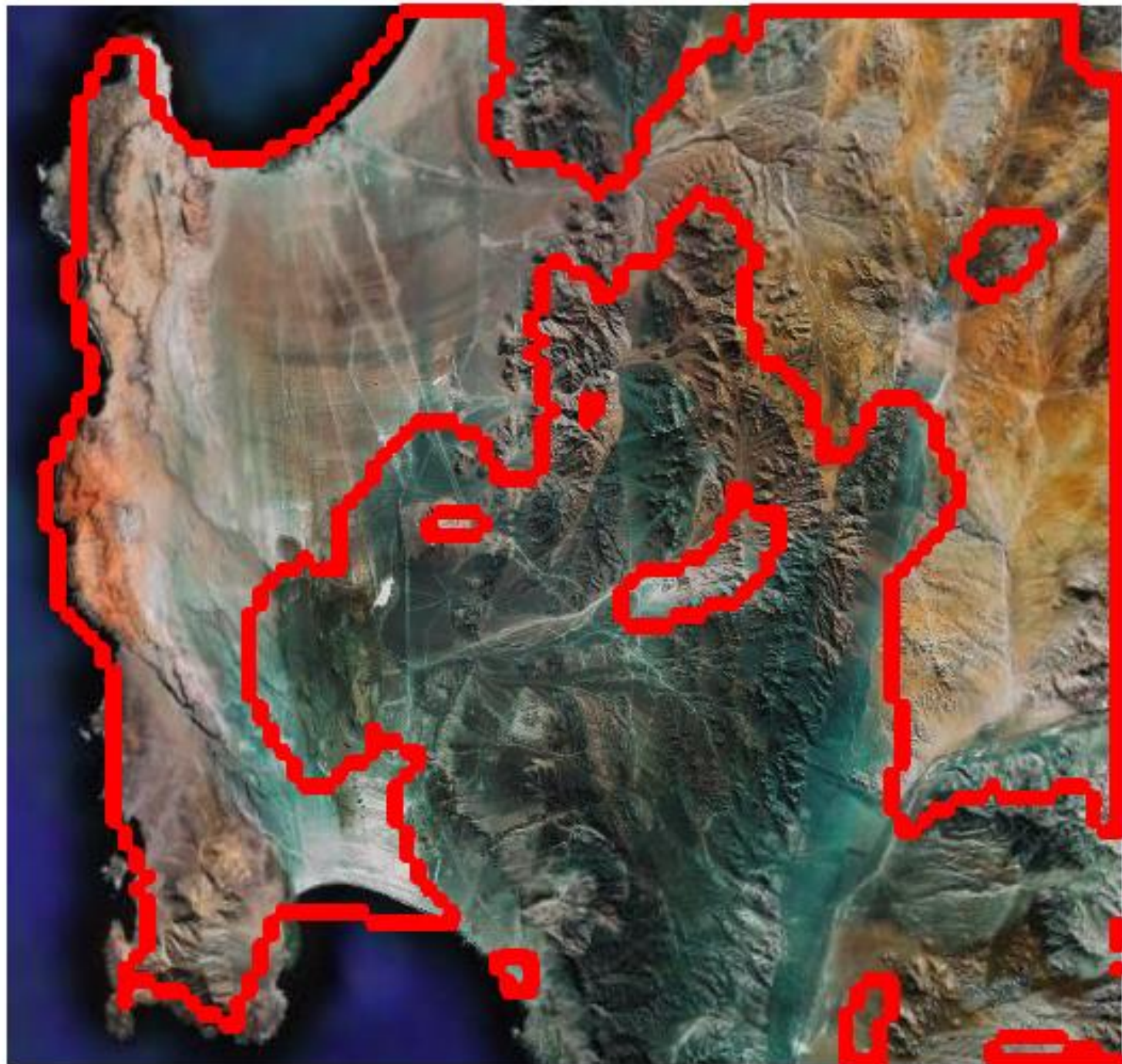


Sum
Variance
> Threshold



Sum
Variance
> Threshold

= zonas desérticas





T7: Sum Variance



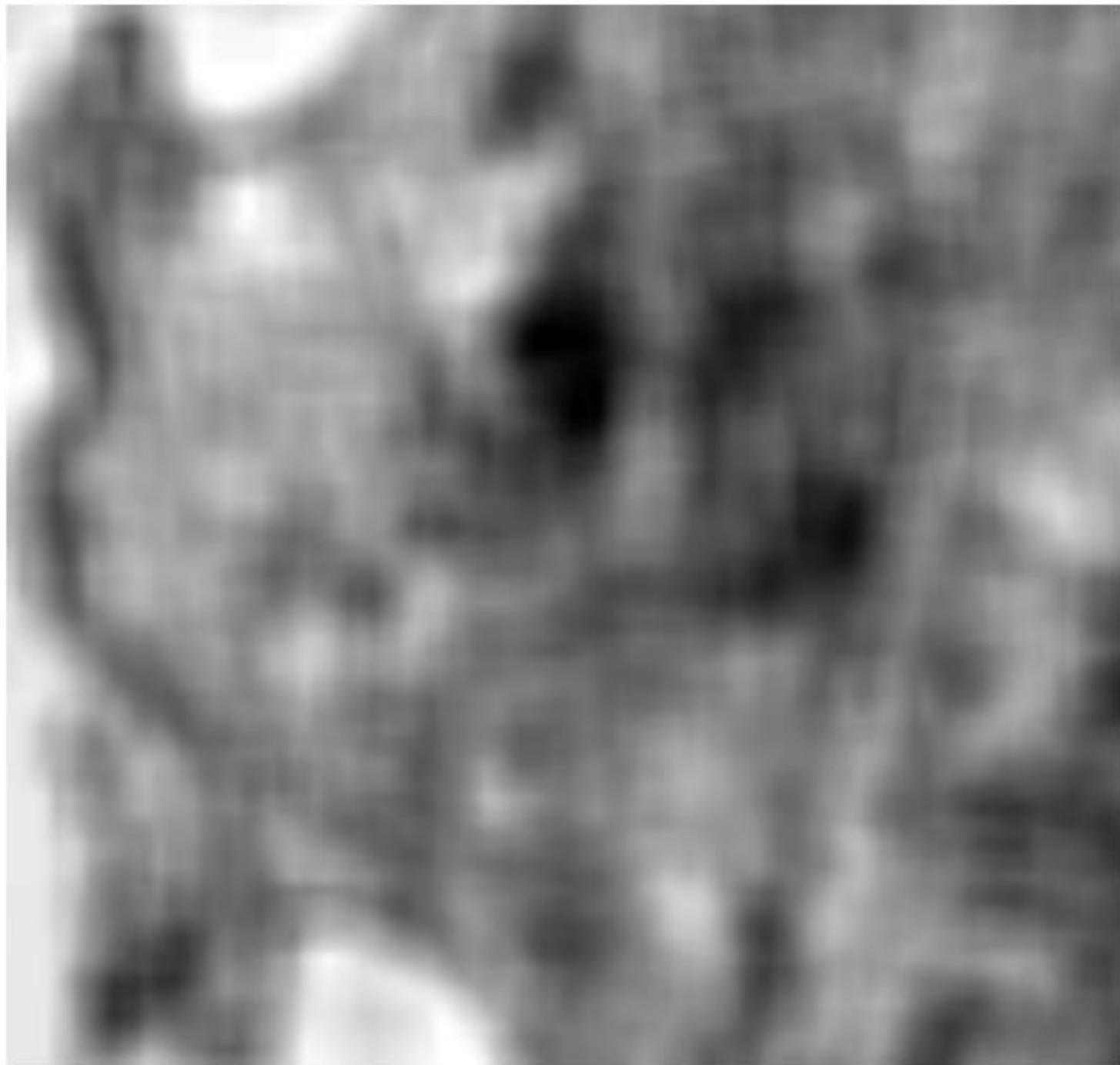
Thresholding of T7 using Otsu



desert areas

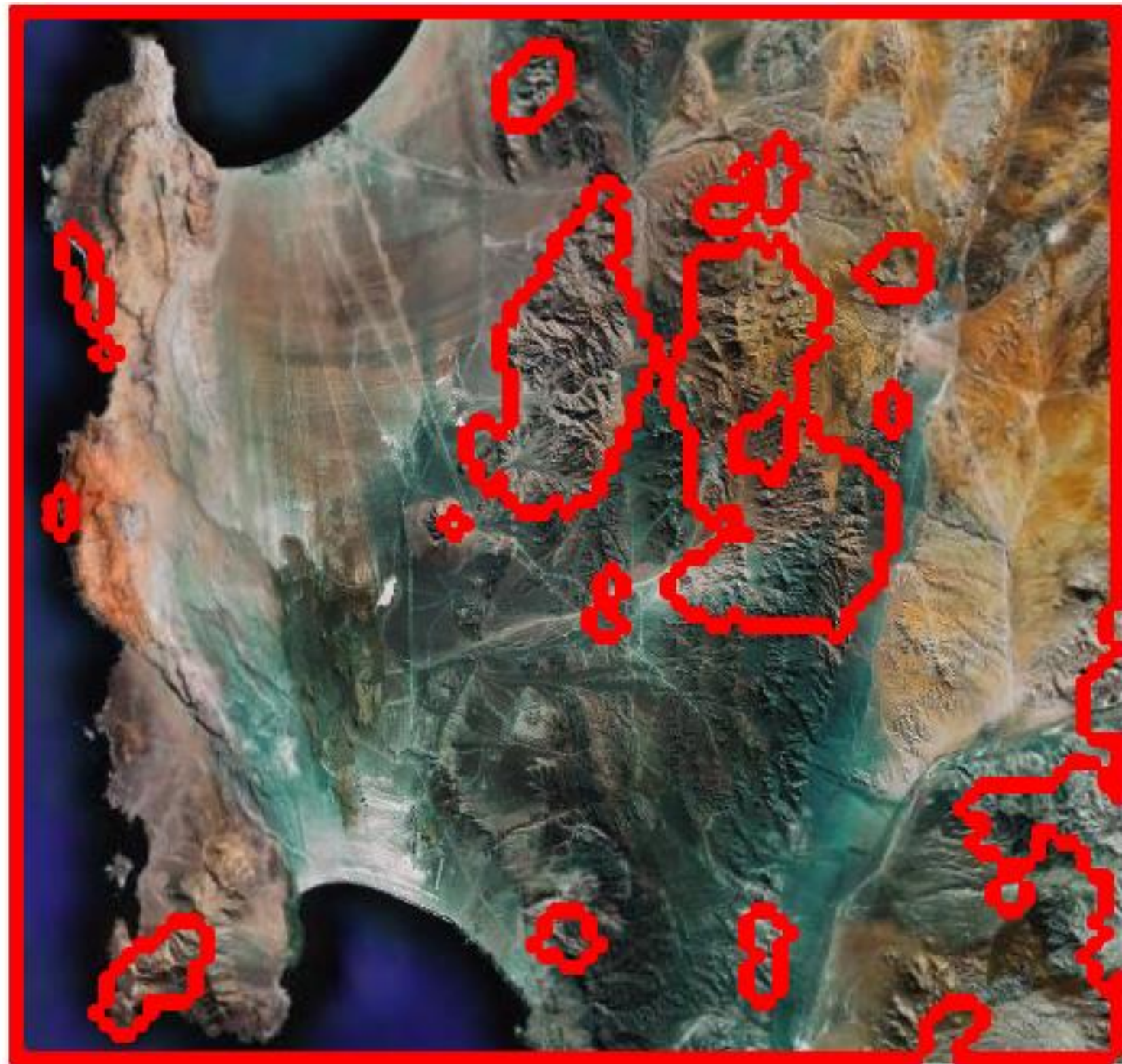


Inverse Moment Difference



Inverse
Moment
Difference
< Threshold

= Montañas





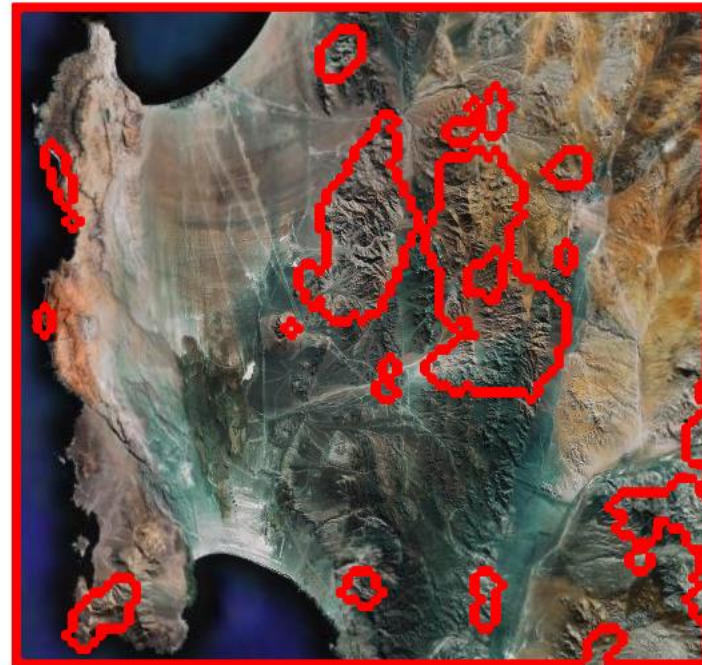
T5: Inverse Difference Moment

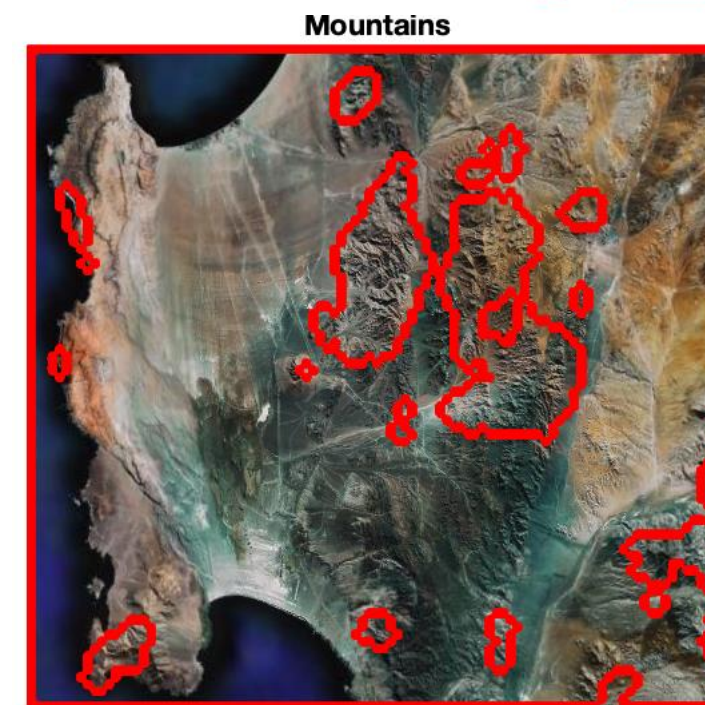
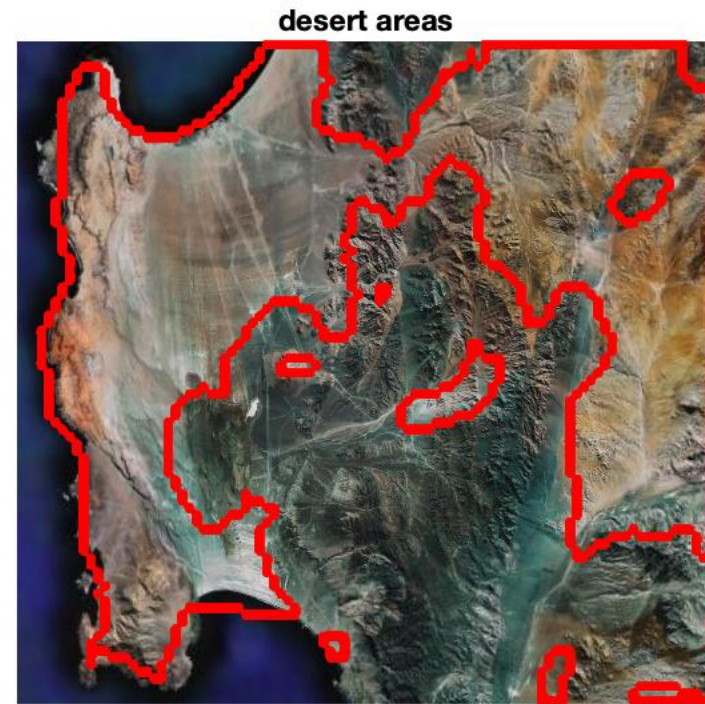
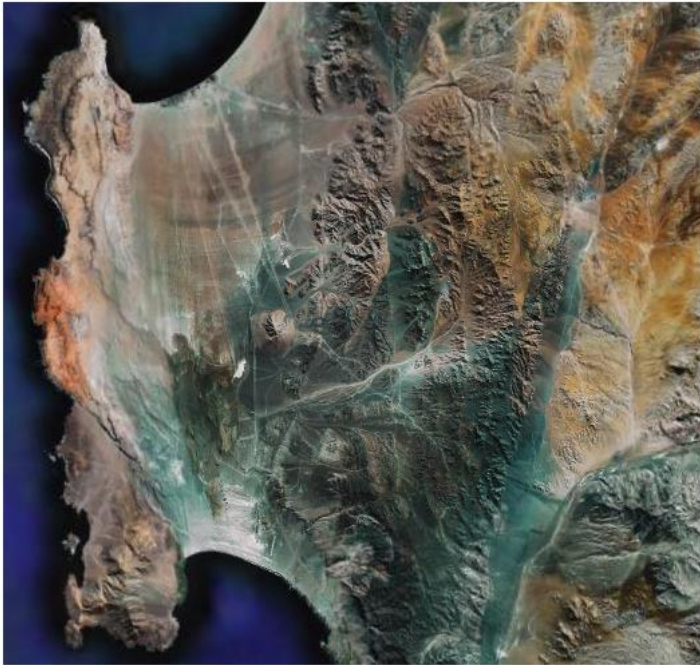


Thresholding of T5 using Otsu



Mountains





Características de Intensidad

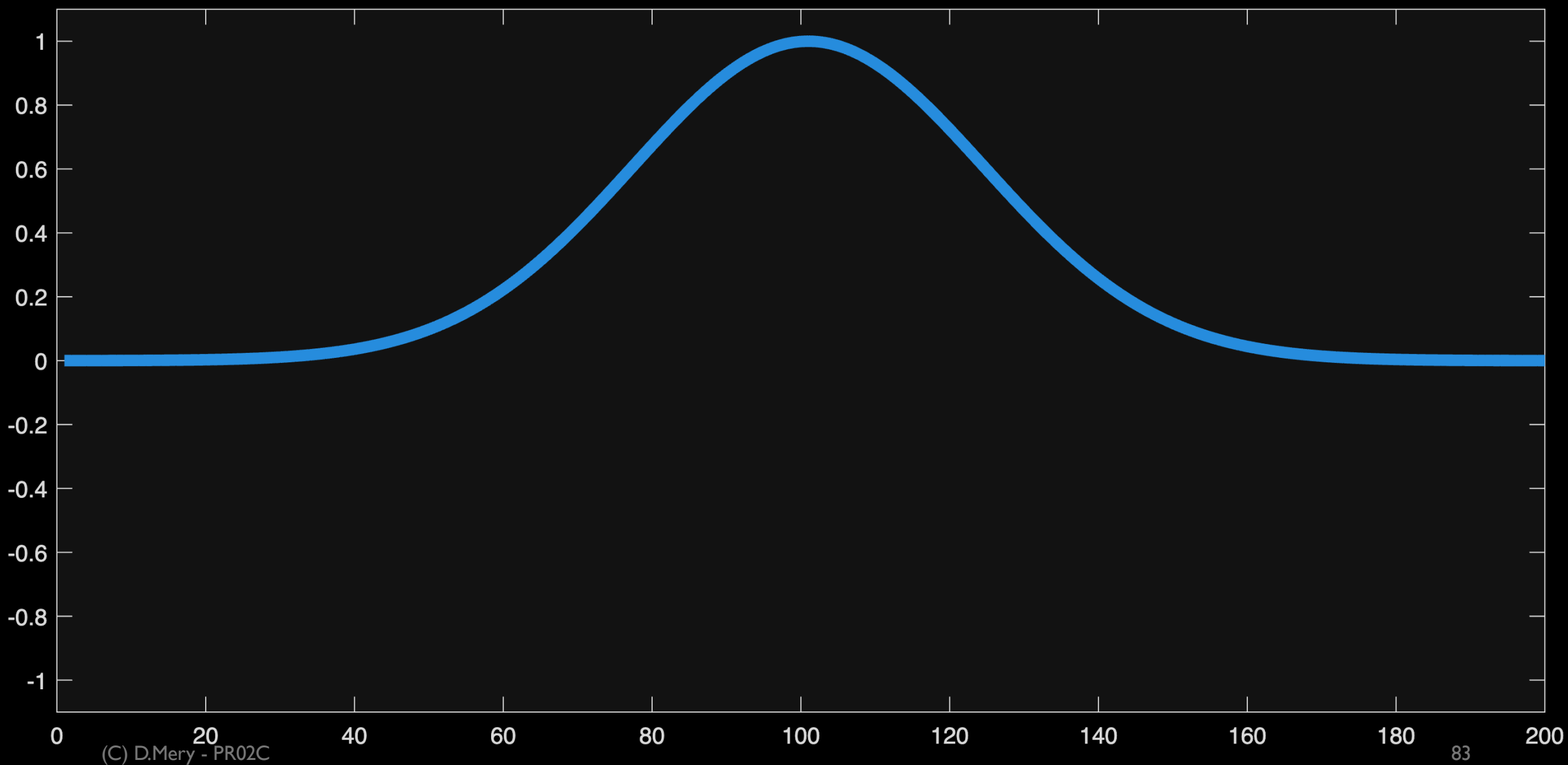
Características de Textura según Gabor

Haralick, R. M. (1979). Statistical and structural approaches to texture. *Proceedings of the IEEE*, 67(5), 786-804.

Características de Gabor

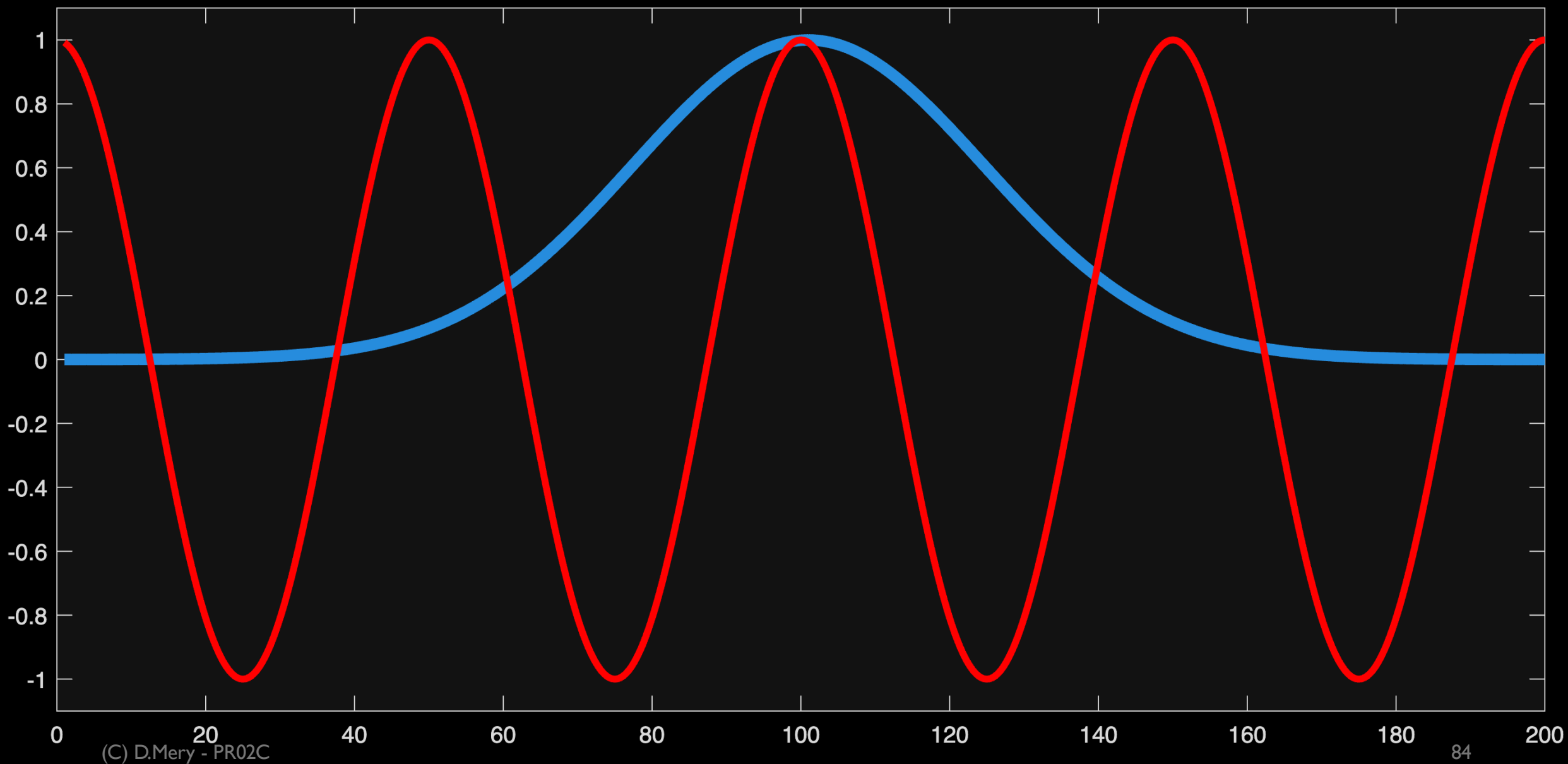
Funciones de Gabor en ID

GAUSSIANA

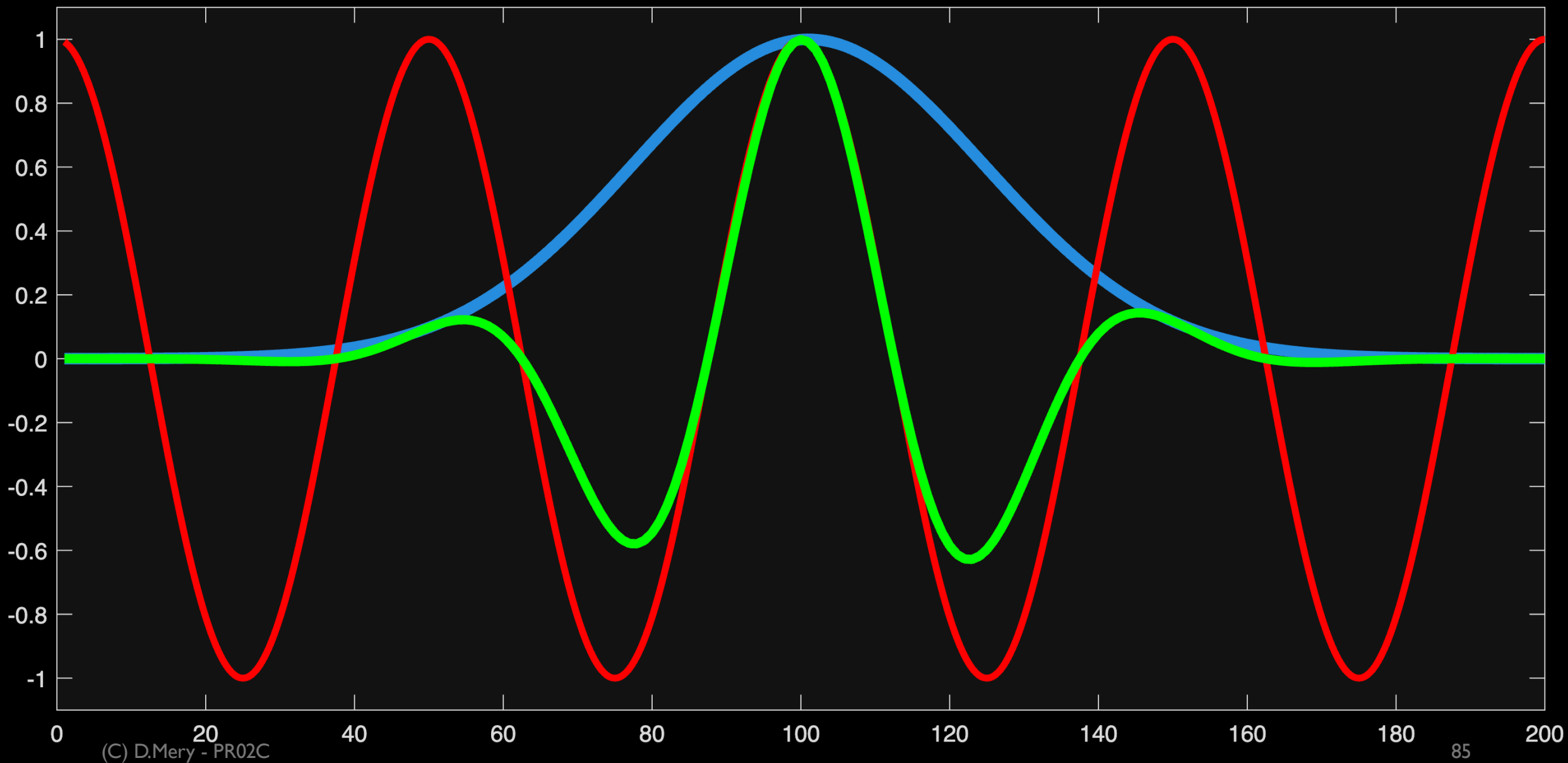


GAUSSIANA

SINUSOIDE

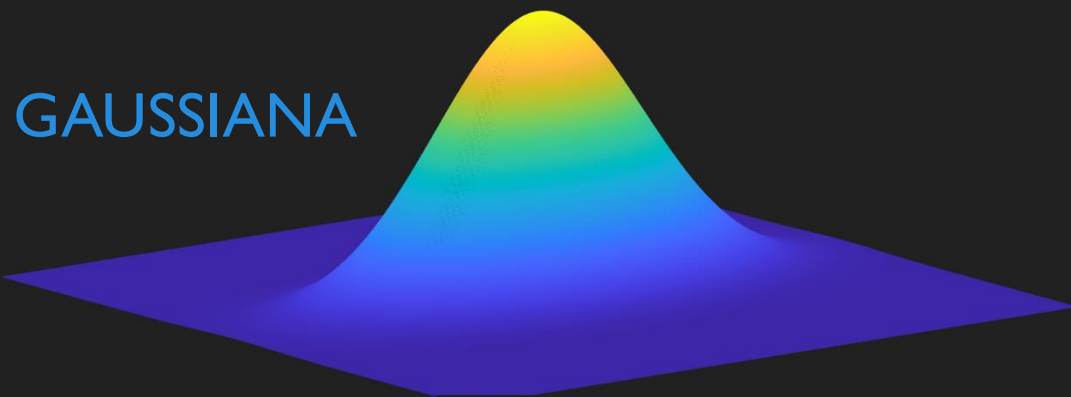


$$\text{GABOR} = \text{GAUSSIANA} \times \text{SINUSOIDE}$$



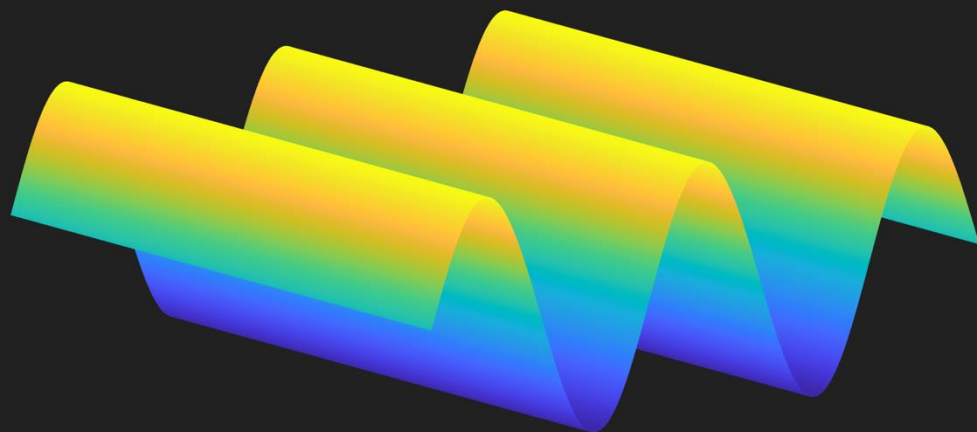
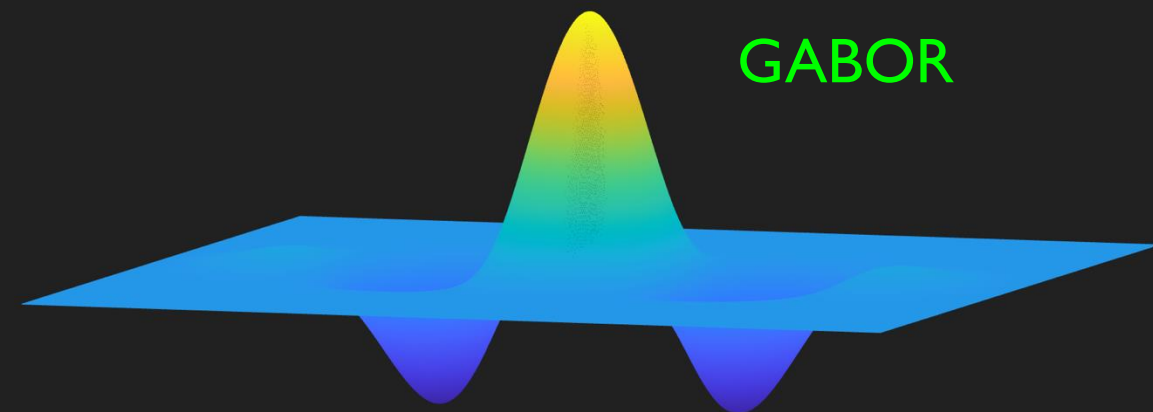
Características de Gabor

Funciones de Gabor en 2D

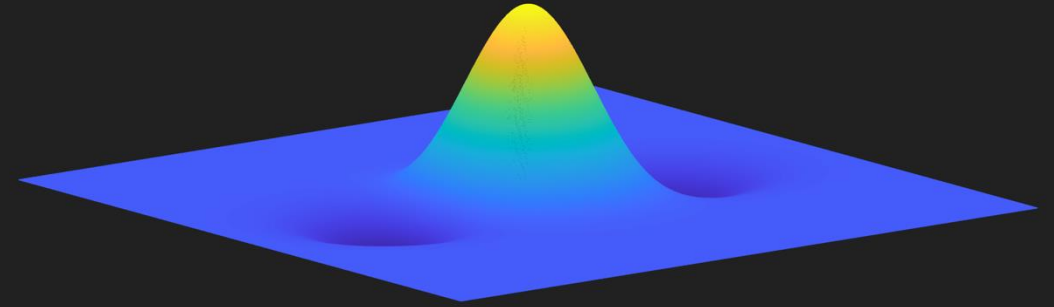
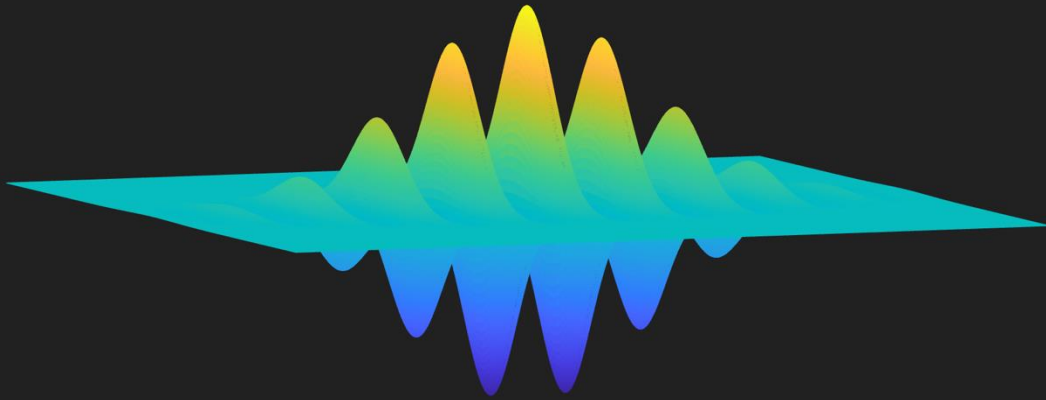


X

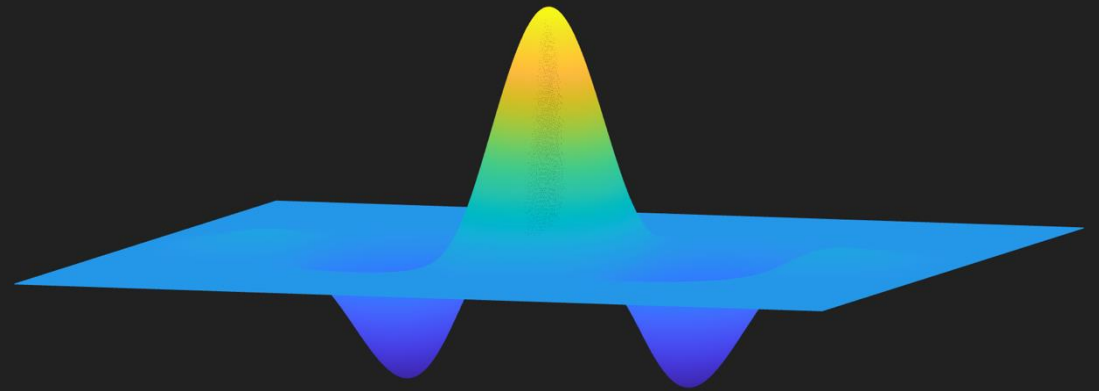
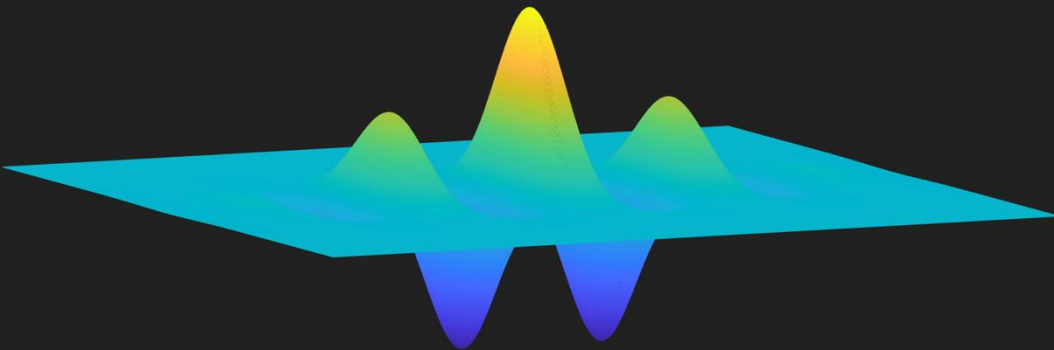
=

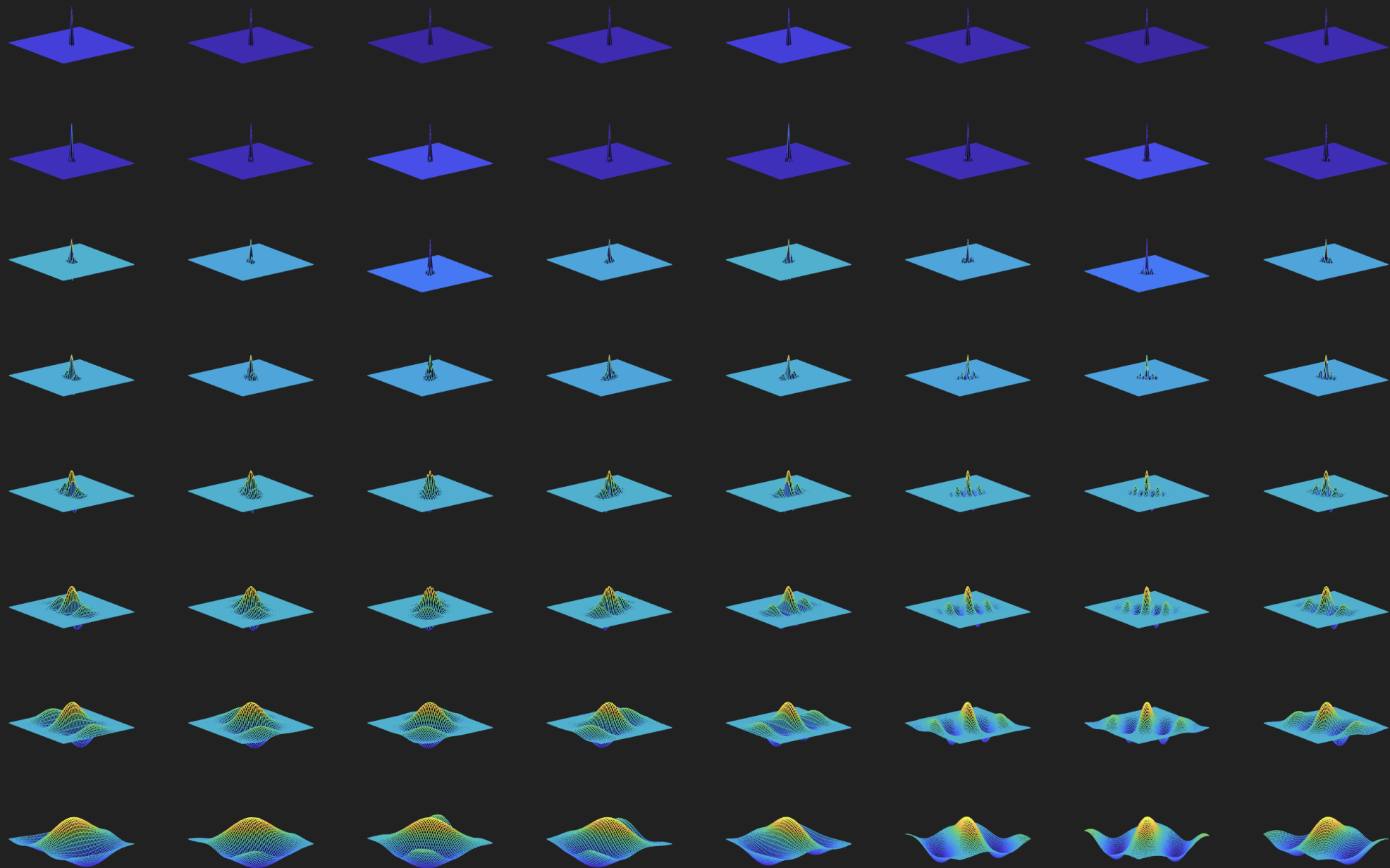


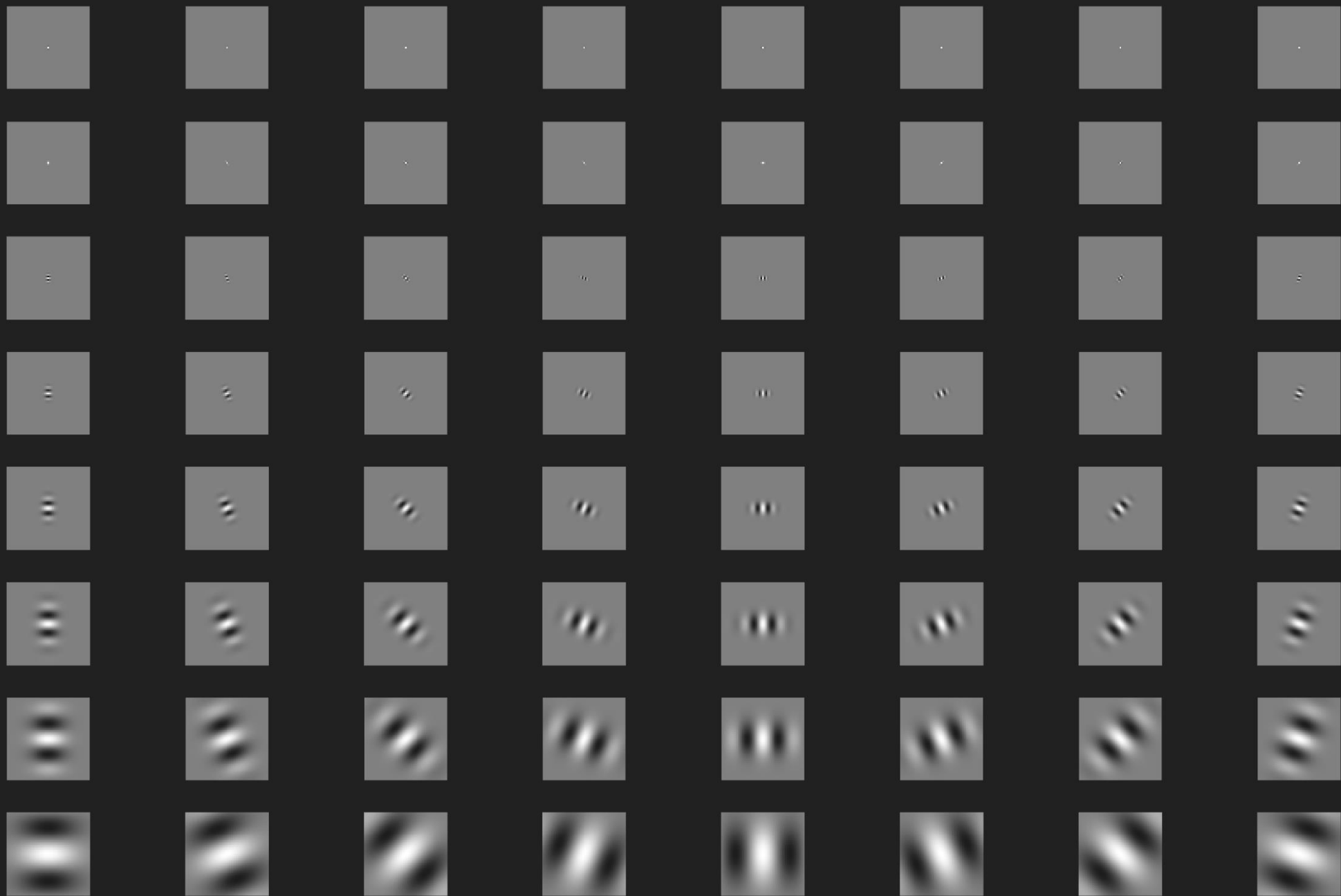
SINUSOIDE



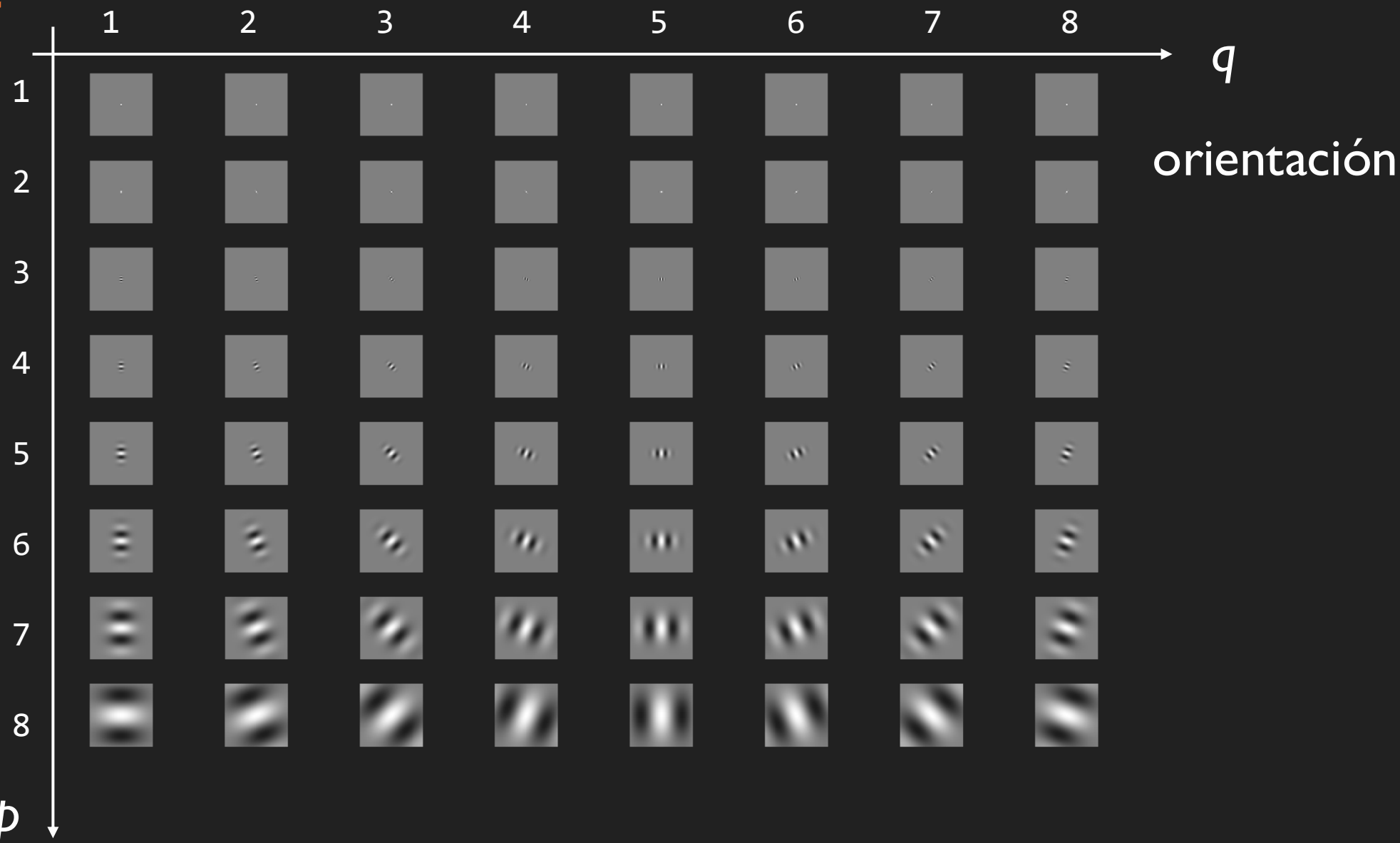
Cambiando la frecuencia de la senoide, el sigma de la Gaussiana, y la rootación...







Kernels de Gabor



Características de Gabor

Definiciones Matemáticas

Manjunath, B. S., & Ma, W. Y. (1996). Texture features for browsing and retrieval of image data. *IEEE Transactions on pattern analysis and machine intelligence*, 18(8), 837-842.

Funciones de Gabor

The Gabor functions are a complete (but a nonorthogonal) basis set given by:

$$f(x, y) = \frac{1}{2\pi\sigma_x\sigma_y} \exp\left(-\frac{1}{2}\left(\frac{x^2}{\sigma_x^2} + \frac{y^2}{\sigma_y^2}\right)\right) \exp(2\pi j u_0 x) \dots (15)$$

where σ_x and σ_y denote the Gaussian envelope along the x and y -axes, and u_0 defines the radial frequency of the Gabor function.

Funciones de Gabor

self-similar filter bank can be obtained by appropriate dilation and rotation of $f(x, y)$ through the generating function

$$f_{pq}(x, y) = \alpha^{-p} f(x', y')$$

where

$$\begin{aligned} x' &= \alpha^{-p}(x \cos \theta_q + y \sin \theta_q) \\ &= \alpha^{-p}(-x \sin \theta_q + y \cos \theta_q), \\ \alpha &> 1; \quad p = 1, 2, \dots, S; \quad q = 1, 2, \dots, L. \end{aligned}$$

The integer subscripts p and q represent the index for scale (dilation) and orientation (rotation), respectively. S is the total number of scales and L is the total number of orientations in the self-similar Gabor filter bank. For each orientation q , the angle θ_q is given by

$$\theta_q = \frac{\pi(q-1)}{L}, \quad q = 1, 2, \dots, L.$$

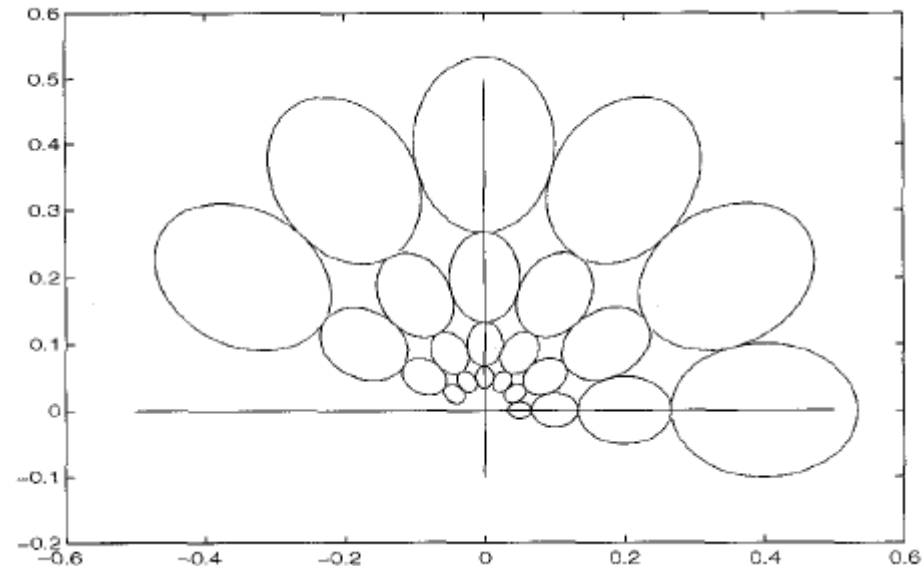
Funciones de Gabor

$$\alpha = \left(\frac{f_h}{f_l} \right)^{-(1/(S-1))}$$

$$\sigma_x = \frac{\sqrt{2 \ln 2}(\alpha + 1)}{2\pi f_h(\alpha - 1)}$$

$$\sigma_y = \left[2 \ln 2 - \left(\frac{2 \ln 2}{2\pi \sigma_x f_h} \right)^2 \right]^{1/2} \cdot \left[2\pi \tan\left(\frac{\pi}{2L}\right) \left(f_h - 2 \ln\left(\frac{1}{4\pi^2 \sigma_x^2 f_h}\right) \right) \right]^{-1}$$

Se asegura así que la respuesta en frecuencia de los filtros Gabor escogidos no se traslapen.



Funciones de Gabor

$$I_{pq}(x, y) = \left\{ [f_{pq}(x, y)_e * I(x, y)]^2 + [f_{pq}(x, y)_o * I(x, y)]^2 \right\}^{1/2} \quad (6)$$

where “*” denotes 2-D convolution operation, and $f_{pq}(x, y)_e$ and $f_{pq}(x, y)_o$ represent the even and odd parts of the Gabor filter separated from (3).

for $p=1 \dots s$ for $q=1 \dots r$ (‘s’ scales and ‘r’ orientations)

$$g(p, q) = I_{pq}$$

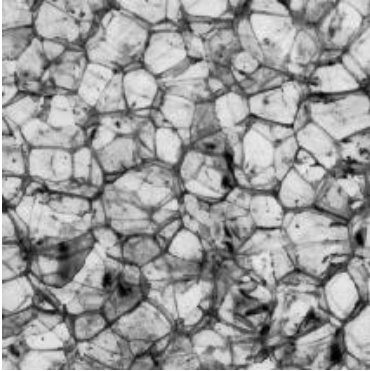
$$J = (g_{\max} - g_{\min}) / g_{\min}$$

Typically, $r = s = 8$

Características de Gabor

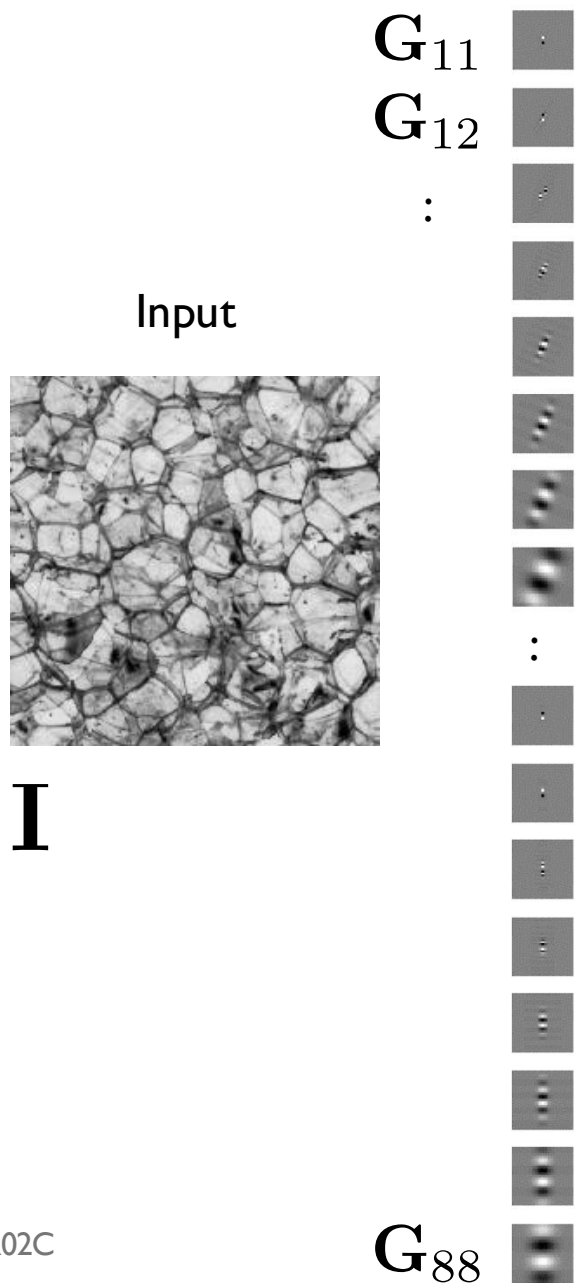
Uso en Reconocimiento de Patrones

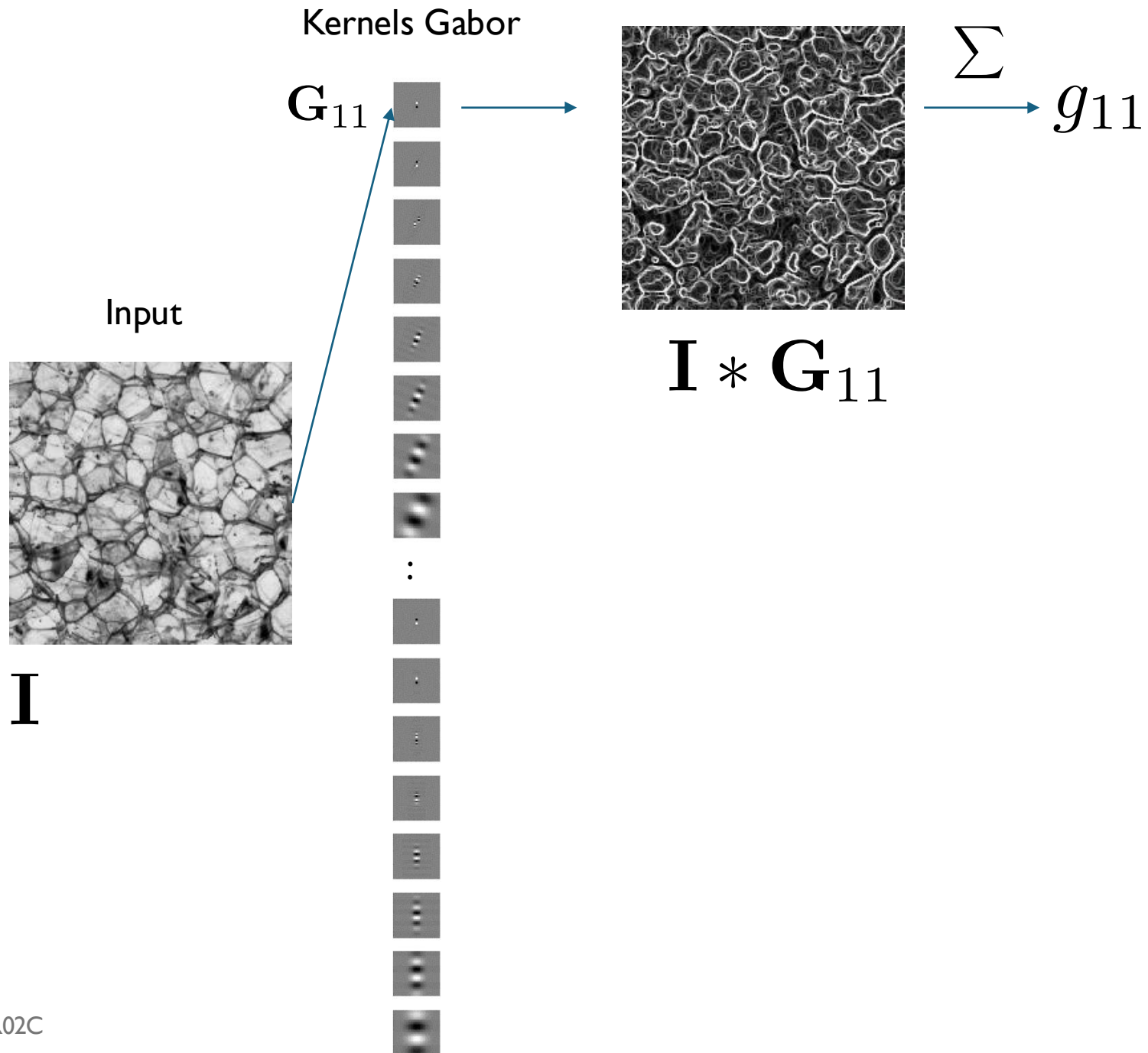
Input

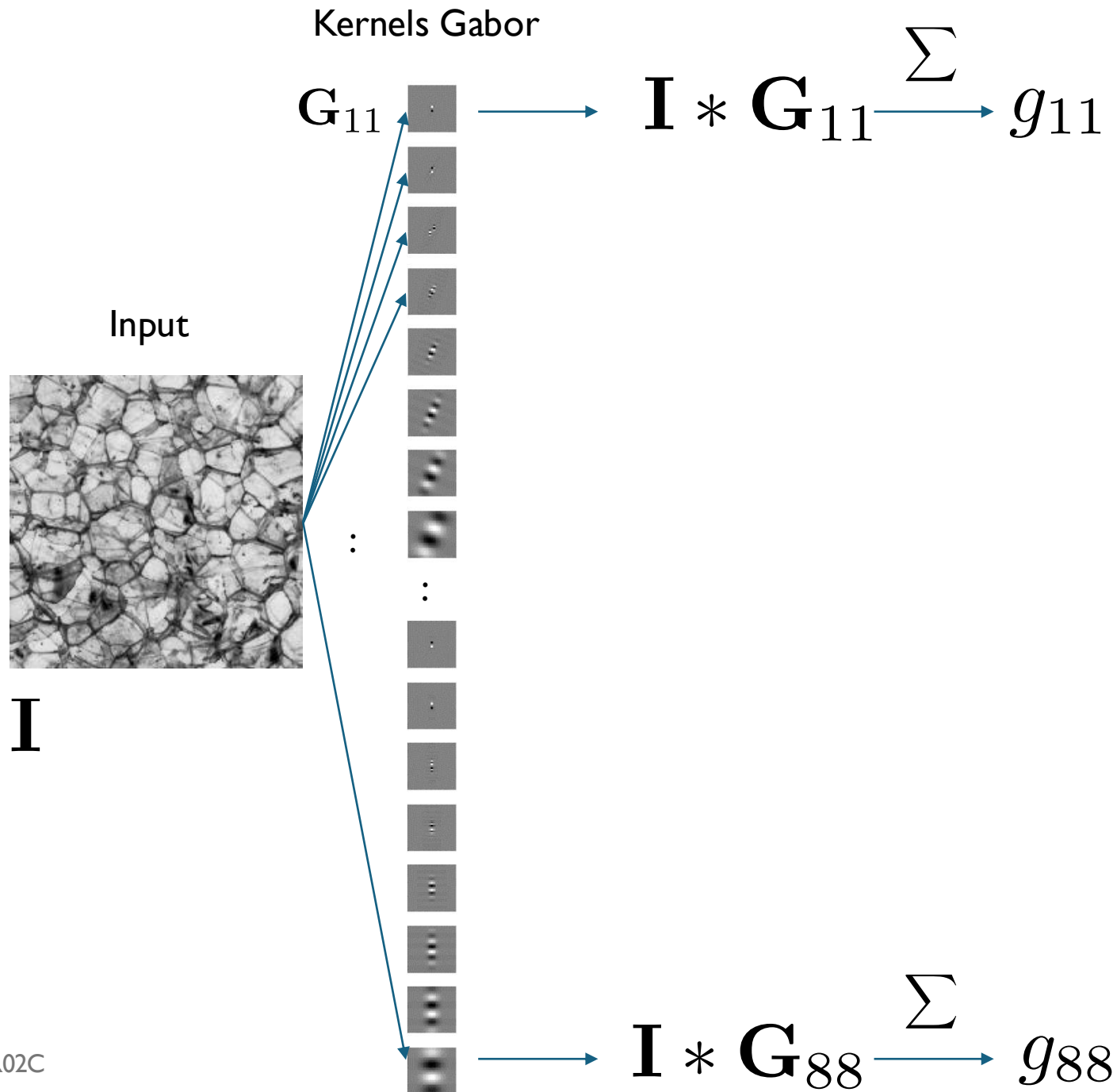


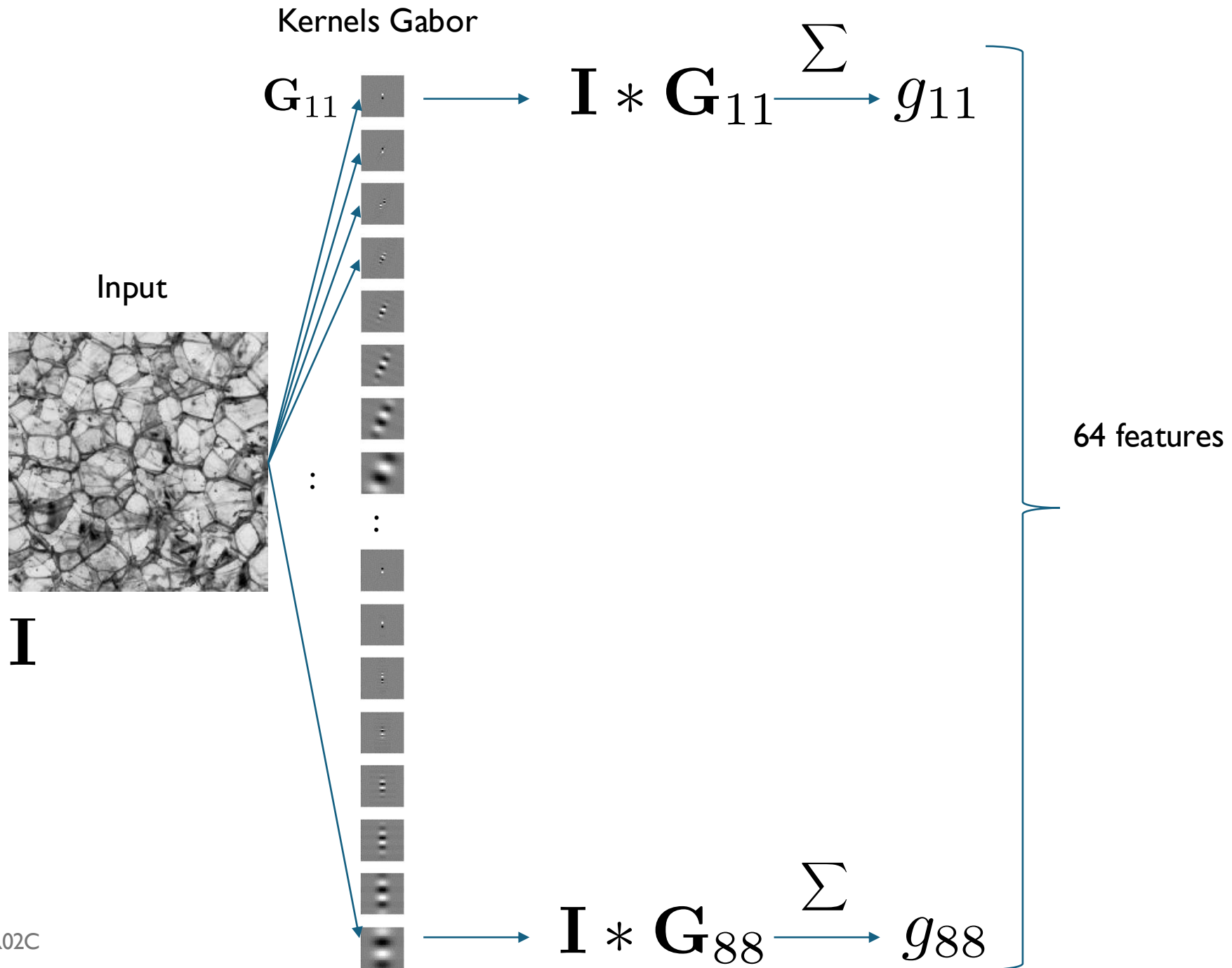
I

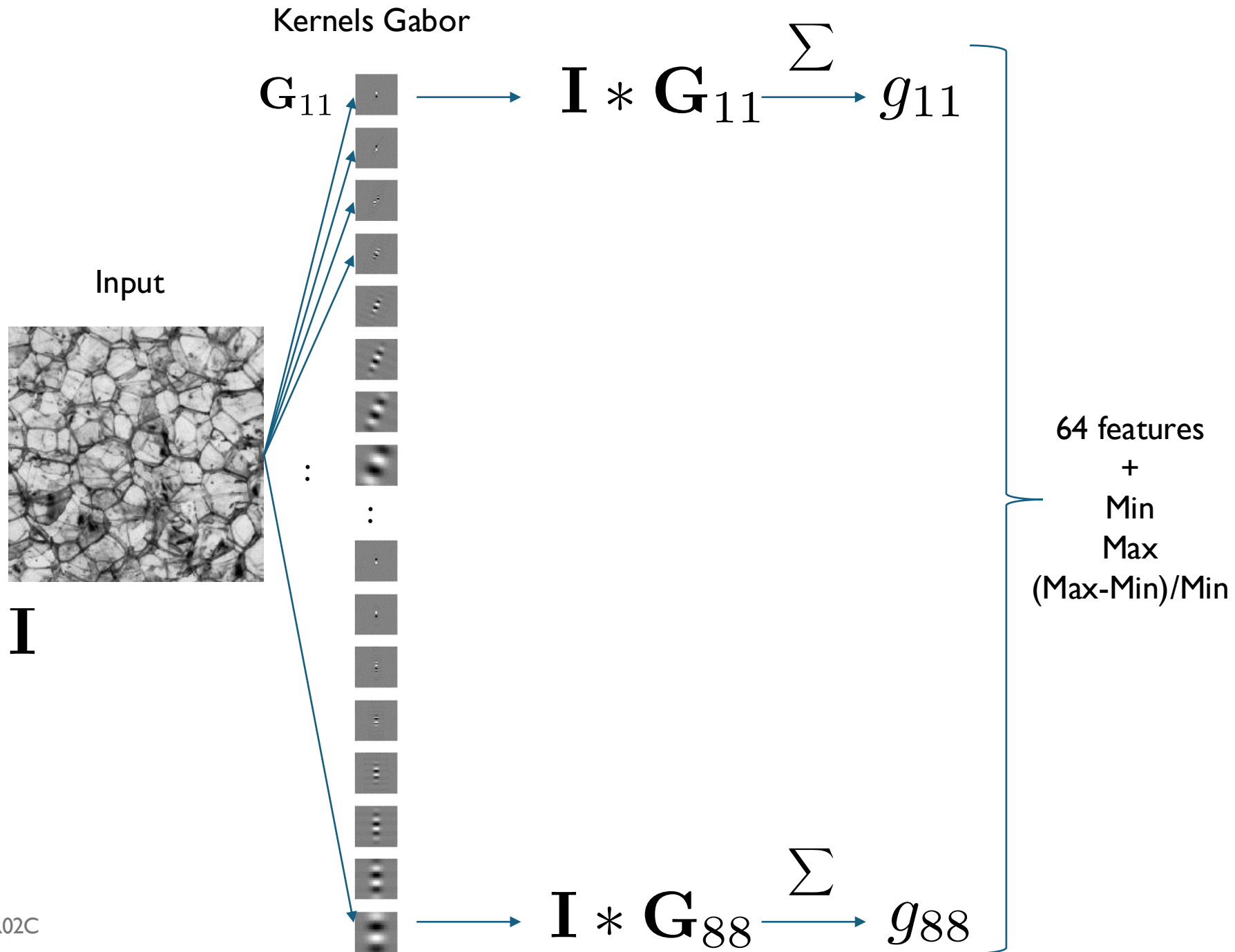
Kernels Gabor





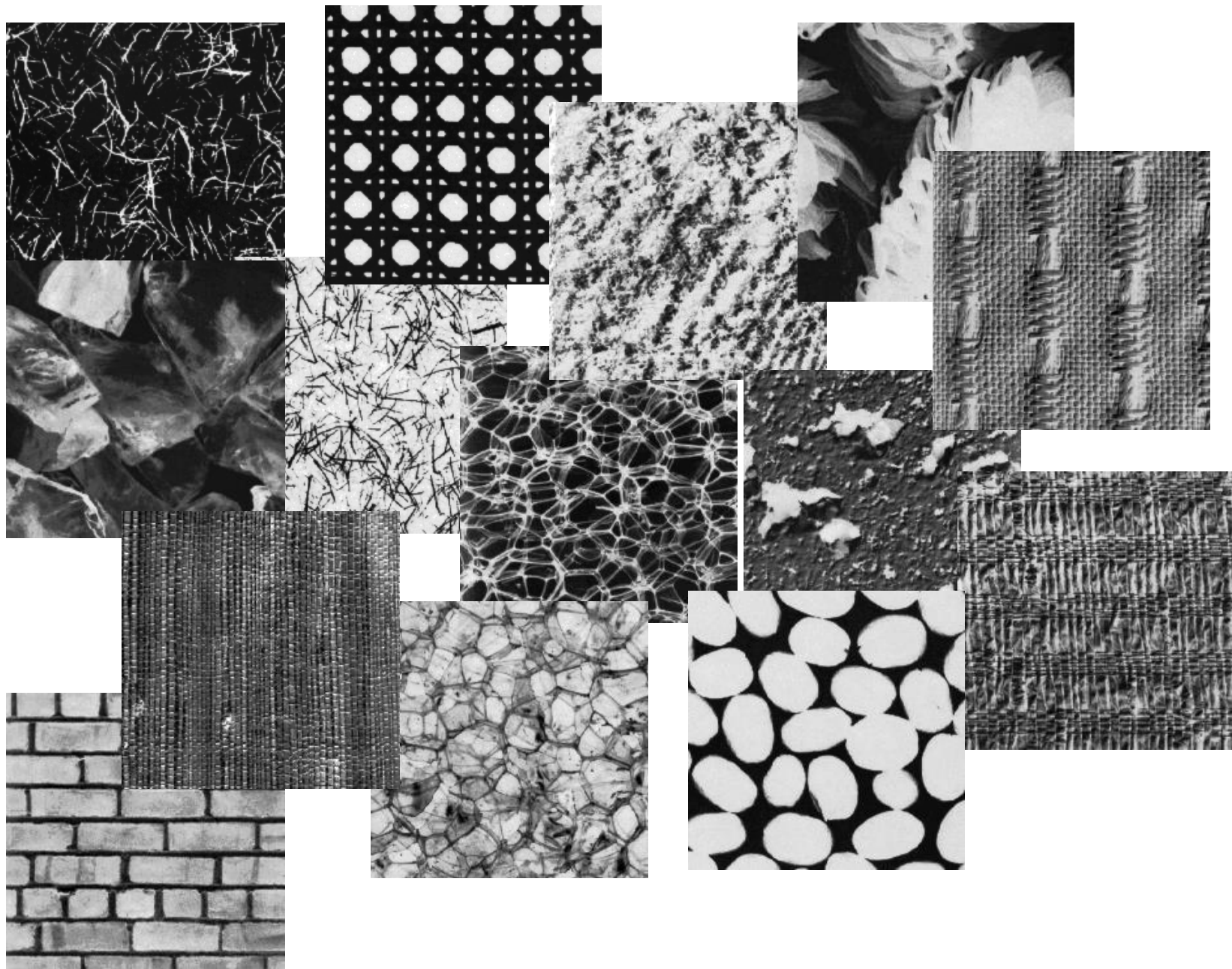


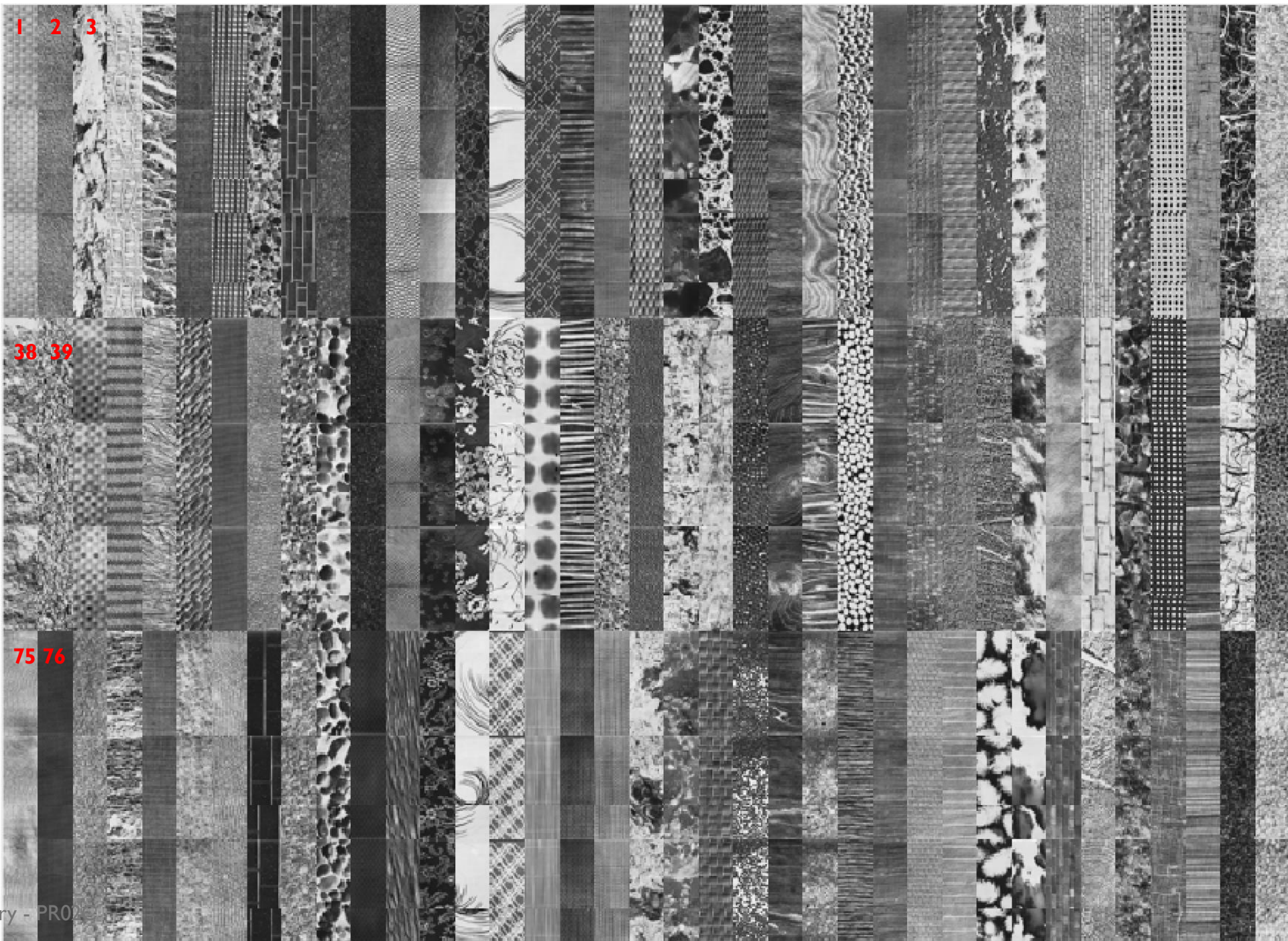




Ejemplo

Base de Datos: Brodatz
| | | texturas
9 imágenes por textura





Resultados usando Texturas de Gabor las III

texturas se reconocen con un 92% de accuracy

